



**KTH Industrial Engineering
and Management**

Illustrative scenarios of biofuel policies for transport sector in a developing country; A case study of Zambia

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 KTH Industrial Engineering and Management Master of Science Thesis EGI 2012:72MSC EKV903 Illustrative scenarios of biofuel policies for transport sector in a developing country; A case study of Zambia		
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Abstract

The increasing demand for transportation fuels has led most countries around the world to look for alternatives to the predominant fossil fuels in the sector. Furthermore, the needs for security of energy supply, social – economic and sustainable development are among the factors driving the search for alternative fuels with regard to the fact that most oil producing countries are politically unstable. Developing countries are not an exception and are also formulating policies that will enhance development of alternatives to fossil fuels.

This study analyses the opportunities and consequences of the development of the biofuels industry in Zambia from a local, regional and global perspective with regard to policy, technical, socio – economic, environment and sustainability considerations. From analysis of global, regional and local situation, the development of biofuels in Zambia is simulated using Long – range Energy Alternatives Planning system (LEAP) under different policy assumptions with consideration of current and future demands. Furthermore, the socio – economic and environmental consequences are modelled in LEAP using these policies and other assumptions which include demand – cost projections. The environmental loadings are also simulated for the period with the assumption that biofuels are zero net emitters of green house gases. Land requirements are then estimated from these results for the simulated period. A model of illustrative scenarios for implementation of different biofuels policies is developed which can be used to monitor and analyse their consequences by adjusting the major parameters depending on varying circumstances.

The results of this study show that there is considerable potential for feedstock, land, water and mature technology for production of both biodiesel and bioethanol to meet

blending demands for the simulated period. There is considerable potential for production of bioethanol from sugar cane and sweet sorghum using the well established fermentation processes. Biodiesel can be produced from jatropha and other oil bearing plants. Most of the current fleet of vehicles in Zambia was manufactured after 1990, thus there is little or no modification required on engines to run on low biofuel blends considered in the simulations. It is clear from the results of the simulation that biofuel demand for blends of 5% in 2015 rising to 10% in 2030 with current yields required for both can be achieved with the available land and water resources. To achieve the 10% blending target set in the simulated period for bioethanol and biodiesel from sugar cane and jatropha, land required is 4,480Ha and 47,000Ha respectively for growing this feedstock at current yields for the region. The GHG reduction would be equivalent to the 10% in 2030 if we assume that biofuels are zero net emitters. However, policies have to be formulated and implemented which should include mandatory blending targets and incentives on tax and investments.

Key Words: Biofuel, bioethanol, biodiesel, environment, sustainability

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Nomenclature

CEEEZ	Centre for Energy and Environmental Engineering Research in Zambia
DME	Dim Ethyl Ether
ESCOs	Energy Supply Companies
ZEMA	Zambia Environmental Management Agency
ERB	Energy Regulation Board
EU	European Union
E5, E10, E20	Gasoline – Ethanol blend at 5%, 10%, 20% respectively
FNDP	Fifth National Development Plan
FAME	Fatty Acid Methyl Ester
GHGs	Green house gases
GMOs	Genetically Modified Crops
GRZ	Government of the Republic of Zambia
HFO	Heavy fuel oil
IEA	International Energy Agency
ILO	International Labour Organisation
IPP	International Parity Pricing
IPCC	International Panel on Climate Change
Kw	Kilowatt
LEAP	Long – range Energy Alternatives Planning System
LSFO	Low sulphur oil
LFO	Light fuel oil
LPG	Liquefied petroleum gas
MW	Mega Watt
MEWD	Ministry of Energy and Water Development
NEP	National Energy Policy
NISIR	National Institute for Industrial and Scientific Research
OMCs	Oil Marketing Companies
PV	Photovoltaic
RTSA	Road Transport and Safety Agency
SAAP	Southern Africa
SADC	Southern Africa Development Organisation
Syngas	Synthetic gas
TAZAMA	Tanzania Zambia Mafuta
UNDP	United Nations Development Program
USA	United States of America
WTO	World trade organization
ZESCO	Zambia Electricity Supply Corporation

1. INTRODUCTION

1.1 Background

On the global scale, the problem of increasing energy demand arising from limited fossil fuel resources has become of great concern in the world. Fossil fuels currently provide 80% of current energy demand and the apparent effect of their emissions on the atmosphere are driving international and national action plans to find sustainable energy solutions. Biomass contributes only 11 % of the total global energy demand, while nuclear and solar contribute 7% and 2% respectively (STEM, 2003). Technological improvements and innovations are being sought to resolve the energy crisis which is looming worldwide. Today, increasing efficiency in energy consumption and finding sustainable alternatives have increasingly gained importance in many nations. Other fuels such as biofuel and hydrogen are being considered as alternatives to fossil fuels for different reasons. Among these alternatives, biofuels have been identified as one of the viable options for displacing transportation fossil fuels (IEA report 2006). This is largely due to their being renewable and less polluting compared to fossil fuels. Biofuels are renewable and considered to be a clean source of energy when produced in a sustainable way. However, most of the technologies for conversion of biomass to biofuel are still in their early stages of development and a lot of challenges on its sustainability as an alternative to fossil fuels still remain unresolved (IEA report 2006). The costs of producing biofuels using some technologies are high compared to the well established refining processes for crude oil to obtain gasoline and diesel.

The need to reduce carbon dioxide as well as other greenhouse gas emissions and increasing demand for energy has driven many countries to turn to liquid biofuels as a means to sustainably meet their transport fuel needs. However, today this is not the only motivation for increasing production of liquid biofuels. Some countries like Brazil and Malaysia are looking at the opportunities for export to Europe, USA and Japan. Other countries such as China, South Africa and India are motivated by ecological

aspects and the need to reduce dependence on oil imports (IEA Report, 2006). A global survey carried out by the world watch institute shows that as a basis for the formulation of international policy on liquid biofuels, country specific considerations of policies, biomass resources, markets and technological capacity have to be taken in to account. The security of energy supply, potential for specific feedstock, technological capacity and sustainability issues for each specific country have to be considered.

The global energy crisis has not spared developing countries and Zambia is not an exception. In the vision 2030, developed by the government of Zambia in 2006 which seeks to achieve middle income status for the country by the year 2030, biofuels are given an important place in the future energy system of the country. Thus, in the Fifth National Development Plan (FNDP) the government of Zambia intended to have an energy system that allows utilisation of renewable energy inclusive of liquid biofuels. This is clearly outlined as an objective to see the country become an energy exporter rather than a net importer.

In this policy document there are intentions to set-up a biofuel development program whose goal will be to facilitate the development of the biofuels industry. The plan is to develop a strategy for promoting the utilisation of biofuels as these fuels are currently not being used. This program will also look at developing a biofuels implementation strategy which should ensure security of energy supply. The other program that is being proposed is renewable and alternative energy development program whose main objective will be providing and disseminating up to date information on renewable and alternative energy resources. This is expected to enhance effective planning and awareness, development, management and utilisation by undertaking comprehensive assessments of renewable energy potential in selected parts of the country in order to develop a resource map and bankable project proposals. Under the same program awareness campaigns are to be conducted to educate the public on biofuels production and utilisation. Finally the government intends to create an energy development fund which shall provide funds for different types of energy projects including promotion of gasification technology, biofuels and other renewable energy.

In this regard, the government is working towards developing a national approach to biofuels in order to reap their potential benefits, while ensuring sustainable development, and avoiding the creation of new risks. The current energy policy includes biofuel blending in the transportation fuel mix of the country (NEP 2008).

The decision to include biofuel in the transportation fuel mix has an interdisciplinary character combining issues from the fields of renewable energy systems, sustainable development, energy markets and climate change. Other issues include food security, national, regional and global development agenda. Biofuels use and technology is in its infancy in Zambia, thus the country has much to learn from experiences of countries already producing and using biofuels. This study intends to contribute to the development of a sustainable biofuel industry by providing answers to the following questions;

- Does Zambia need to look for alternative transportation fuel?
- Are biofuels a feasible alternative to transportation fossil fuel in Zambia?
- What are the prospects for a sustainable biofuel industry in Zambia?

In this study, the global biofuel trends are analysed and experiences are used to answer these questions and in formulating a frame work model for the development of a sustainable biofuels industry in Zambia.

1.2 Objective of thesis

The main objective of the study is to provide pre-feasibility guidelines on the prospects and opportunities for a sustainable biofuel industry in Zambia which is in line with the global development agenda.

1.2.1. Specific objectives

The specific objectives are as follows;

- Give an overview of the potential for biofuels production and utilisation in Zambia
- Analyse different policy scenarios with regard to future fuel demand, technology, environmental and social - economic consequences quantitatively and qualitatively.
- Evaluate the different policies, feedstock and technology for biofuels production and utilisation as well as identify the more sustainable options for Zambia.

1.3 Scope of work

The scope of the study covers;

- A review of the global biofuels situation with regard to existing policies, feedstock, production, processing technologies, consumption trends and motivations for development of the biofuels industry around the world
- Analysis of energy situation in Zambia with respect to ; security of energy supply, availability of biofuel resources, processing technology, end-use applications, government policy, demand - supply scenario for petroleum, infrastructure and investment requirements.
- Discussion on issues regarding biofuels development, with respect to social - economic and environmental sustainability,
- Conclusions on future outlook of policy options, institutional models and technology with regard to the local, regional and global development agenda.

1.3.1 Method of attack

The strategy and process used in the study are outlined below;

- Firstly, data was collected through literature survey of research papers, various reports, government documents, conference proceedings and internet on the biofuels sector at global, regional and local levels. Discussions with key players in

biofuels industry which included regulators, producers and processors. Among these were farming community, research institutions, Non – Governmental Organisations, industries such as National Institute for Scientific and Industrial Research(NISIR), Zambia Environmental Management Agency(ZEMA), Energy Regulation Board (ERB), Tanzania Zambia Mafuta Pipeline(TAZAMA) , INDENI, Ministry of Energy and Water Development(MEWD) and many others.

- The external and internal energy environment of Zambia is examined in order to understand aspects of energy security from both inside and outside the country. Data on regional and global studies is analysed and used to develop a conceptual model of the future of the biofuels industry in Zambia.
- Data on energy demand for Zambia was collected and used to analyse future energy patterns with the inclusion of biofuels in the transportation fuel mix using Long – range energy analysis and planning (LEAP) software program. Future energy demand patterns, environmental and some social – economic consequences for varying scenarios are simulated. Recommendations on policies, feedstock and technological options for biofuels production and utilisation are then made based on results from literature and scenario analysis.

2. OVERVIEW OF BIOFUEL PRODUCTION TECHNOLOGY

The term biofuel is used to refer to liquid or gaseous fuels for the transport sector that are produced from biomass. A variety of fuels can be produced from biomass resources including liquid fuels, such as bioethanol, methanol, biodiesel and Fischer–Tropsch diesel. Gaseous fuels such as hydrogen and methane can also be produced (Demirbas A., 2008). Biofuel has in recent years been identified as one of the most sustainable alternatives to fossil fuels for transportation. They are compatible with current fossil fuel engines and can thus share the long established infrastructure with little or no modification (EIA report, 2006). The main biofuels are bioethanol and biodiesel and are already being used as low - percentage blends with fossil fuels as well as in neat form.

Bioethanol production

There are several feedstock and technologies for converting biomass into Biofuels. Some feedstock and technologies are still in there infant stage of utilization and development while others are well established. Feedstock such as cereals, grains, sugar crops and other starches are used for production of ethanol by fermentation. Production of ethanol from biomass involves pre-treatment of feedstock, fermentation and ethanol distillation to neat form. The ethanol so obtained following distillation is ready for use in its neat form or blend as maybe required. A summary of the main steps in ethanol production are shown in Figure 2.1 below.

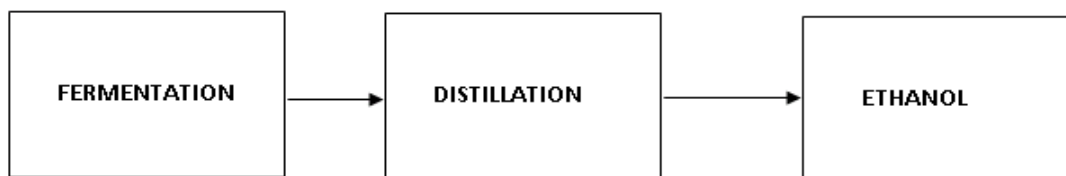


Figure 2.1: Main steps in bioethanol conversion

Cellulosic materials, including grasses, trees, and various waste products from crops, wood processing and biomass from municipal wastes can also be converted to ethanol. Biodegradable agricultural and forest wastes can also be converted in to ethanol. Conversion of cellulosic biomass is much more complex as the cellulose first has to be broken down to simpler sugars before the process above is carried out.

Biodiesel production

Oil from different types of crop seeds can be extracted by a variety of techniques and subsequently converted into methyl esters. Oil crop seed that may be used as feedstock include jatropha, rape, soybean, sunflower and cotton seed. The oil produced can be used directly or converted into methyl esters through a process known as transesterification. The product of esterification is of lower viscosity reducing the problems associated with application of fossil diesel in an engine. The biodiesel obtained can be used directly without blending or in blended form at appropriate ratios. Organic waste oil such as used cooking oil, lad and others can also be used as feedstock for production of biodiesel through the transesterification process following a pre-treatment process. Figure 2.2 below illustrates the basic conversion process for biodiesel.



Figure 2.2: Main steps in production of biodiesel

2.1.1 Bioethanol production technology

Ethanol is produced from any feedstock that contains a considerable amount of sugar or material that can be converted into simple sugars such as starch and cellulose. Sugar

beet and sugar cane are examples of feedstock together with corn, wheat, sweet sorghum and other cereals which can easily be converted into simple sugars. Other types of feedstock include some trees, grasses, sawmill and agricultural wastes as well as many other cellulosic and hemicellulosic substances which can be converted to starch with more advanced techniques.

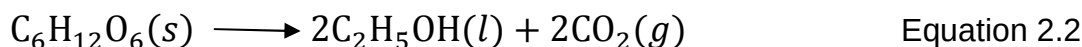
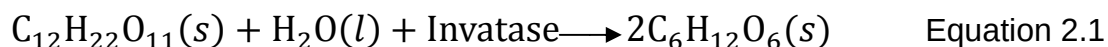
Biomass material from feedstock such as certain trees and grasses contain cellulose that can be hydrolyzed to simple sugars. Because cellulose and hemicellulose molecules bind strongly to each other, it is more difficult to break them down to simpler polysaccharides that can easily ferment to produce ethanol. The cost of pre-treatment significantly increases the cost of production using current technologies.

Most ethanol is produced by well established fermentation methods that rely on yeasts to convert glucose to alcohol. In Brazil and most tropical countries that produce ethanol, sugar cane is the most common feedstock due to the high content of sucrose which can easily be hydrolysed to glucose. In Europe, starch from grain crops is converted to ethanol although an additional step is required before the process for glucose to ethanol can be followed.

2.1.1.1. Glucose to ethanol conversion

Glucose-to-ethanol is the least complicated way of producing alcohol. This is done by using biomass (e.g. sugar cane, sweet sorghum) that contains organic structures from which we can obtain six-carbon sugars that can be fermented into ethanol. This involves removal of the sugar solution usually sucrose from the sugar containing feedstock through crushing, soaking and chemical treatment. In production of alcohol from sugarcane, the cane is first washed, crushed and then filtered to extract the juice leaving bagasse. Yeasts and other microbes are then used to reduce the complex sugars to monosaccharides (glucose) and ferment it into ethanol. The equations below represent the chemical breakdown of polysaccharides (e.g. sucrose, fructose.) to glucose and fermentation in the production of bioethanol. Equation 2.1 and 2.2

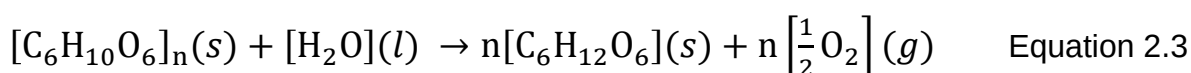
represents the basic chemical breakdown of sucrose into glucose and its fermentation to ethanol respectively.



The fermentation process takes about three days at a temperature of between 250°C and 300°C. Ethanol (usually 95% ethanol and 5% water) is obtained from the product of fermentation by fractional distillation.

2.1.1.2 Starch to ethanol conversion

The conversion of starch to ethanol involves utilisation of starch containing component of feedstock such as corn, wheat or barley. Firstly, the starch component is cleaned then ground either in wet or dry form. The starch is then converted to a six-carbon sugar using enzyme processes at temperatures of between 27°C and 38°C. The basic reaction for hydrolysis of starches to sugar is represented by Equation 2.3 below.



Once this is done, the procedure follows similarly to that of glucose to ethanol described in the previous section. This process uses only the starch component and so most of the other plant matter is left as waste.

2.1.1.3 Cellulose to ethanol conversion

Cellulosic biomass to ethanol conversion is carried out on cellulosic and hemicelluloses components. These non - sugar matter make up most of any plant material. Cellulose

and hemicelluloses can be converted to alcohol by first converting them into simpler sugars. Currently there is no commercial production of alcohol from this feedstock, however, a lot of research is being carried out in many countries around the world (IEA report 2006). The use of cellulose and hemicelluloses is seen to have a lot of potential benefits, such as increased potential feedstock e.g. biodegradable municipal solid wastes, agro and forest waste (e.g. Bagasse, sawdust) which usually just goes to waste as compost. Furthermore, use of this feedstock avoids the conflicts associated with bioenergy systems such as the food versus energy option. This is because these do not compete with food requirements unlike feedstock such as grain, soybean and others which are preferred for food. The net “well to wheels” which measures the total investment cost and benefits including environmental consequences of producing biofuels from biodegradable waste is also comparably less than that for feedstock strictly grown for fuel production. This is because there is no direct use of fossil fuels and artificial fertilizers which directly contribute to environmental degradation in one way or the other. These potential benefits make development of these technologies a priority in many national biofuel policies.

The first step in the cellulose to ethanol conversion process involves cleaning and breaking up the materials. Acid hydrolysis is applied to split the biomass into cellulose, hemicelluloses and lignin components. Cellulose is then hydrolysed into sugars by saccharification which is the conversion of complex sugar (Polysaccharide) to simple sugar (monosaccharide). In this method a dilute acid is used under high temperature and pressure or a more concentrated acid is used at low temperatures and atmospheric pressure. A decrystallised cellulosic mixture of the acid and sugars reacts in the presence of water to complete the hydrolysis process. The product is then neutralised and yeast is added to ferment the hydrolysed cellulose to ethanol. The largest set back is the dilute acid that makes the process harsh and leads to toxic products being produced during fermentation.

The common method of acid hydrolysis is expensive and so most of the current research involves development of biochemical methods that involve enzymes that can

breakdown cellulose and hemicelluloses. One biochemical method that has been introduced involves simultaneous saccharification and fermentation (SSF), which has recently been improved to include co - fermentation of multiple sugar substrates. This improves efficiencies and reduces on the number of vessels bringing down the cost (Sreenath et al, 2001). Researchers are now looking at the possibility of producing all required enzymes within the same reactor vessel. The strategy is use of the same “microbial environment” to produce both enzymes that help breakdown cellulose to sugars and to ferment the sugars to ethanol. The production of enzymes and alcohol in the same vessel, “consolidated bio processing” is seen by many as the logical end point in the evolution of biomass conversion technology. In this technique there is excellent potential for improved efficiency and cost reduction (Hamelink et al, 2003).

A lot of investment is going into this research area with USA having the largest of these programs, but even other countries like China are also actively involved. The other task is reducing the cost of the enzymes as this makes the production cost higher.

In Figure 2.3 below shows a summary of the different bioethanol conversion pathways

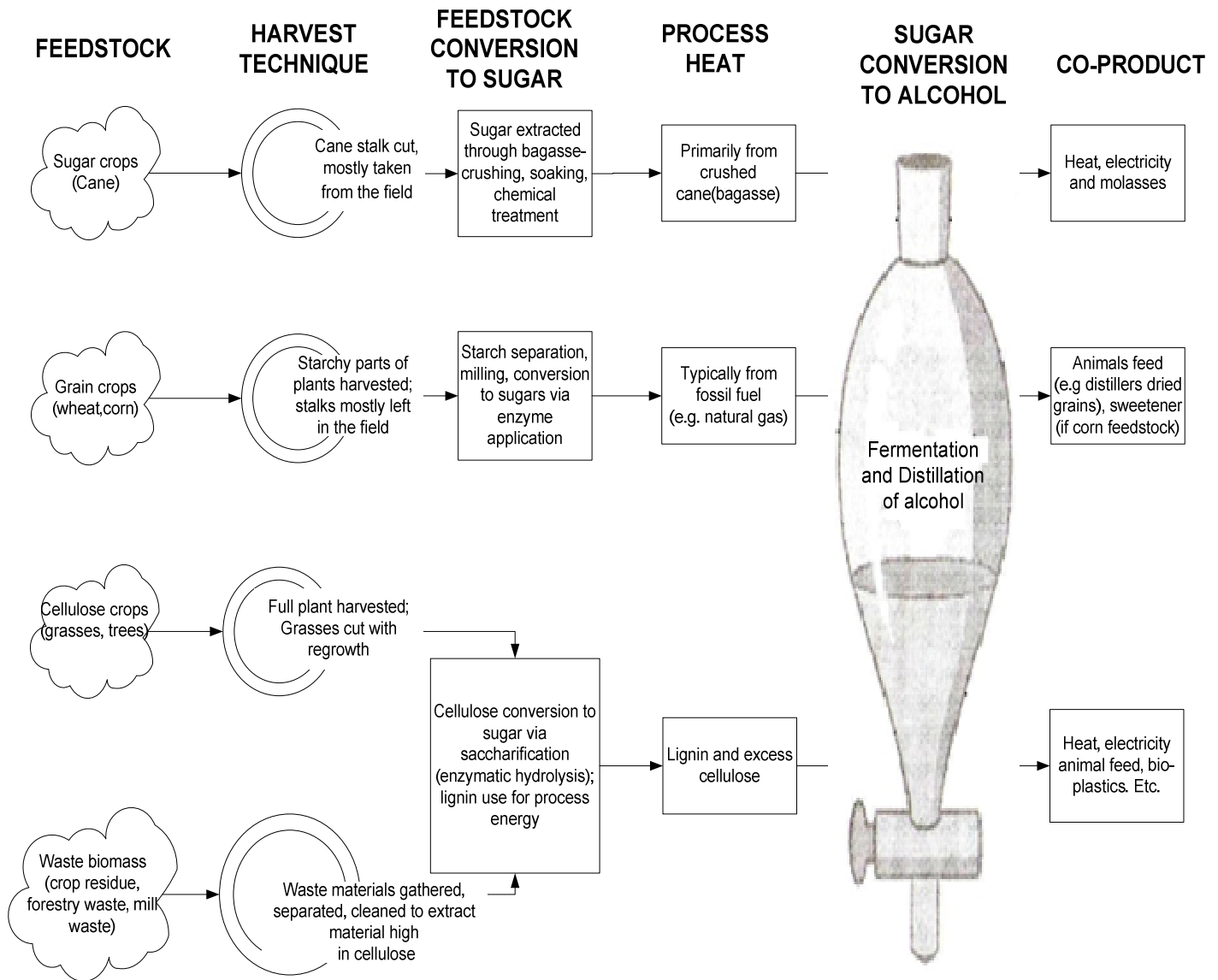


Figure 2.3: Ethanol conversion steps by feedstock and conversion method

[Adopted: IEA Report 2004]

2.1.2 Biodiesel production technology

Biodiesel is a biofuel produced from fats and oils which can be used in fossil fuel diesel engines in neat form or blended in certain proportions with fossil diesel. The feedstock

can be glyceride bearing substances such as oils derived from seed crops (e.g. soy, sunflower, rape, jatropha, etc) as well as waste cooking oil and animal fat.

Biodiesel sometimes called “Fatty Acid Methyl Ester”, or FAME is produced by a variety of transesterification technologies, a chemical process in which an alcohol reacts with a feedstock oil or fat in the presence of an acidic or alkaline catalyst. The transesterification reaction is a reversible reaction of glycerides with alcohol (Methanol or ethanol) to form fatty acid alkyl esters and glycerol. It can be alkali, acid, or enzyme – catalysed. However, currently most of commercialized technology resides in transesterification using alkali catalysed reaction (Ma F Hanna; 1999). Most biodiesel is derived from vegetable oils using sodium or potassium hydroxide as catalysts in catalytic transesterification (Demirbas A., 2009).

2.1.2.1 Production of biodiesel

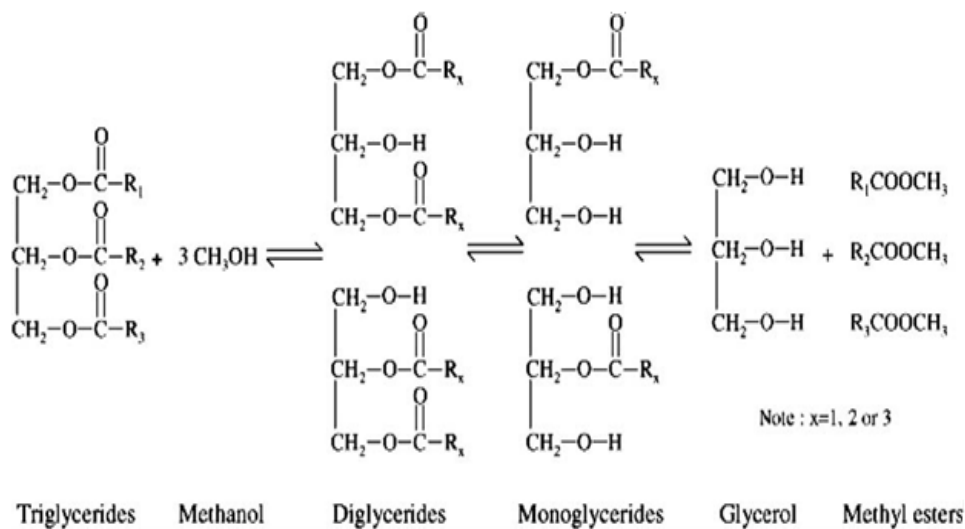
In the production of biodiesel from plant oil the first step is extraction of oil by one of several different techniques from the oil bearing seed, filtering and pre – processing to remove water and other contaminants. There are several different methods of extracting the oil from the seed with the common one being the screw and hydraulic press extractors. Water is then removed because its presence causes hydrolysis of glycerides giving salts of the fatty acids (soaps) instead of esters.

Prior to the esterification of the feedstock, a sample of the oil extract is titrated with a standard base solution to determine the concentration of fatty acids (carboxylic acids) present in it. This amount of the fat acid content is used to determine the amounts of reactants to ensure complete transesterification. The carboxylic acids are esterified to bond into glycerides, transesterified to biodiesel or removed through neutralisation.

The glycerides (usually triglycerides) are then reacted by slowly adding an alcohol in which a calculated quantity of base (usually sodium hydroxide) acts as a catalyst. Three

equivalents of the alcohol are required to react with triglycerides. Usually an excess of alcohol is added to ensure that the reaction is complete (Sandum D, et al, 2007).

The chemical reactions thus involve breaking down the triglycerides into diglycerides then monoglycerides till the final product which is glycerol and methyl ester (Biodiesel). In the transesterification reaction the hydroxyl part of alcohol takes the part of the hydrogen in the fatty acid forming the biodiesel (Equation 2.4).



Equation 2.4

[Source: *Environmental resources Letter*, 2007:2:1-6]

The alcohols used in transesterification reaction are generally short chain alcohols such as methanol, ethanol, propanol and butanol (Speight JG, 2008). Ethanol is the preferred alcohol in the transesterification reaction compared to methanol because it is biologically less objectionable to the environment. However methanol is more commonly used due to low cost and its physical and chemical advantages of being polar and is the shortest alcohol (M. Balat and H. Balat, 2010).

2.1.2.2 By - products of biodiesel production

The products of transesterification do not only include the biodiesel and glycerol but also, by products, soap, excess alcohol and water. These must be removed to obtain purified biodiesel oil. The removal of the by products is process dependent. However, the higher density of glycerol in comparison to the biodiesel is used to separate them. The excess alcohol is either removed by distillation and reused or washed out using water. Soaps can be removed and converted into fatty acids.

In the production processes for biodiesel from plant seed another by - product, cake is obtained in the initial stages. For non – poisonous seeds the cake obtained from the crushing process is sometimes used as animal feed. At times the economics of biodiesel production is improved when co – products are sold to the cosmetic and medical industry. However the market for the by products tends to be limited with increased production of biodiesel as medical and cosmetic industry only require a small amount of these.

2.1.2.3 Oil plants and biodiesel yields

Different oil yielding crops have different oil yielding capacities in litres of oil per hectare. Yields are also dependent on land, technologies and climate in specific regions. The Table 2.1 below gives some oil yield per hectare of land for some biodiesel yielding plants at time of harvest. It is worth noting that maturity periods in table 2.1 differ with respect to plant species and climate conditions (journeytoforever.org/biodiesel-yield.html#yield).

Table 2.1: Biodiesel oil yields

Crop	Kg Oil/ha	Litres oil/ ha
Linseed (Flax)	402	478
Hazel nuts	405	482
Euphobia	440	524
Pumpkin seed	449	534
Coriander	450	536
Mustard seed	480	572
Sesame	585	696
Sunflower	800	952
Tung oil plant	790	940
Cocoa	863	1026
Jatropha	1590	1892
Avocado	2217	2638
Jajoba	1528	1818
Peanuts	890	1059
Oil Palm	5000	5950
Olives	1019	1212
Rapeseed	1000	1190
Opium Poppy	978	1163

[Source: <http://journeytoforever.org/biodiesel/>]

2.1.3 Other biofuel production technologies

The other biofuel production process involves biomass gasification and conversion to liquid fuels. Ethanol can be produced in this way. However, other fuels are more easily produced and at lower cost using this approach than ethanol (IEA 2004 Report). The production of ethanol by gasification involves passing hot air past the biomass in a gasifier and reacting the product gas to the final liquid product. Synthetic diesel and gasoline can be produced using these approaches. There are a variety of processes that are available both for gasification and conversion of product gas to final fuel. The

IEA's implementing agreement on bioenergy is helping to coordinate research in this area.

2.1.3.2. Biomass gasification

Gasification is the process by which synthetic gas (Syngas) is produced by combustion of biomass in a limited amount of oxygen. The gasification process produces a variety of gases ranging from methane, carbon monoxide and carbon dioxide, nitrogen and hydrogen depending on the method applied.

There are four main stages involved in the gasification process; drying of the fuel, pyrolysis, reduction and combustion. It is usually convenient to consider these reactions as being confined to several zones within the gasifier (reactor). A reactor for gasification can thus be divided into these four zones as illustrated in Figure 2.4 below. A fuel must pass through all the four zones to be completely gasified in a counter flow direction to air/steam stream.

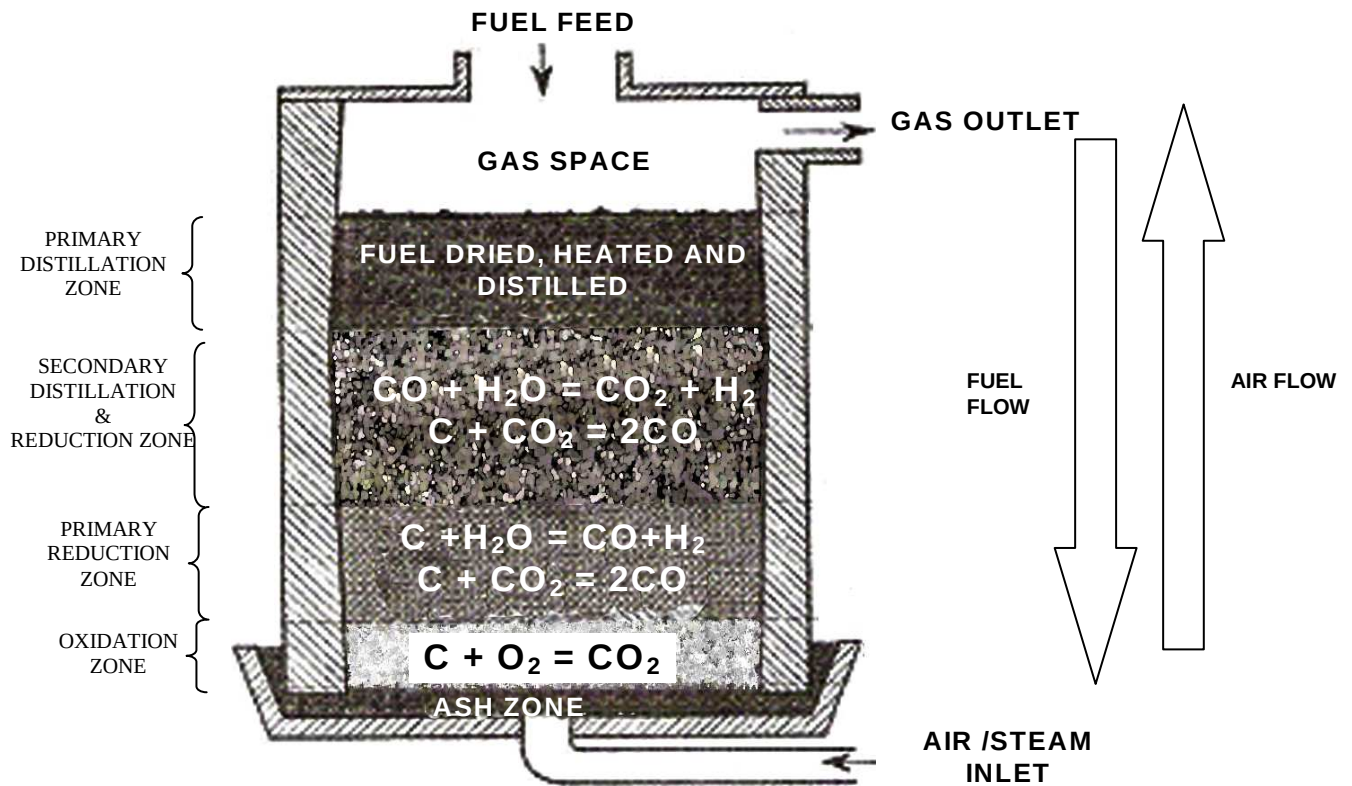


Figure 2.4: Illustration of gasifier for production of syngas

[Source: The efficient use of fuel, Ministry of Technology, United Kingdom]

The combustion (oxidation) or ash zone is generally situated near the base of the gasifier (see Figure 2.4). This is where air and steam is fed into the gasifier allowing the oxidation reaction of the fuel to take place as in an ordinary furnace. The required air draught is usually provided by a suction pump or by appropriate arrangement of fans/blowers. The main feature is that the air supply is restricted so that the combustion does not spread to the whole fuel load. If this occurred the gasifier would become a simple furnace, producing heat and incombustible gases.



The basic chemical reaction taking place in the combustion zone is the combination of oxygen in air with the carbon component of the fuel to produce carbon dioxide and incombustible gas(Equation 2.5).This is an exothermic(heat releasing) reaction, and

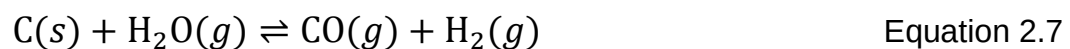
thus leads temperatures in the combustion zone to reach levels between 900°C and 1300°C or higher in some gasifiers. If any hydrogen is present in the oxidation zone, it reacts exothermically producing water vapour.

The hot gases produced in the combustion zone move up into the reduction zone. This is always adjacent to combustion zone, although it may depend on the configuration of the gasifier. It may either be on top as in Figure 2.4 for vertical configuration while in the horizontal configuration it will be besides the combustion zone. No air is admitted to the reduction zone, hence there is no free oxygen thus giving rise to reduction reactions. These reactions play an essential role in gasification as they convert some of the combustion gases (CO_2 and H_2O) emerging from the combustion zone in to products that can react. Carbon dioxide reacts with carbon resulting in the production of the combustible gas carbon monoxide (Equation 2.6).



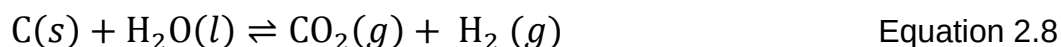
This reaction is endothermic, that is it absorbs heat, and temperatures in excess of 900°C are required for it to take place. The carbon monoxide produced is the major component in the final mixture of gases drawn from the gasifier.

Another important reaction that takes place in the reduction process is that between steam (H_2O) reacts with carbon. This is also endothermic and requires temperature above 900°C. In this reaction the products are carbon monoxide and hydrogen. This is commonly called the “water gas” reaction. Since both products are combustible it tends to increase the calorific value of the product gas. This reaction is represented in Equation 2.7 below.



During these endothermic reactions, heat is absorbed from the gas stream. This reduces the temperatures in the reduction zone as the gas is drawn further from the

combustion zone. As temperatures drop to between 500°C *and* 600°C , the reactions between water vapour and carbon produce hydrogen and carbon dioxide as illustrated below(Equation 2.8).



If excess water is present in the reduction zone, an unfavourable reaction known as the water gas shift reaction can take place as well. In this reaction the carbon monoxide reacts with water to produce carbon dioxide and hydrogen. This reaction is undesirable as it is exothermic and thus reduces the calorific value of the producer gas. Excess moisture in the fuel is therefore to be avoided.

The third is the pyrolysis zone which is between the primary distillation and secondary zone and generally above the reduction and combustion regions. There is no air drawn in this zone but it draws heat from the adjacent zones which are counter current to the gas stream from the reduction zone. The temperatures here are about 400°C thus enabling self sustaining exothermic reactions to take place in which the natural structure of the biomass feedstock fuel breaks down. These reactions lead to formation of water vapour, methane (Equation 2.10), acetic acid and a considerable quantity of heavy hydrocarbon tars are produced. Most of the volatiles are given off in this stage and the solid material that remains after pyrolysis is carbon in form of charcoal. This passes down the gasifier, and is combusted in the combustion zone and reduction zones.



The drying zone is situated at the top of the other zones. Here temperatures of about 150°C are not high enough for any chemical reactions to take place. However, in this zone fuel is dried and any moisture contained in the incoming feedstock fuel is driven off in form of water vapour.

However, the gasification techniques that are currently being used are all relatively more expensive than the traditionally established enzymatic hydrolysis of cellulose to sugar followed by fermentation. The producer gas has low thermal value due to the effects of nitrogen in the combustion air. Using pure oxygen in the combustion zone, instead of air, leads to a higher thermal value but increases the cost of production. The cost is further increased by need for energy addition and low efficiency of gasifiers. The production costs of fuel by gasification have to be halved in order for them to compete with fossil fuel at the current world oil prices (EIA 2006).

2.1.2.3 Composition of producer gas

The energy content of the producer gas is dependent on the production process and feedstock used. The typical composition of the more prominent biomass feedstock for production of producer gas, wood and charcoal are given in Table 2.2 below.

Table 2.2: Typical composition of producer gas

Gas	Wood Percentage on Dry basis	Charcoal Percentage on Dry basis
Nitrogen	50 – 54	60 – 63
Carbon dioxide	9 – 11	3 – 7
Carbon monoxide	20 – 22	23 – 33
Hydrogen	12 – 15	4 – 14
Methane	2 – 3	-

[Source: Foley G. And Bernard G. (1983)]

The energy content of the producer gas determines that of the synthetic product that will be formed. Furthermore, the gasification method used is of great importance in determining the energy content of producer gas. Product gas from wood gas has energy content of $5500 \text{ kJ}/\text{m}^3$ while charcoal gas has $4100 \text{ kJ}/\text{m}^3$. Therefore, synthetic fuels from wood gas would generally higher energy content than those from charcoal gas.

3. USE OF BIOFUEL AS TRANSPORT SECTOR FUEL; PROSPECTS AND CHALLENGES

The production and use of biofuels has both advantages and disadvantages which are diverse in character. They have technical, social, economic, and environmental as well as sustenance aspects attached to them which are dependent on particular systems. This leads to several issues that have taken centre stage in the debate on viability and sustainability of biofuels. Thus, the impacts of biofuels production and utilisation have to be weighted on a case by case basis although there are general observations.

The impacts of biofuels production and use are discussed here regarding technical, social and economic including the issues that are being debated regarding their environmental sustainability. The notable consequences (both negative and positive) of producing and using biofuels are highlighted together with the possible mitigation measures for would be negative impacts.

3.1 Issues pertaining to gasoline - alcohol blends

Alcohols are blended with gasoline in low blends such as 5% ethanol to 95% Gasoline (E5) 10% ethanol to 90% gasoline (E10) and 20% ethanol to 80% gasoline (E20) in most cases without considerable modification to conventional vehicle engines. However, alcohols may degrade some types of plastic, rubber and other elastomer components. Since alcohol is more conductive than gasoline, it accelerates corrosion of certain metals such as aluminium, brass, Zinc and Lead (Pb). This is because ethanol has more oxygen than gasoline and has a higher reduction potential. The resulting degradation can damage ignition and fuel system components like fuel injectors and fuel pressure regulators (Otte et al., 2000). The corrosive effect of a fuel increases with ethanol concentration, thus some materials that may not corrode with an E10 blend may degrade with higher blends. The swelling and embrittlement of rubber fuel lines and oil rings can over time lead to component failure. These problems are eliminated by using compatible materials, such as Teflon or highly fluorinated elastomers (Vitornis; EU-

DGRD, 2001). Corrosion can be avoided by using some stainless steel components for fuel filters. In blend levels above E10, some engine modifications may be necessary, though the exact levels at which modifications are needed varies with local conditions such as climate, altitude and driver performance criteria (EAIP, 2001). In Brazil, cars with electronic fuel injection, including imported cars built for the Brazilian market with minor modifications (such as tuning and the use of ethanol-resistant elastomers) have operated satisfactorily on 20% to 25% ethanol blend since 1994. There have been few reported complaints about drivability or corrosion (Moreira, 2003). Other studies have shown that ethanol blends have relatively poor hot – fuel handling performance, due to high vapour pressure. Fuel formulation to control vapour pressure is necessary to ensure smooth running in warm climates and higher altitudes (Beard, 2001).

3.2 Issues pertaining to ethanol - diesel blends

Ethanol can also be blended with diesel forming an emulsion for use in compression ignition engines. This brings about the possibilities of expanding the use of biofuels into the diesel fuel “pool”, since ethanol has greater output per unit (8480lt/ha) of land devoted to growing crops compared to biodiesel with yields less than 2500lt/ha. The use of these blends has been low till recently due to the low cetane number which makes it very difficult to ignite by compression in diesel engines (Murthy, 2001). The blending of ethanol and diesel has been found to have some pleasant benefits such as solubilising effect, detergency and low temperature properties. These properties reduce the problems of cold starting in diesel engines when temperatures are low. Furthermore, because of the low cetane number of ethanol (and therefore poor auto ignition properties) the additive package must also include a cetane enhancing additive such as ethylhexyl nitrate or di-tert butyl peroxide (McCormick and Parish, 2001). Fleet testing on E-diesel blends has been done in Europe with (Sweden, Denmark, Ireland), Brazil, Australia and the United States (Nevada, Illinois, Nebraska, Texas and New York city) taking the lead. In Sweden, a variant of E-Diesel tested for many years on urban buses has had great success, showing significant improved emissions performance (RAE,

2002). Although studies have shown that this is a viable option, countries that want to embark on this path have to do further research.

3.3 Issues concerning use of biodiesel

Biodiesel (FAME) in its neat form (B100) can be used in standard compression – ignition (Diesel) engines designed for petroleum – based diesel fuel. Germany, Australia and Sweden have promoted the use of 100% pure biodiesel trucks with only minor fuel system modifications. In France, biodiesel is blended at 5% and at 30% in some fleet applications while in Italy, it is commonly blended at 5% in standard diesel fuel (EU, 2001). Low level (e.g. 20% or less) biodiesel blends can be used as a direct substitute for diesel fuel in virtually all heavy – duty diesel vehicles without any adjustment to the engine or fuel system (EC, 1998 and NREL, 2000).

Pure FAME biodiesel acts as a mild solvent; B100 is not compatible with certain types of elastomers and natural rubber compounds and can degrade them over time (NREL, 2001). On the other hand, the solvent properties of biodiesel have also been noted to help keep engines clean and well running. In some cases, standard diesel leaves a deposit in the bottom of fuel lines, tanks and delivery systems over time. Biodiesel can dissolve this sediment. However, the deposits may then build up in the fuel filter. Initially the filters may have to be changed more frequently with biodiesel. The higher the blend level, the more potential for degradation. In particular, the use of B100 may require rubber hoses, seals and gaskets to be replaced with more resistant materials, other non rubber seals or biodiesel compatible elastomers. The quality of biodiesel has also been found to be an important factor in its effects on vehicle fuel quality requirements and is considered an important step in choosing a fuel.

3.4 Environmental costs and benefits of using biofuels

The impacts of biofuels on the environment are either negative or positive. The negative impacts are usually referred to as “environmental costs” and are quantified in terms of

lost environmental benefits such as carbon sinks and land degradation. The environmental costs and benefits are some of the aspects that make comparison of biofuel and fossil fuel complicated to quantify on the whole “well to wheels” basis. The full “well to wheels” cycle which measures the total investment cost and benefits including environmental consequences is not easy to quantify. However, it has been possible to measure the emissions related to end use in the fuel utilisation chain. It has been acknowledged that biofuels when used in “neat” form or as blends with fossil fuel can reduce certain vehicle pollutant emissions which create air quality problems. The extent to which this can be achieved depends on the type of biofuel and blending proportions in certain cases. An ethanol - gasoline blend has a different effect on emissions to those of a biodiesel - diesel blend and these are independently affected by blending percentages as well.

Besides pollutant emissions, biofuels offer an alternative to current waste disposal practices, if the wastes are used as feedstock for their production. Among potential waste products are biodegradable municipal waste, crop residues, forest waste and waste cooking oil. Use of these would avoid problems of landfills and in certain cases increase net output from industries such as agriculture and forestry.

However, as mentioned earlier there are some effects that are overlooked when analysing the pollutant emission aspects if the full “well to wheels” circle is not taken into consideration. These other aspects depend on the way in which the biofuels are produced with regard to the growing of energy crops and actual conversion technology. These factors include application of fertilizers, effect of monoculture on soil and water resources as well as technology for conversion such as chemical use in hydrolysis and other processes.

3.5 Environmental sustainability of biofuel

Increasing use of biofuels is controversial. Ongoing debates include net energy, diverting agricultural resources to fuel instead of food production, water use, and the

potential impacts of land use and indirect land use change brought about by the cultivation of biofuel feedstock (Rogerio et al 2010). Recent studies have questioned the contribution of bioenergy to climate change mitigation (Schubert and Blasch, 2010). It has been shown that from a life cycle perspective the carbon emission balance of biofuels is not always positive but can also be negative. For instance, in the net energy debate Pimentel and co-workers (D. Pimentel, 2003; T.W. Patzek, 2005) have produced a series of articles arguing that it takes as much as 1.7 times the fossil energy to produce a gallon of ethanol from corn as contained in the ethanol. Most recently they extended their results to include cellulosic and sugarcane-derived ethanol (T.W. Patzek and D. Pimentel, 2005).

A number of GHGs are emitted during crop production; nitrous oxide (N_2O) generated directly by nitrification and denitrification and indirectly via runoff of various N-species, CO_2 from soil carbon transformations, methane (CH_4) from poorly drained soils and CO_2 emission-associated production and transportation of agricultural inputs (fertilizers, herbicides, etc.) and farm equipment use. In most LCA's N_2O is calculated using standard International Panel on Climate Change (IPCC) factors based on added nitrogen to determine the emissions. Recently, Crutzen et al. have suggested that these factors might underestimate N_2O production by agriculture and that a 3–5% of nitrogen application may be more appropriate. Although these impacts might affect the conclusions of many biofuels, as suggested by the authors, for sugarcane the impact is minimal.

Thus the sustainability of biofuel depends on feedstock, management of land resources when growing the feedstock, the kind of land use changes introduced by cultivation and finally on the conversion and processing methods (Fritsche and Wiegmann, 2008; Zah et al 2007).

Some proposals however have been made on the sustainability of bioenergy in general by several researchers. Fritsche et al. 2009 proposes that sustainability criteria for bioenergy typically includes requirements for reduction in greenhouse gas emissions compared to fossil alternatives, sustainable land use and social sustainability.

Bioenergy products which only marginally compete with other land uses such as food production are found to induce least direct and indirect land changes and usually show the most favourable GHG balance (Frische and Wiegmann, 2008).

Schubert et al, 2009 also propose a minimum standard for bioenergy sustainability which is more comprehensive. Life circle greenhouse gas emissions are supposed to be reduced by 30ton CO_2 per Tera joule (TJ) of raw biomass used in comparison with those from the equivalent fossil fuels displaced. However, it is still not easy to compute the full GHG emissions reduction for many biofuels as in many cases the actual inputs during production are difficult to account.

It is also suggested that indirect land - use changes must be avoided. This entails that no land that is already being used for other purposes such as food production or natural forests is used to produce biofuel crops. This would render biofuel production unsustainable thus it encourages use of waste land when available for sustainable biofuel production. No use of biomass feedstock cultivated on land with high carbon stocks and high biodiversity (i.e. primary forests, wetlands, biodiversity grasslands, savannahs). Furthermore soil quality and fertility should be maintained by reducing the use of agrochemicals and fertilizers and by leaving sufficient post harvest residues on the fields. Forest ecosystems must be conserved by using wood residues sustainably, this entails producing second generation biofuels from the waste. Water resources must also be managed sustainably in production of biofuel. Diversion of natural water stream which could lead to destruction of natural river frontage should be avoided. There must be controlled use of genetically modified organisms (GMOs) to avoid invasive species and other undesirable effects that might arise. The labour and social standards must be in compliance with basic International Labour Organisation (ILO) norms. These proposals among others are the guidelines currently available as sustainability criteria for biofuels globally. On the overall it has been agreed that second generation biofuels offer a more sustainable fuel for transportation (Biofuels and Sustainability).

3.6 Food versus fuel crisis

The increased production of biofuels has been blamed for the increase in food prices between 2003 and 2008 because most of the feedstock used for its manufacture was largely food crops mostly sugarcane and sugar beet, corn and oily seeds such as rapeseed (Mitchell, 2008;Schimidhuber, 2007;UNCTAD, United Nations Conference on Trade and Development, 2008). In the second half of 2008 the Food and Agriculture Organization's food - prices index substantially dropped, while in 2009 there was a steady increase (OECD/FAO, 2009). In January 2010 the indexes for cereal grains, oils, and sugar prices were, respectively, 70%, 69% and 76% higher than those for 2002 – 2004 average values (FAO, 2010). This has brought a new debate arena on biofuel policies and has caused some countries to momentarily halt or reduce biofuels support programs (Sorda et al, 2010).

The debate around “food or fuel” option has increased as an important concern on the sustainability of biofuel production. It is certainly important to have enough food to feed the world's population as well as provide sustainable energy solutions, and currently there are two broad opposing opinions existing with regard to this requirement. There are those that are against the current biofuel strategies on moral and ethical basis that think that biofuels should only be grown on lands where food cannot be effectively produced (Rogério Cezar de Cerqueira Leite, 2007).

This approach would limit biofuel feedstock to wastes, agricultural residues or energy crops produced on marginal lands. In contrast, biofuels proponents, who find a moral imperative in addressing environmental and security issues, say biofuels production, has little impact on food prices or availability. Any price increases or volume reductions will be temporal and ultimately be offset by market forces, which will encourage others, particularly in less developed countries, to plant more grains. Increases in yields over time will also reduce land use needs, for both fuel and food production.

This debate is on going and no conclusion has been reached. The consequences of biofuel production on food will to a large extent be dependent on regional conditions and policies.

4. STATE OF THE ART; BIOFUELS FROM A GLOBAL PERSPECTIVE

During the past 25 years, worldwide petroleum consumption has steadily increased, due to higher standards of living, increased transportation and trucking, and increased use of plastics and other petrochemicals. In 1985, total worldwide petroleum consumption was 2807 million tons, but, the figure reached 3928 million tons in 2008, with an average annual growth rate of almost 1.5% according to BP's annual Statistical Review of World Energy of 2009. At the end of 2008, the world proven oil reserves were estimated at 1.7×10^{11} tons with a reserve-to-production (R/P) ratio of 42 years as reported in the BP review. Furthermore, petroleum-based products are one of the main causes of anthropogenic carbon dioxide (CO₂) emissions to the atmosphere. The transportation sector worldwide is almost entirely dependent on petroleum-derived fuels with an insignificant amount of electricity and biofuels being used. One-fifth of global CO₂ emissions are created by the transport sector (Goldenberg J., 2008), which accounts for some 60% of global oil consumption (IEA 2008). There are active research programs to reduce reliance on fossil fuels by the use of alternative and sustainable fuel sources, and thus to increase the time over which fossil fuels will still be available (Namasivayam et al, 2009). As an alternative to petroleum-based transportation fuels, biofuels can help to reinforce energy security and reduce the emissions of both greenhouse gases (GHGs) and urban air pollutants. There are several other motivating factors influencing different countries to embark on biofuel programs.

By displacing petroleum fuels, countries can reduce import expenditure and dependence on fossil fuel and will thus improve security of energy supply from within the countries that produce their own biofuel. Biofuels are more environmentally friendly compared to fossil fuels, hence there is also an opportunity for some countries to reduce emissions and meet their Kyoto targets. However, it is important to note that the environmental impacts of biofuels also depend on the techniques used to produce them. The growing of energy crops in some regions is expected to increase agricultural diversity and thus improve livelihood in some rural areas by creating employment.

Biofuel is obtained from renewable energy resources, hence with sustainable production mechanisms put in place, it offers a viable option to fossil fuel. These are some of the motivations driving biofuels production in different regions.

Over the past ten years biofuels production and consumption has increased tremendously. Ethanol output has experienced an increase from 16.9 to 72.0 billion litres in the period 2000 to 2010 while biodiesel production has increased from 0.8 to 14.7 billion litres in the period (Figures 4.1a and 4.1b). This increase in production of biofuels has been created by different government interventions. These interventions have varied from one nation to another and the results have been different.

In the USA, which is the world's largest producer of fuel ethanol, there are government driven financial incentives guaranteed for biofuel producers while in the European Union, the world's largest biodiesel producing region biofuel production is mostly driven by mandatory blending targets both in France and Germany. Biofuel production is unprofitable and needs to be promoted by government interventions such as tax exemptions, subsidies or other forms of financial support with the exception of Brazil ethanol (Rajagopal and Zilberman, 2007). The USA production of fuel ethanol which is currently 10 billion litres is 20 times more than any other IEA country. In Figure 4.1a below, the world leading nations in ethanol production are shown as well as their ethanol production levels for 2007 and 2008. The leading nations in biodiesel and biodiesel production are given together with the production quantities for 2007 and 2008(Figure 4.1b).

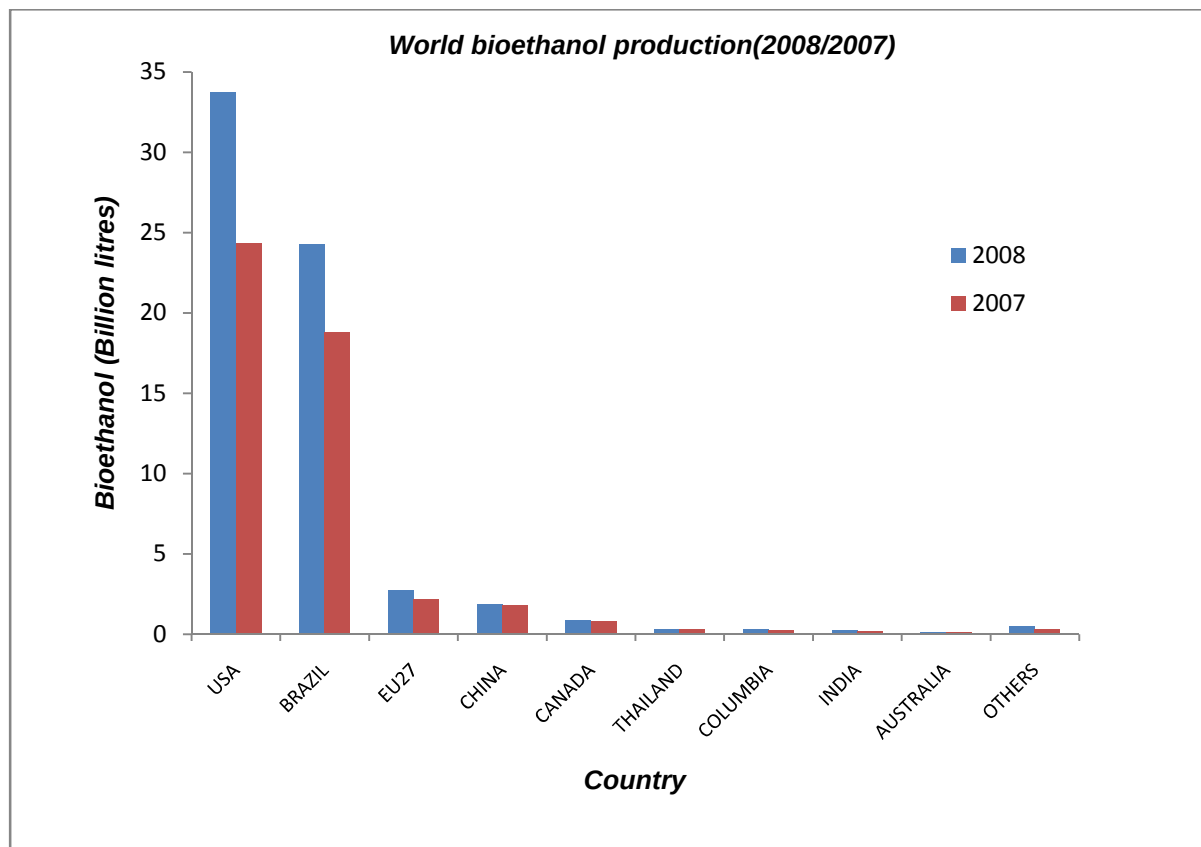


Figure 4.1a: World fuel ethanol production by country (2008 -2007) in billion litres.

[Source: RFA, Renewable Fuel Association statistics , Annual World Ethanol Production by Country (based on F.O. Licht estimates) (RFA, Renewable Fuel Association, 2010). Original version in US gallons and here converted in liters.Available at <http://www.ethanolrfa.org/pages/statistics#> as of 29/06/2010]

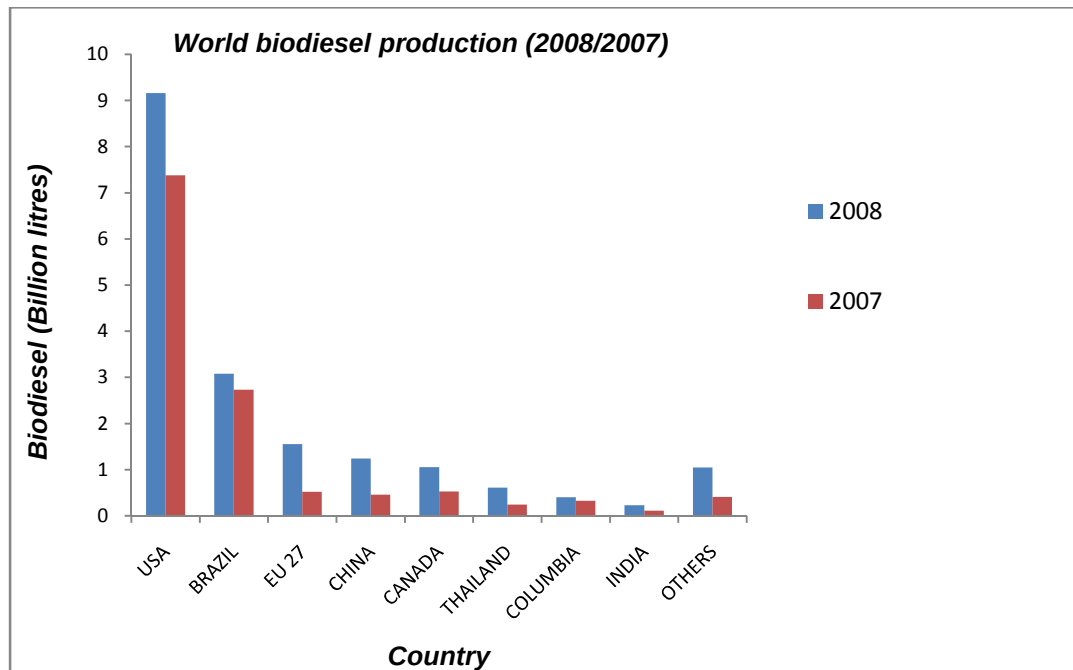


Figure 4.1b: World biodiesel production by country (2008 -2007) in billion liters

[Source : OECD/FAO, Agriculture Outlook 2009 -2018. In particular Agriculture Outlook 2009 -2018 data base, where data was originally published in tonnes. Conversion rate: 1 liter of biodiesel = 0.88Kg(OECD – FAO Data base , 2009).

Data available at: http://www-outlook.org/document/6/0.3343,en_36774715_40969158_1_1_1_1.00.html. as of 29/06/2010]

This section gives an analytical review of the current state of biofuels on a global perspective. The production trends, motivations for development of biofuels industry in different regions of the world, policy measures and developments are reviewed for selected individual countries by region. Considered here are the leading biofuel producing countries as well as new entrants in the production of biofuels found in North America, South America, Europe and Asia including Africa. Individual countries are included in the discussion to provide an overview of the national policies. Finally, the most outstanding motivations and policy measures that have led to the growth of the biofuel industry are discussed.

4.1 North America

4.1.1 USA

In the USA, ethanol production and use as a vehicle fuel can be traced to the early 1990's. The 1990 Clean Air Act amendments, and "oxygenated fuels program", established a requirement that gasoline sold in the "carbon monoxide non-attainment areas" must contain 2.7% oxygen to increase cetane number. Also, the energy policy of 1992 encouraged the use of "alternative fuels". In an effort to encourage alternative fuel a certain percentage of state fleet was required to run on alternative fuel which included biofuels. A move to phase out Methyl butyl ether (MTBE) in several states also tended to be a driver for ethanol demand, so as to replace MTBE as oxygenate. Furthermore, the car fleet manufactured after 1992 in the US can run on neat biodiesel without need for engine modification. This followed the 1992 Energy Policy Act (EPACT) which encouraged use of alternative fuels. The various programs that were introduced led to the increase in ethanol consumption. More legislation was passed by congress in 2003 that would support domestic ethanol production. For ethanol, there has been a federal tax credit of 1.4 cents per litre of E10.

In July 2010, the updated Renewable Fuel Standard (RFS2) went into effect, thus finalising the proposals made with the Energy Independence and Security Act of 2007. In this proposal a target of 136 billion gallons of renewable fuel must be used for transport fuels by 2022. This target is to be achieved in stages, from 2015 onwards; the volume of conventional ethanol (Corn ethanol) should be 57 billion litres. Cellulosic biofuels are expected to amount to 0.37 billion litres in 2010 and surge to 61 billion litres by 2016, while the total contribution of advanced biofuel (i.e. biomass based diesel, cellulosic and non - cellulosic biofuels) should not be less than 80 billion litres by 2022 (Sorda et al, 2010).

The RFS2 further requires that producers of advanced biofuels reduce life – cycle GHG emissions by at least 50%, while standard biofuels have to achieve a reduction in life

cycle emissions of 20%. However, the 20% threshold only applies to renewable fuel from facilities that commenced production after December, 2007, with an additional exemption from the 20% threshold for ethanol plants that commenced construction in 2008 or 2009 and are fired by natural gas, biomass or any combination thereof (Federal Register, Regulation for fuels and Fuel additives: Changes to Renewable Fuel Standard, March 26, 2010, pp. 14687). Furthermore, all renewable fuels should certify the type of feedstock used and comply with regulations on land use.

The obligatory consumption of given biofuels volumes in the US was first implemented with the inclusion of Renewable Fuel Standards (RFS1) in the energy Policy Act of 2005. Its objective was to have 15 billion litres of renewables in transport fuels in 2006 and increase the share over the years. Thus the new regulations (RFS2) further increase the targets, give incentives for second generation biofuels and implement new criteria to ensure more environmentally sustainable production (Sorda et al, 2010). The guiding principle has been a reduction in the country's dependence on oil.

In the recent past, the main source of financial support for biofuels has been the Volumetric Ethanol Excise Tax Credit (VEETC). This was enacted in 2004 by the American Jobs Creation Act and is scheduled to expire in 2010. VEETC was guaranteed for every domestic or imported gallon of ethanol blended with other fuels. Blenders of petroleum fuel with biofuel can claim 51 cent credit per gallon used (EIA, Energy Policy Administration, 2008). The energy administration estimated that the credit cost which was 2.5 billion dollars in 2006 could cost government 5 billion dollars in 2010 if ethanol production exceeds 11 billion litres (EIA, Energy Policy Administration, 2007). The 2004 American Jobs Creation Act guarantees tax credit for biodiesel as well under VEETC. In the case of biodiesel this subsidy amounts to US\$1 per gallon of biodiesel produced from virgin oils (the main feedstock) or fats and US\$0.50 per gallon for recycled oils.

The US uses import tariffs measures to protect the local industry from more competitive foreign manufacturers (above all Brazil). There are two taxes levied against imported

ethanol; i.e. an ad valorem duty of 2.5% and a tariff of US\$0.54 per gallon applied under the Most Favoured Nations (MFN) scheme.

4.1.2 Canada

Canada started blending ethanol with gasoline in the 1980's. The federal government embarked on a program to more than double the 2003 production levels by putting through a number of initiatives to boost production – potentially with 35% of gasoline to contain 10% ethanol by 2010(Canada, 2003). The federal government allocated substantial amounts of money under its climate change plan to encourage construction of new ethanol plants and development of cellulose-based ethanol. Henceforth, there has existed a National Biomass Ethanol Program which encourages firms to invest in the local ethanol industry, partially as compensation for a planned reduction or elimination of a 10cent per liter excise tax exemption on fuel ethanol. Some states have introduced mandatory vehicle testing whose positive results have apparently led to increased ethanol-gasoline sales as public awareness on the benefits keeps increasing.

The tax systems have also been changed time and again to facilitate the increased production and utilization of bioethanol. In 2008 the Canadian Environmental Protection Act Bill C – 33 mandated a 5% renewable content in gasoline by 2010 and a 2% renewable content in diesel fuel and heating oil by 2012 respectively (EcoAction, 2008). For this target to be met, a minimum of 1.9 billion litres of ethanol need to be produced considering the current consumption trends of gasoline. The biodiesel mandate of 2% can be met if production capacity is increased to 520 million litres by 2012(Dessureault, 2008).

Since 2008, there have been direct incentive payments for biofuel production besides the mandates. Ethanol manufacturers have enjoyed a maximum incentive rate of CANS\$0.10 per litre for three years from 2008 to the end of 2010 through the EcoEnergy for Biofuels Program. This payment should then start to decline by CAN\$0.01 every year until it reaches CAN\$0.04 in 2015 and 2016. For biodiesel, the

maximum incentive rate has been CAN\$0.20 per litre since 2008 and should start decreasing after 2010 at a rate of CAN\$0.04 every year until it reaches a value of CAN\$0.06 in 2016. There are several other fund schemes apart from the EcoEnergy program that are in place to increase production by targeting an enlargement of production facilities (EcoAgriculture Biofuels Capital Initiative) granting support to research clusters (Agriculture Bio – Products Innovation Program and Sustainable Development Technology Canada) and accelerating the commercialization of innovative agriculture products services (Agri – Opportunities Program).

In Canada, ethanol is predominantly produced from cereal grains. About 69% of all the ethanol was produced from corn while 30% was contributed by wheat in 2009. Animal fat is the main feedstock for biodiesel production. In 2009, estimates show that tallow grease was the leading biodiesel feedstock (49%) followed by yellow grease (37%). The Canola plant's relevance in biodiesel production has been increasing and accounted for 14% of total output in 2009 (Dessureault, 2008). The production of biofuel has an aggregate positive impact on the Canadian economy of CAN\$2 billion annually according to the Doyletech report (Doyletech, 2010).

4.2 South America

4.2.1 Brazil

The earliest and most successful ethanol programme is the Brazilian alcohol Program (Proálcool), which was launched in the 1970's during the first energy crisis. Brazil produces about 33% percent of all the ethanol in the world (Licht F.O, 2007). The Proálcool program involved cooperation between the government, farmers, alcohol producers and car manufacturers. The objective was to limit energy supply constraints, provide a stable internal demand for excess production of sugar cane and counter weight variations in international sugar prices (Walter and Cortez, 1999). It has succeeded in demonstrating the technical feasibility of large scale production of alcohol

as a transportation fuel and its use in high blending percentages as well as dedicated ethanol vehicles.

The target for the initial five years was to displace 20% and increase to 25% of gasoline within that period. The first task was achieved without major engine modifications in light duty vehicles. Further targets were set after the second oil crisis in 1978/79, it was to use hydrated, “neat” ethanol typically 96% to 4% water. The car industry (e.g. Volkswagen, Volvo - Brazil, e.t.c.) agreed to modify vehicles so that they could safely run on neat fuel. The government facilitated this by providing soft loans and tax reductions that made ethanol attractive to customers. By December 1984, the number of cars running on neat ethanol reached 1,800,000 that was 17% of the countries car fleet (Ribeiro, 2000). However, in 1985 a shift in government policy and sharp decrease in oil prices made fuel ethanol expensive. Proálcool was abandoned and this led to a decrease in neat ethanol vehicles, from almost 100% of new car sales in 1988 to less than 1% by the mid-1990's, cars had dropped by over 50% at 4.4million in 1994 to 2.1million in 2002(DATAGRO, 2002). Overvaluation of Brazilian currency between 1994 and 1995 further increased ethanol production costs which reduced sales of ethanol powered cars to 1% by the late 1990's.

The government tried to limit these draw backs by implementing legislation in 1993 that required 22% ethanol blending with gasoline. Further deregulatory policies in the energy and fuel markets were implemented during the 1990's. In 1998 the government liberalized the price of hydrated alcohol to be used in fuels and in 1999 it stipulated that hydrated ethanol sales should be carried out through public auctions (World Trade Organisation, Trade Policy Review – Brazil 2008, 2005). In 2002, the Proálcool program was revived; industrial production tax was reduced for manufacturers of ethanol vehicles, as well as subsidies for purchases of new ethanol cars. Credits for the sugar industry were also introduced to cover storage costs, in order to guarantee ethanol supplies. The government of Brazil is in a 10 year deal in which Germany will purchase carbon credits as part of the Kyoto protocol commitments and in turn, subsidise taxi drivers and car hire companies by US\$ 300 per vehicle on the first 100, 000 vehicles

sold (Kingsman News, 2003). Gasoline blending with 25% anhydrous ethanol was made mandatory in all motor gasoline sold in Brazil in 2003. The amount of ethanol production and demand started increasing with a growth of 2.9 billion litres per year, almost 20%, as at 2005 (F. O. Lichts, 2003). The consequences have been accelerated growth in technical innovations that allow motor vehicles to run on these fuel blends.

The high oil prices that occurred during the period spanning from 2003 to 2008 brought ethanol back to its initial success in Brazil. It made ethanol competitive with petroleum gasoline and profitable without being promoted by tax exemptions, subsidies and other forms of government interventions in Brazil unlike other countries (Rajagopal and Zilberman, 2007). Furthermore, the introduction of flex – fuel engine technology, which allows vehicles to run on gasoline or on neat ethanol, has contributed to the growth of the ethanol industry. In 2006, 83% the cars sold in Brazil were Flex – Fuel Vehicles (FFV) and the country has since achieved oil independence (Colares, 2008). It is estimated that FFV's could make up 27% of the car fleet in Brazil in 2010 and 43% in 2015 (Almeida's et al, 2008).

Brazil is the world's second largest producer of ethanol and has the best technology for use of gasoline ethanol blends. Its ethanol is also recognized as the most cost – competitive biofuel in the world. Production costs of ethanol in the centre – south region of Brazil have been estimated to be US\$0.23 per litre by Macedo and Nogueira (2005). According to Koima and Johnson (2006), fuel ethanol average costs lie between US\$0.23 and US\$0.29. These values would make ethanol competitive with oil prices at US\$30 per barrel (G. Sorda et al, 2010). The calculated average production cost of new ethanol products to be around US\$0.37 per litre (de Almeida's et al, 2008). This case would make an oil price of US\$42 per barrel cost competitive with ethanol.

The low production costs of ethanol in Brazil are due to several factors. Firstly, all the feedstock, in this case sugar cane is comparatively cheap. The land used has high productivity and there is no need for irrigation. Secondly, the production process in the mills does not require external energy input as the co – generation power plants use

bagasse to provide all the required energy (Almeida's et al., 2008). Finally, the support that government gave to the industry allowed large investments in research and technology developments that have transformed processes and lowered manufacturing costs in the long run.

Unlike the US, today Brazil does not give any direct subsidies to the ethanol industry although preferential treatment is given compared to gasoline. There is no excise tax on ethanol since 2004 and other federal duties are much lower compared to those imposed on gasoline. Estimated tax incentives on ethanol amount to US\$977 million according to de Almeida's et al (2008). The importance that sugar and ethanol play in Brazil is evident in the economy. The two industries account for 3.6 million jobs and 3.5 % of GDP, while ethanol alone consumes 50% of total sugar cane supply (Almeida's et al., 2008). Brazil is now looking at the possibility of establishing international export markets around the world mostly in Europe.

Following the success of ethanol program, Brazil is now investing in biodiesel. In 2005, the National Program on Biodiesel Production and Usage (PNBP) was launched. The initial requirement of the PNPB was replacement of 2% petroleum diesel with biodiesel from 2008 to 2012 and an increase to 5% from 2013 onwards (Colares, 2008). By July 2009, the government had mandated a 4% biodiesel blending share and indicators are that the 5% target will be achieved before 2013 as specified in the current legislation. In order to meet projected demand, total capacity must be increased to 4 billion litres. Currently there are 65 operating plants and another 12 plants awaiting official authorization.

The main feedstock for biodiesel production in Brazil is Soybean, although other important vegetable oil plants such as castor bean, palm tree and jatropha are also available. In contrast to ethanol, biodiesel is not cost competitive and is thus subsidised. Among the major support schemes are the auctions organized by the government at which the National Petroleum Agency (ANP) buys given quantities of biodiesel to ensure targets are met. Under this scheme the prices at which the biodiesel is bought

are higher than the production costs in order to stimulate supply. The other support scheme is assistance through tax exemptions with a focus on regional location of production and on provenience of feedstock (“Social Fuel Stamp” scheme). A minimum percentage of raw materials have to be purchased for family farmers to qualify for the fiscal benefits. The percentage of feedstock purchased from family farmers varies across regions. These tax incentives vary from 73% to 100% of the existing federal levy (Barros, 2009).

4.2.2 Argentina

All gasoline and diesel in Argentina was required to contain a 5% biofuel share from January 2010(Argentina Biofuel Law 26.093 of April 2006, implemented in February 2007 under decree 109/2007). The technical requirements and quality of bioethanol were specified in Resolution 1295/2008 made in November 2008 while the biodiesel standards were formalized in February 2010 with resolution 6/2010.In 2008, a pricing scheme for ethanol that was internally marked was established under resolution 1294/2008 and the biodiesel domestic price was set up with resolution 7/2010.

There are financial incentives for both bioethanol and biodiesel sold locally but none for exports. Manufacturers of biofuel can choose between a refund of the value added tax or accelerated depreciation on capital investments. Furthermore, the government guarantees the purchase of the biofuel output for a 15 year period that corresponds to the term of the Biofuel Law. The financial incentives are revised annually, hence they are not guaranteed and furthermore the government determines the prices (Rutz et al., 2009).

The main feedstock for biodiesel production is soybean and is centred on the soybean processing plants in Rosario. Soybean plays a very important role in the Argentinean agriculture sector as the country is the third largest soybean producer (Joseph, 2009). The biodiesel output and productive capacity has increased in recent years, just between 2007 and 2008 the increment has been 33% (Argentina Renewable Energies

Chamber, 2009). In 2009 its production capacity was expected to reach 880 million liters with an output of 2.4 billion litres (Joseph, 2009). On the other hand ethanol production is less developed with a production of 45 million litres in 2009 and is dependent on the sugar industry.

Most of the biofuel produced was exported until 2009. Biodiesel exports receive favourable export tariffs in comparison to its feedstock which are soybean and soybeans oil (Sorda et al, 2010). Soybean exports are taxed at 35%, soy oil at 32% while biodiesel effective levy is 17.5% (a 20% export charge and a 2.5% tax rebate).

However, the new trading regulations in one region for importing are likely to hinder exports of the other. One such regulation is the EU's Renewable Energy Directive of 2009 which requires a GHG emissions reduction of 35% from biofuels. The same document assumes that biofuel from soybean only reduces emissions by 31%, thus preventing its use in Europe.

4.2.3 Columbia

The government in Columbia mandated a 10% blend in cities with population above 500, 000 inhabitants (Law 963). The Law went into effect in late 2005 (Pinzon, 2007). By 2009, 75% of all gasoline consumed in the country had 10% ethanol content (Rutz et al., 2009). It was aimed that all gasoline consumed in 2010 would contain 10% bioethanol and further set a target of 25% ethanol content by 2020 (ProExport Columbia, 2008). Ethanol is primarily derived from sugar cane and its prices are fixed by the government on the basis of international sugar prices and it is exempt from value added tax.

On the other hand promotion of biodiesel and consumption started in 2004 with passing of law 939 and Resolution 1289 of 2005. The resolution established an obligatory mandatory blending of 5% biodiesel by 2008 (Pinzon, 2008). It is intended to increase

the blend to 10% in 2010 and 20% in 2012 (Rutz et al., 2009). Palm oil is the feedstock of choice for biodiesel production and it enjoys sales and fuel tax exemptions.

Other incentives are given to new agro – industrial projects. For instance, income obtained from crops used as biodiesel feedstock is also exempted from taxation. Further, in 2009 the government required with enactment of Decree 1135 that from 2012, 60% of all new vehicles sold in Columbia must support a E85% flexi fuel technology and the quota is to increase to 100% by 2016 (Rutz et al. , 2009).

4.3 European Union

In the European Union, policy dealing with biofuels had been contained in directives adopted in 2003. Directive 2003/30/EC sought to have biofuels, natural gas, hydrogen and other alternative fuels provide 20% of automotive fuel by 2020. This directive set a 5.75% target of biofuels penetration by 2010, although the quota was not binding and individual countries were allowed to present their own national action plans. The second directive 2003/96/EC addressed the tax treatment of biofuels within the context of overall energy products taxation. A proposal to adjust taxes so as to allow favourable tax reductions on biofuels was passed. Agricultural policy was considered for review so that the entire EU would have a “common agricultural policy” which will make it compatible with world trade organization (WTO) rules, among other things. This was expected to enhance environmental and agricultural practices. Although, countries were asked to aim at an indicative share of 2% biofuel share by 2005, the share was only 1% of transport fuels by this time.

In April 2009, as part of EU Directive 2009/28/EC on renewable energy the parliament of the European Union endorsed a minimum binding target of 10% for biofuel in transport by 2020. Furthermore, the directive specified a minimum of 35% reduction in GHG emissions to be achieved by biofuels during their life cycle. This target is meant to increase to at least 50% starting from 2017. Sustainability criteria for indirect land use

changes are also included such that no biofuel – feedstock shall originate from primary forests, highly bio - diverse grassland, protected areas and carbon rich regions.

In order to achieve these targets, several countries have introduced tax reduction or exemptions in support of biofuel production and consumption. Directive 2003/96/Ec on Energy taxation specified the incentives allowed for the promotion of biofuels and for the achievement of the targets set by the common agenda. These tax exemptions can be carried out by single countries after approval by the EU Commission. The exemptions are given several parameters to avoid overcompensation and were limited to six years thereafter to be reviewed (G. Sorda et al).

4.3.1 Germany

The most prominent user of biofuels in Europe has been Germany which is the largest producer and consumer of biodiesel in the world. The German government fully supported the indicative target set by the EU of 2% share of biofuels by 2005. Germany has not been a fore runner in ethanol support but has taken the lead in biodiesel promotion; Volkswagen is the first car manufacturer to extend warranties that cover biodiesels. In Germany, biofuels were made completely exempt from excise duty and this led to even greater growth of the industry. In 2005, Germany had surpassed the target set for that year. However, these tax exemptions that made Germany a top producer of biodiesel are now being phased out in a plan to equalise biodiesel with the tax on conventional diesel. A new biofuel obligation of minimum 4.4% biodiesel and 2% ethanol in diesel and gasoline respectively is expected to ensure sustained production and use of these fuels.

In 2009, the German government approved a revision of its mandatory biofuel targets according to the “Draft Law on Amendment of Promotion of Biofuels” of 2009. The overall mandatory share of biofuel in the transport sector was changed to 5.25% for 2009 and it will be held at 6.25% from 2010 until 2014(all percentages are measured in

net calorific values). As from 2015 onwards, the biofuel quotas will be determined on the basis of GHG emission reductions (Sorda et al. 2010).

In January 2007, Germany eliminated fuel exercise tax exemptions and replaced them with quota obligations and tax rebates. Previously the total revenue loss due to biofuel tax exemptions reached 1.98 billion per year (Kutas et al. 2007). Due to budget constraints German government opted to abolish excise duty exemptions as a form of subsidies.

The main feedstock for biodiesel is rapeseed with its production capacity being 4.8 million tonnes per year according to the Federation of the Germany Biofuels Industry. Ethanol production is based on sugar – beet and grains. However, it is estimated to be 2.7 million for 2010(Flasch, 2009). Ethanol production is much lower than biodiesel production. Ethanol production capacity amounts to 880,000 tonnes per year while actual output is estimated to reach 500,000 tonnes in 2010.

4.3.2 France

France is among the largest producers of biofuels in the European Union. The biofuel policies include mandatory blending of bioethanol and biodiesel as well as others. In 2005 the aggregated share of biofuel was set at 1.2% and is expected to increase to 7% in 2010. The shares were calculated with respect to calorific values. The initial plans of the government to reach a 10% target of biofuels integration have been halted and currently there are no specific biofuel goals.

Only a specific quota of biofuels production qualifies for tax rebates. The production quotas are allocated through public tenders (Kutas et al., 2007). Tenders for these quotas are open to all manufacturers of biofuels within the EU and so far few foreign companies near France's border have qualified as well. These quotas are valid for six years and failure to reach the targeted incorporation rate attracts a penalty. Tax rebates

are reviewed annually and have been reduced over the years. The fiscal rebates are expected to be phased out by 2011(Sorda et al., 2010).

Although there are these reductions on incentives there has been an increase in biofuel production. The feedstock for bioethanol is wheat, corn and sugar – beet while biodiesel is obtained from vegetable oils (mainly rapeseed). France is the leading ethanol producer in the EU with a forecasted 700, 000 tonnes of ethanol in 2010(Flach, 2009).

4.3.4 Spain

Spain is the largest producer of fuel ethanol in the EU, with plans to increase capacity and biodiesel consumption. The national and regional governments provide subsidies for plant construction and for promoting ethanol use. A target of achieving 9-12% renewable energy by 2010 was set however this has not yet been achieved.

4.3.4 Netherlands

Netherlands introduced a partial tax exemption to compensate for the high cost of biofuels production but limited it to a maximum of 2%. Analysis of potential for biofuels in the Netherlands showed that the 2010 indicative EU target could be achieved by using local feedstock though this would include crops used for food.

4.4 Scandinavians

4.4.1 Finland

Finland is the world leader in the production of wood-based bioenergy and biomass technologies. However, only recently has it shown interest in biofuels. At the time of the EU directives Finland had almost zero biofuel production and use. By 2005, only few small scale ethanol and biodiesel projects existed. Today, several plants are at various

stages of development. Finland has very little agricultural land and would have to depend on development of 2nd generation biofuels, however it has put some tax exemptions starting in 2006 on certain biofuels as well as set a target as percent share of biofuels by 2010. It recently began a pilot project with an ethanol plant of E5, to produce about 12million litres per year. Finland produces most of its biofuel from biodegradable sewerage waste as the little woody biomass resource is used for heat and power generation.

The major turning point was in 2006 when a national task force was put in place to conduct a detailed study into possibilities for biofuel. Based on this an obligation law was imposed in 2008 in order for Finland to achieve a minimum biofuel share of 4% in 2009, 5.75% in 2010.

It was estimated that 2-3% of transportation biofuel could be produced from Finnish biomass in 2010 with consideration of cost levels and technology (Bioenergy in Europe report, 2008).

4.4.2 Sweden

Sweden uses both high and low blends of ethanol, however most of this fuel is imported. Other countries like the UK also have a number of initiatives for promoting alternative fuels and low carbon vehicles although support for biofuels is limited. Most of the other EU members are having biofuels on an experimental level only.

On the whole, in Western Europe biofuels production and consumption would have to double every year if the 2010 target of 5.75% biofuel in the fuel mix had to be met. Using residues in the biofuel production chain can improve the environmental benefits of first generation biofuels. A major challenge however is achieving the amount of biofuel produced in an environmentally friendly manner.

4.5 Eastern Europe

In Eastern Europe, the unexplored potential for biofuels is large. Most countries have recently taken interest in biofuels with the driving motivation being largely the need for EU countries to achieve the biofuels directives from imports coming from this region. According to a 2003 study by the Institute for Prospective Technology Studies the potential contribution of the 12 CCs to the EU-27's biofuel consumption would be relatively modest – but not insignificant – around 1% to 3% for bioethanol and 1% to 2% for biodiesel, under various scenarios (IPTS, 2003).

4.5.1 Poland

Poland produces 50 million litres of biofuels, from a high of 110 million litres in 1997 due to a shift in policy on biofuel subsidy. Following the EU directive, however the polish parliament adopted a strategy for further development of biofuels by 2010. Proposals to have all fuel sold in Poland blended with 4.5% of ethanol, raised to 5% in 2006 have been made. The other driving factor is the need to stimulate development in rural areas. The policy decisions towards biofuels have faced a lot of opposition from oil and gas fuel producers including car manufacturers that feel their market share will be reduced. The Ministry of finance is also not in support because biofuels excise tax exemption will decrease budget revenues coming from excise and VAT taxes.

4.5.2 Czech Republic

In the Czech Republic Ministry of agriculture provides subsidies for the production of biodiesel from rapeseed oil and bioethanol. The biodiesel program commenced in 1991 and today, biodiesel is sold in most filling stations. The government has also spent some money on supporting production of 650 thousand litres of bioethanol (UNFCCC, 2003).

4.5.2 Other Eastern European Countries

In the Ukraine, ethanol production is growing rapidly for the domestic and possible export to European markets. The Ukraine parliament passed a law which allowed a high-octane oxygenate to the tune of 6% blend with gasoline. Also excise tax was reduced by 50% for 6% blended gasoline in 2000, however this tax has since been removed.

Hungary has also embarked on a program to develop the domestic ethanol market, and has already removed taxes on ethanol at the pump. The main gas and oil company has expressed interest increased production of ethanol and blending with gasoline.

4.6 Asia

4.6.1 India

In Asia, India is the largest producer of ethanol and in 2000 it produced 1.7 billion litres which was more than total production in the EU. Energy security, socio – economic and environmental benefits are cited as the main reasons for the government's support for biofuels. Other reasons include the possibilities for new employment opportunities in raising, reaping and processing of biofuel crops (Mandal and Mitirha, 2004). There has also been the need for addition to renewable energy options for decentralized generation(DDG) of electricity and for motive power applications such as (water pumping, milling etc.) in energy deficient rural India(MNES, 2004). Furthermore, there is the need of greening waste lands and regeneration of degraded forest lands, thereby helping in restoration and prevention of further land degradation (Mandal, 2004).

India's biofuel program is largely based on bioethanol derived from sugar cane molasses and biodiesel derived from non - edible oil seeds, e.g. jatropha and pongamia. The government of India took two important policy initiatives for the promotion of

biofuels; firstly it put in place mandatory 5% ethanol blending from 2003 in several states. A national mission on biofuels (NMB) was planned to be put in place under which 0.4 million hectares of waste land was to be planted with jatropha (Planning commission, 2003). This was intended to increase to 11.2 million hectares of waste land by 2012 and achieve a 10% blending target. Furthermore, India signed a Memorandum of understanding with Brazil related to ethanol sales and technology transfer. The oil extraction industry in India consists mostly of small scale establishments in villages and large scale units based on solvent extraction technology mostly devoted to production of edible oils. According to an estimate, the oil extraction industry in India is operating at 30-40% of its installed capacity (Tandon, 2004). Biodiesel is being produced mostly on a small scale in pilot projects, however plans for the first major commercial processing plant with capacity of 30 tonnes of biodiesel per day expandable to 100 tonnes per day are underway. The fiscal incentives for biodiesel production include the exemption from the central excise tax (4%), however most state administrations have maintained the state excise duties.

In 2005, India had 122 bioethanol plants with capacity of 1.2 million kilolitres/ annum (ethanol India, 2005). Although for bioethanol production there are no direct financial tax incentives, subsidised loans are offered by the government for sugar mills constructing ethanol plants. The government of India also ensures a price to sugar millers at \$0.33 for every litre of ethanol they produce; this represents a sizeable subsidy over production costs estimated to be as low as \$0.15 per litre.

In September 2008, the Indian government approved the National Policy on Biofuels which set ambitious biofuel targets. By 2017, a 20% share of biodiesel and bioethanol shall be blended with fossil diesel and gasoline, respectively. In this policy, a key feature is the requirement to produce biodiesel from non – edible oil seeds which are to be cultivated on waste and marginal land. Also a continuously revised Minimum Support Price (MSP) for biodiesel oil seeds is guaranteed by the government. Furthermore, a Minimum Purchase Price (MPP) for bioethanol and biodiesel is also being implemented (Altenburg et al., 2009)

4.6.2 China

Biofuels are relatively new in China and have been considered due to several concerns. Firstly, the need to reduce GHG emissions and other pollutants from vehicles and other agricultural equipment. The country also has to maintain its rapid economic growth without dependence on petroleum imports. Furthermore, is the need to provide employment to the large numbers of people leaving on traditional agricultural land by increasing crop diversity.

The ethanol production program was launched in 2002 to make use of excessive maize stock piles and shortly after that the State Scheme of Pilot Projects on Bioethanol for Gasoline Automobiles followed. In 2004 the National Development and Reform Commission (NDRC) extended the scope of the pilot project to national level and initiated the State Scheme of Extensive Pilot Projects on Bioethanol Gasoline for Automobiles (SSEPP). Ethanol production and distribution was state controlled by the government that also imposed their own distribution company. In 2006 extensive projects were carried out in five provinces and 27 cities had achieved 10% ethanol blending targets (Dong, 2007).

As part of the eleventh five year development plan an ambitious goal of 6.6 billion litres of biofuel output by 2010 was proposed by the NDRC in December 2006 although it was not approved by the state council due to rising food prices (Latner et al, 2007).

The Chinese government has been taking careful consideration of the potential impact of biofuel production on the food markets and so there has been a shift from the initial ambitions. However, in August 2007 the National Development and Reform Commission announced Medium and Long Term Development Plan for Renewable Energy. In this plan the share of renewable energy in total primary energy was expected to rise to 10% by 2010 and to 15% by 2020. Ethanol production is projected to reach 2 million tonnes by 2010 and 10 million tonnes by 2020 while biodiesel consumption should correspond to 200,000 tonnes by 2010 and 2 million tonnes by 2020 (GSI,

Global Subsidies Initiative, 2008). The government of China is not likely to grant production permission to new factories using corn or maize as feedstock nor authorise capacity expansion (GSI, Global Subsidies Initiative, 2008). Instead, the attention is to promote non – grain feedstock such as cassava, sweet sorghum and sweet potatoes. However, the limited non – grain feedstock supplies are a constraint to the growth of the biofuels sector whose ethanol production was forecasted to reach 1.7 million tonnes in 2009 (Beckman and Junyang, 2009).

The government controls the price of fuel ethanol at a level that would make ethanol production unsustainable without external financial support. An initial subsidy of US\$ 200 per tonne of ethanol (US\$0.158 per litre) which was granted in 2007 has since been replaced by payments based upon the yearly evaluation of individual plants' performance (GSI, Global Subsidies Initiative, 2008). Also, licensed ethanol producers are exempted from the 5% consumption tax and the 17% VAT. However, there are no direct subsidies granted for biodiesel. Furthermore, there is no biodiesel standard thus preventing its blending with fossil diesel and distribution across the country (Sorda et al, 2010).

4.6.3 Malaysia

Malaysia being the highest producer of palm oil hoped to take advantage of growing interest in biodiesel. It launched its National Biofuel Policy (BNP) in 2005 with a proposed mandatory 5% biodiesel (B5) blend. Furthermore, Legislation to regulate and assist the industry was formulated by the Biofuels Industries Act of April 2007. However, the 5% biodiesel mandate has not yet been officially implemented.

Biodiesel is produced in two variants, envodiesel which is obtained from the direct blending of petroleum diesel with raw palm oil and Palm Methyl Esters (PME) biodiesel manufactured by transesterification of palm oil. Envobiodiesel is for domestic consumption while PME is largely destined for foreign markets and makes up the larger

share of biofuel production. In 2007, Malaysia exported 95,000 tonnes of PME, equivalent to 75% of total biodiesel production (Lopez and Laan, 2008).

There are some incentives which biodiesel producers are eligible to in Malaysia. These are attainable by claiming either Pioneer Status (PS) in the biofuel sector or by applying for an Incentive Tax Allowance (ITA). These support schemes are all based on the promotion of Investment Act of 1986(Hoh, 2009). Tax reductions on statutory income of as much as 70% are allowed under the PS scheme for periods of up to five years on statutory income obtained from biodiesel production. In the ITA, companies with high investment costs on equipment and machinery are targeted. These are given allowances spent on fixed assets which are deducted from taxable statutory income for five year periods. Furthermore, there is no export duty on processed palm oil or biodiesel while there are export charges on crude palm oil.

The inclusion of biodiesel was intended to lower the budget spent on petroleum due to the high subsidies that it enjoys in the country. However, the increasing cost of palm oil (caused by increased demand for biofuel) has contributed to higher production costs for biodiesel production in comparison to petroleum fuel has hampered the growth of the industry.

4.7 Africa

In Africa biofuel industry is not yet well developed compared to other regions of the world although most of Sub Saharan Africa has great potential to produce biofuels. It is envisaged that Southern Africa will be an exporter of biofuels in the near future.

4.7.1 Southern African Development Community

Several countries among them, Malawi and Zimbabwe in the Southern Africa development community (SADC) started ethanol blending as far back as 1982, however currently South Africa appears to have taken a more serious move towards setting an

ambitious national biofuels program. In the past decade some policies have been proposed and implemented such as blending obligations with a view to favour import substitution by diversification of energy resources. In other countries within the region biofuels are in the early stage of development with negligible production quantities although there is reasonably high potential.

The proposed South African biofuels industrial strategy (December, 2007), sets a target to displace 2% of transport fuels (Petrol and Diesel) by 2013. In this strategic plan, Maize and Jatropha are left out to avoid the fuel versus food conflict. Fulfilment of this target requires use of 1.4% arable land; this is possible considering that South Africa still has a considerable amount of arable land. The strategy is intended to address issues of poverty and economic recovery by promotion of farming in underutilised arable land. The feedstock included in this draft proposal includes Sugar cane, Sugar beet, Soybean, Sunflower and Canola while maize and Jatropha are excluded. The proposed blends are B2 and E8 for biodiesel and ethanol respectively. However no blending obligations are cited in this program, although proposals for 100% tax exemption and increased fuel levy exemptions of 40% to 50% have been made. Furthermore, a producer incentive scheme will be given to ethanol produced from feedstock grown on targeted land. Lately, increasing oil prices favour the biofuels penetration and development without substantial fiscal burdens.

In Malawi, Ethanol Company of Malawi (ETHCO Ltd.) has been producing ethanol from molasses from the sugar industry as far back as 1982 to comply with a 10% – 20% blending ratio that was intended to reduce foreign exchange expenditure. The main energy crop has been sugar cane although potential for maize, cassava, sunflower and jatropha exists. Malawi has the advantage of reduced risks related to drought due to availability of enough water for irrigation from Lake Malawi. However, there is the risk of land use dynamics and food security with the increased use and production of biofuels. The ethanol program is still expanding and in 2007, ETHCO acquired a Flex fuel vehicle from Brazil for tests in Malawi.

Triangle limited in Zimbabwe started producing ethanol from different sources and varying reasons as early as 1980 with blending targets of 8% – 13%. However, droughts experienced in the early 1990s affected sugar and consequently ethanol production. Although most of the ethanol is produced from sugar cane the country has potential to grow maize and cassava, however these also have the risk of compromising food security and land use dynamics. After the drought, attempts to start again were unsuccessful, due to market and competitiveness constraints, such as lack of government support and the preference for the rectified spirits market instead of internal transportation market (Amigun et al, 2006; Johnson and Matsika, 2006).

Mozambique is another country that has great potential due to relative abundance of land and water resources as well as favourable climatic conditions and low population density (Batidzirai et al, 2006; Ecoenergy, 2008). However, only recently has the government identified biofuels as a promising opportunity to promote agricultural development, improve energy security, boost exports and attract foreign investments while tackling poverty (Lucia L. Di., 2010). Although still in its embryonic stage, a study commissioned by the government shows that 2.5 billion litres of ethanol from sugar cane and sweet sorghum, and biodiesel from Jatropha and coconut could be produced annually under moderate agricultural technical inputs from an area of just 1.6 million hectare (Ecoenergy, 2008). This corresponds to roughly 9% of the projected EU consumption of biofuels for 2010, and could have been of substantial benefits to Mozambique. The government of Mozambique is now actively supporting the production of biofuels. It adopted a National Policy Strategy in 2009 (Government Mozambique, 2009) which emphasizes the importance of large scale production of biofuel for exports as well as for domestic markets as a means of advancing rural development, poverty reduction, export revenue and energy security. However, most of these initiatives are still at strategic levels and yet to be implemented fully.

Another country with significant potential yet little production and utilisation of biofuels is Tanzania. According to a study funded by GTZ in 2005, the country was found to have substantial potential to produce biofuels from several feedstocks due to its abundant

arable land, considerable water resource and favourable climatic conditions. The drivers for the production and consumption of biofuels in Tanzania have been the increasing prices of fuel on the international market, availability of energy crops and possible need to improve air quality. The feedstock available and being considered in Tanzania are Sugar cane, Sugar by - products, lignocelluloses waste from sisal for ethanol, while Jatropha, oil palm and many other oil yielding plants can be used for biodiesel. Up to date no targets have been set for biofuel use in Tanzania.

On the whole the SADC region has potential which is not yet fully exploited with only a few countries having actually implemented some kind of biofuel policies. The interest in biofuel has been aroused by either public and private interests, availability of feedstock (by products from Cane industry) or the need to ensure security of energy supply. A summary of the biofuel situation in the SADC region is given in Table 4.1 below with regard to policies, potential, and some sustainability and socio – economic aspects associated with biofuels in the region.

Table 4.1: Biofuels in the SADC Region by Country, Policies, Potential and Social Aspects

Country	Policy Aspects	Energy Crop	Sustainability and Socio – economic aspects
Angola	Public Private interests	Cassava, Sugar cane , Maize, Oil palm, Sunflower, Jatropha	Land use dynamics(deforestation), Food security
Botswana	Government interest	Jatropha	Drought related risks, Food security, Land use dynamics
Democratic Republic of Congo	Government interest	Cassava, Oil palm, Soy bean	Food security, threat on biodiversity
Lesotho	Limited interest	Maize, Sorghum	Impact on water resources, Food security, Land use dynamics
Madagascar	Public Private interests	Sugar cane, Oil palm, Cassava, Jatropha	Land use dynamics, Food security, threat on biodiversity
Malawi	Blending targets set	Sugar cane, Maize, Sunflower, Jatropha	Land use dynamics, Food security
Mauritius	Availability of bagasse feedstock from sugar cane	Sugar cane , Maize	Land use dynamics, Food security
Mozambique	Public and private interest due to potential feedstock production	Maize, Sugar cane, Cassava, Jatropha	Land use dynamics, Food security
Namibia	Government interest in perennial crops in view of biofuel blending	Drought resistant crops such as Jatropha	Drought related risks, Food security
South Africa	Blending targets of 2% by 2013 for liquid transportation fuels i.e. E8 and B2 blending ratios. Fuel levy exemption on biofuel. Incentives to biofuel producers on under utilized land	Sugar cane, Sugar beet, Soy bean, Sun flower, Canola (Maize and Jatropha are excluded from the draft policy)	Land use dynamics, Food security, Drought related risks
Swaziland	Produces ethanol from sugar cane	Sugar cane, Maize	Land use dynamics, Food security
Tanzania	Energy security. Potential for feedstock production.	Sugar cane, Maize, Sorghum, Jatropha	Land use dynamics, Food security
Zambia	Public and private interest. Energy security concerns.	Sugar cane, palm oil, Cassava, Maize, Sweet sorghum, Jatropha	Rich soils and 80% available arable land, Land use dynamics(Deforestation), Food security
Zimbabwe	Blending targets: 8 – 13% ethanol but has had implementation problems.	Sugar cane, Maize, Cassava	Land use dynamics, Food security, Drought associated risks

[Sources: Compiled from various sources]

4.8 Observations from a global perspective

Biofuel production and consumption has grown rapidly in the last ten years. Many countries that initially did not have biofuel programs started their projects after 2000. Production and consumption has experienced an increase in the period starting from 2000 to 2009 with fuel ethanol increasing from 16.9 to 72.0 billion litres while biodiesel rose from 0.8 to 14.7 billion litres in the same period. The motivations and policy interventions differ from region to region although some are predominant and can be highlighted as the more influential in observed growth of the biofuel sector.

The need for security of energy supply was the first driving factor at the beginning of the biofuels era in the 1970's and still is a major motivation for many countries that are now embarking on the biofuel production and consumption pathway. Many nations would like to reduce dependence on fossil fuel imports and thus be energy independent. The other, is possibility of exporting to regions that have high transportation fuel demands although this is likely to face some trade barriers due to environmental standards that are being legislated in certain regions. This is so with regions where there is excess potential for production of biofuel. Another reason for taking up production in many countries is the possibility of providing employment to rural farming communities in the hope of improving quality of life for the people. Recently, environmental concerns have entered the biofuel debate and have become a driving factor for regions that intend to reduce their green house gas emissions. However, the debate as to whether the biofuels will help mitigate environmental depletion is ongoing.

The increased production and consumption of biofuel in the world has largely been due to various interventions of governments. Among the many policy instruments being used are mandatory blending, tax exemptions, subsidies, financial schemes, guaranteed markets and technology transfer coming out as the most widely applied and successful. These are country specific to some extent although they are also influenced by regional trade policies. Among the regions, the EU has imposed regulations that are most restrictive in the standards for biofuels that qualify to be consumed in the region.

All the regions that have substantial growth in production and consumption have implemented mandatory blending as a policy measure with the exception of the US which has been giving grants to biofuel investments. At inception of biofuels inclusion in the transportation fuel mix, targets which are mandatory have been set by different countries. These have ranged from blending ratios of 2% to 10% biofuel blending in fossil fuel at the beginning and increased with time in steps with the highest target projections being around 20% for 2030. However, some countries have had more ambitious targets but there are very few of such cases. Mandatory blending has been used by almost all countries that are consuming and producing biofuel.

Tax exemptions and rebates of varying types have also been employed in order to assist in the growth of biofuel industries in many of the successful nations. These have ranged from complete to partial exemptions on excise duty and value added tax (VAT) in most cases. It has been possible to reduce these incentives after some time as can be observed in the case of Germany without causing a large drawback in the growth of biofuel industry. However, some kinds of tax rebates have replaced the tax exemptions that existed in Germany. Brazil has also managed to make their bioethanol cost competitive with fossil gasoline without major incentives after the many years of aiding the industry with government backed support. Some countries have provided subsidies on biofuel production which are in different forms. The US has been a pioneer in awarding tax credits as far back as the 70's for ethanol blenders. Financial schemes are also being used to enhance production of biofuels in many nations. These have been in many forms ranging from financing research to providing loans for capital investments.

Another policy measure that can be attributed to the growth of biofuels in some regions is the provision of guaranteed markets for biofuel feedstock (example is India). This has in cases given confidence to producers of biofuel feedstock that their produce does not go to waste in case of surplus. Farmers can be sure of returns although in many cases prices are fixed and so in some cases might not offer the satisfaction.

It is seen that nations that have encouraged biofuel technological developments centered on production and consumption have also had achievements in enhancing growth of the industry. These have ranged from enhanced agricultural practices and production of vehicles whose engines are fully compatible with biofuel (Such as FFV's). Brazil, Germany and Finland have been able to enhance biofuel developments by encouraging technological developments in production and utilisation. Finland has been able to meet the EU targets by producing second generation biofuels. Brazil Flex fuel vehicles have assisted in increasing consumption of bioethanol and efforts that have been made to improve compatibility in equipment have also enhanced growth of biofuels use.

5. OVERVIEW OF ENERGY SITUATION IN ZAMBIA

Zambia has a population of just over 13 million (CSO, 2010) and growing at 3.5% annually. It has a wide range of energy resource which comprises woodlands and forest, hydropower, solar, coal and other renewable sources. Renewable energy sources are those that are replenish able at the rate equivalent to there depletion. In Zambia, the renewable energy sources include: Solar, mini/micro-hydro; biomass (agricultural waste, forestry waste, biodegradable industrial/municipal wastes, energy crops and animal waste), geothermal and wind. There is very little data available on the quantity and utilisation of these renewable energy resources except of hydropower. However, there is reasonable potential for the production of electricity and biofuel from alternative renewable energy options according to a study by the Ministry of Energy and Water Development funded by GRZ/UNDP on the environmental frame work for biofuels (2008 Ministry of Energy Report).

5.1 Energy resources

The main source of primary energy is biomass with a share of 78.5%, then hydro at 10.7%, oil at 9.5% and coal providing the remaining 1.3% which totals to 7124 ktoe according to IEA 2005 statistics. This is illustrated in the chart of Figure 5.1 given below

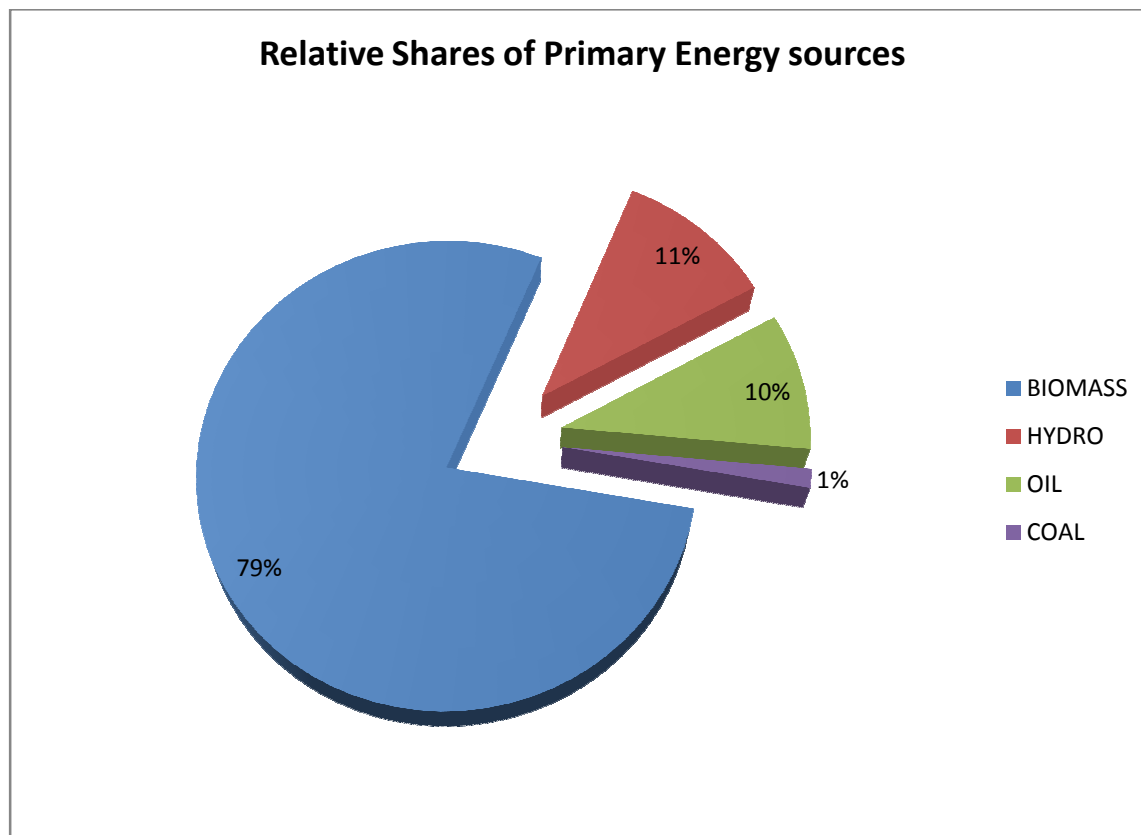


Figure 5.1: Relative shares of primary energy

[Source: IEA 2005 Statistics]

5.1.1 Biomass

Biomass is the main source of primary energy, particularly wood fuel (Charcoal and firewood) although agricultural waste, forestry waste, biodegradable industrial/municipal waste are also used to a less extent. Charcoal is the main source of energy for many households in urban areas while firewood provides most of the energy in rural areas. The use of biodegradable industrial/municipal and agriculture wastes is not very significant. Other biomass energy sources like gel fuels (a fuel made from sugar molasses), biofuels and biogas have just started being promoted by the government, however their contribution is still insignificant.

The woodlands and forests are estimated to cover about 55 million ha, 66% of Zambia’s total area in the National Energy Policy of 2008(NEP 2008). Wood fuel is obtained

mainly from natural woodlands and agricultural lands and it is foreseen to dominate energy consumption now and in the near future. In 2004, it accounted for over 70% of the total national energy consumption with households accounting for 88% of wood fuel consumption according to NEP 2008. Figures within the Ministry of Energy and Water Development show that 60.9% of households used firewood for cooking and 24.3% used charcoal with only 13.8% using electricity. Furthermore, in rural areas, 87.7% use wood fuel for cooking, 9.5% used charcoal and only 1.5% used electricity. The relative shares of wood fuel use in Zambia are given in Figure 5.2 below.

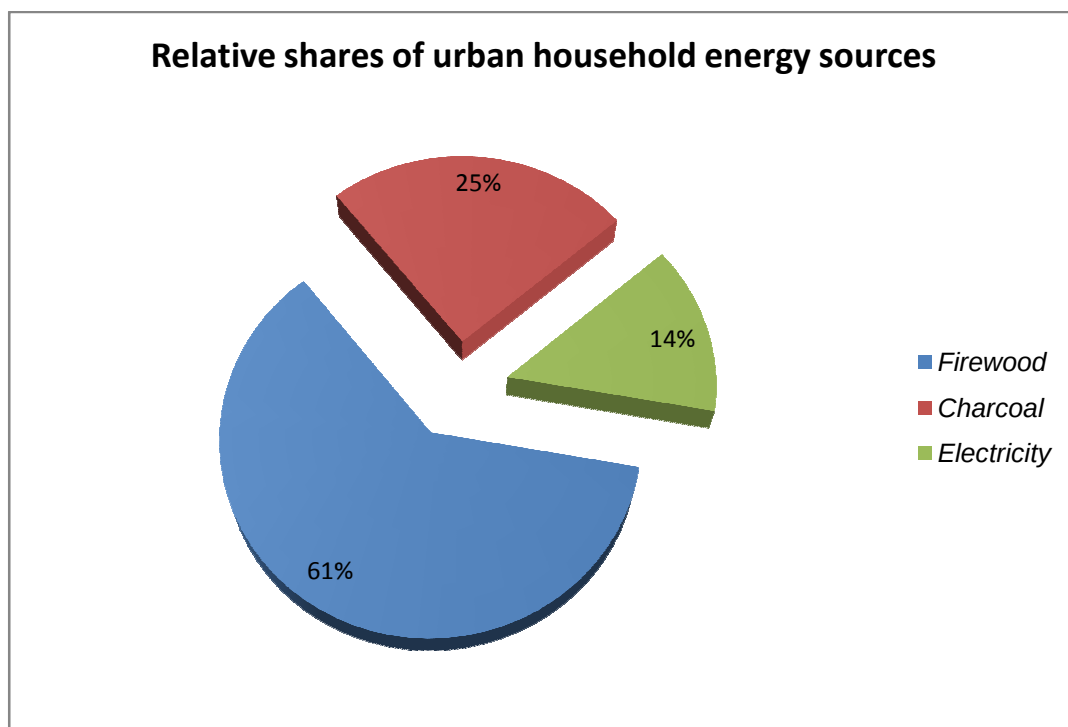


Figure 5.2: Relative shares of urban household energy sources

[Sources: NEP 2008]

If the current trends in utilization of wood fuel continue, this resource which is considered renewable will be depleted and there shall be an energy crisis in the country besides other consequences of deforestation. Most of the woodland near the main cities has been reduced in size due to wood demand for production of charcoal which is the main sources of energy for cooking in the urban areas. In rural areas wood fuel is the predominant source of energy. As at 2002 wood fuel consumption was estimated to be

744000 tonnes per annum (State of the Environment Report, 2004). The deforestation rate is currently about 0.91% per annum and forest area is estimated to have reduced by an estimated 6,672,000 Ha in the period between 1990 and 2005.

5.1.2 Hydropower/Electricity

The second largest quantity of energy used in the country is in the form electricity, which is largely generated by hydropower plants. The total hydropower capacity in Zambia is estimated to be 6000 MW while the installed electricity production capacity is only 1,786 MW considering all installations. In Table 5.1 below installed electrical power capacity is given.

Table 5.1: Current Installed Electricity Generation Capacity

Description	Capacity(MW)	Type
Kafue Gorge	900	Hydro
Kariba North bank	600	Hydro
Victoria Falls	108	Hydro
Lusemfwa & Mulungushi	38	Hydro
Small Hydros	24	Hydro
Isolated generation	10	Diesel
Gas Turbine(standby)	80	Gas
Total Installed Capacity	1,786	

[Sources: Adapted from NEP 2008]

Hydroelectricity plants represent over 95% of electricity production in the country with the major sources being the Kafue Gorge, Kariba North Bank and Victoria falls power stations. Most of the country's electricity is consumed by the mining industry and only 22% remains for the rest of the population. The rural population only account for 2% of the nation's access to electricity (ECZ report, 2005). The Country is currently

experiencing load shedding due to an increased demand in electricity. The electricity consumption by sector is illustrated in Figure 5.3 below.

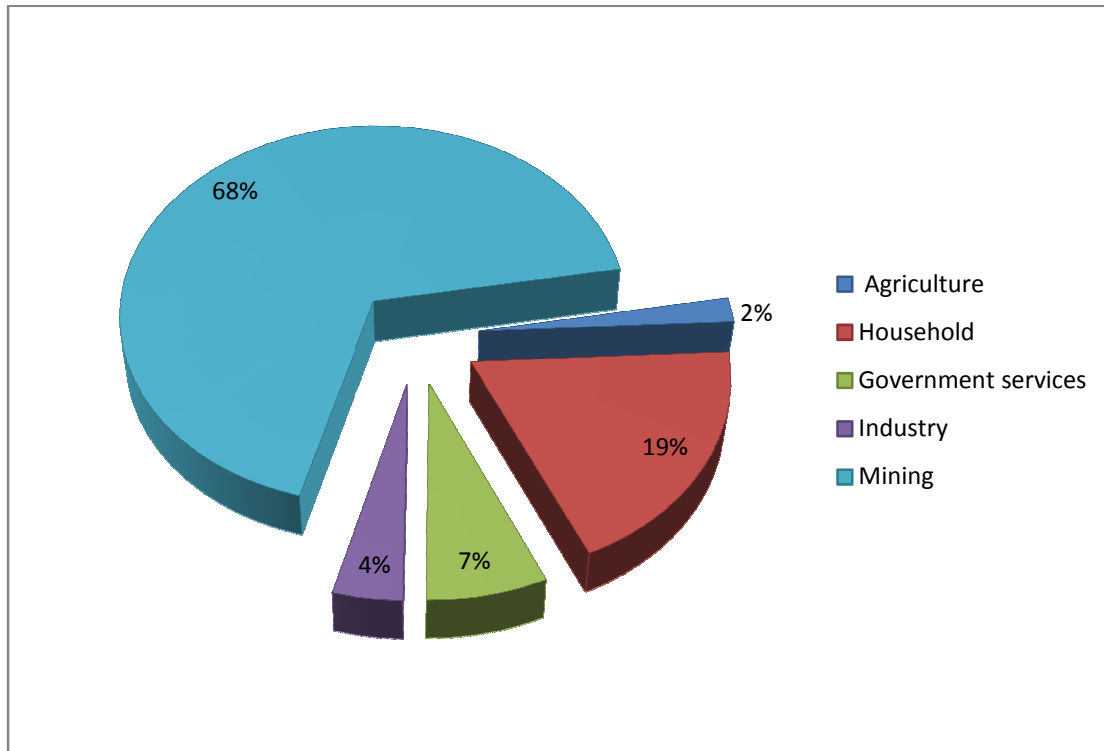


Figure 5.3: Electricity consumption by sector

[Source: NEP 2008]

Zambia’s electricity grid is connected to several parts of the country and neighbouring countries in the region through interconnections at high voltage under the Southern African Power Pool (SAPP). This gives Zambia an opportunity to import and export power to several countries in the region. However, there is need to restructure the energy markets in order to encourage private sector investment in generation and transmission if the grid is to reach other parts of the country. As can be seen in the map in Figure 5.4, most parts of the country are not connected to the grid.

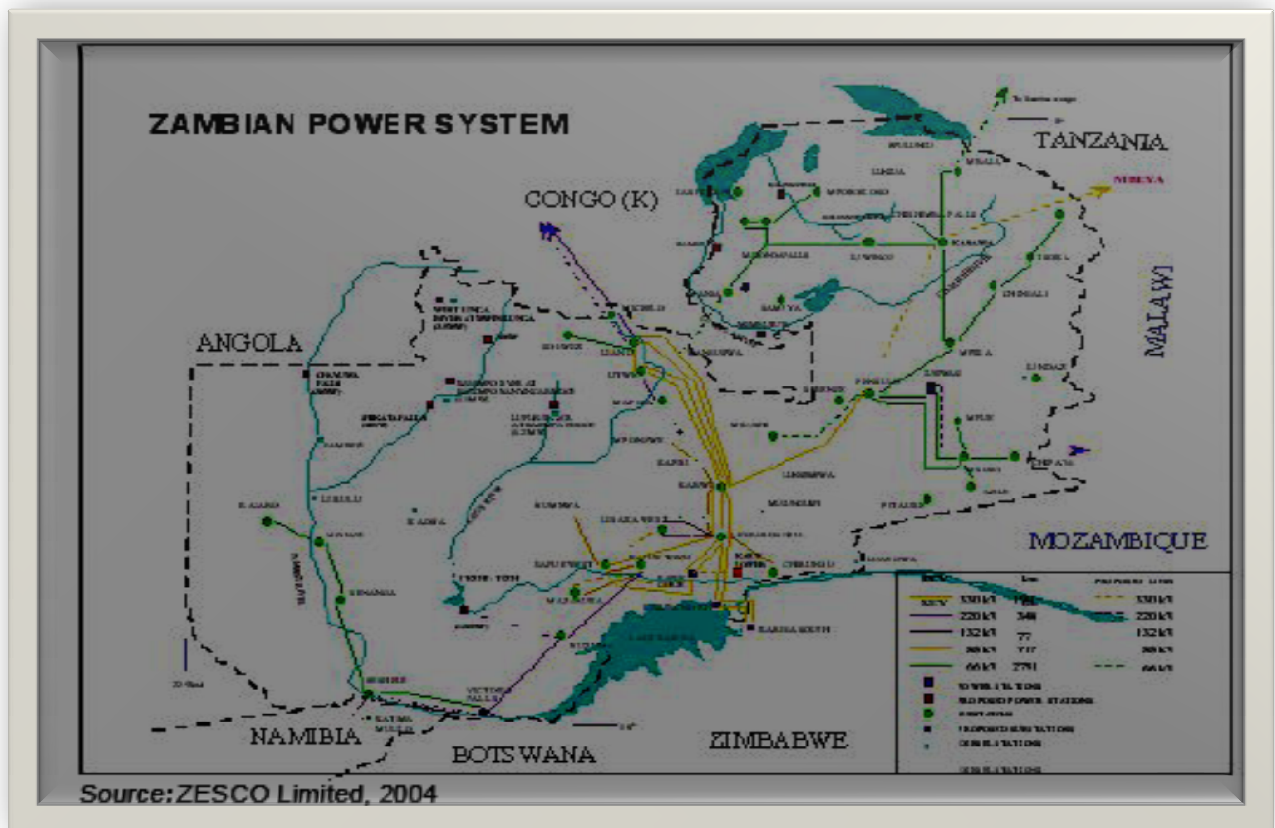


Figure 5.4: Zambia electricity power distribution

[Source: Adapted NEP 2008]

There is also substantial potential for small/mini hydro development. The northern parts of Zambia are most suitable for micro/mini – hydro power developments due to topography and terrain, the geology of the ground and high rainfall figures. Table 5.2 below shows some suitable sites and hydropower potential (were determined) which have been identified by collecting information on rivers with sufficient perennial flows.

Table 5.2: Some small – scale hydro potential

No.	River/Basin	Site	River	Capacity, kW
1	Zambezi	Zambezi falls	Zambezi	To be determined
2	Zambezi	Chavuma falls	Zambezi	10 – 20,000
3	Zambezi	Sachibondo	Luakela	600
4	Zambezi	Mwinilunga	West Lunga	2,500
5	Zambezi	Kapembe	Kabompo	To be determined
6	Zambezi	Chikata falls	Kabompo	3,000
7	Kafue	Kasempa	Lufupa	230
8	Kafue	Mutanda	Lunga	400
9	Kafue	Kelongwa	Lunga	To be determined
10	Chambeshi	Chandaweyaya	Chambeshi	To be determined
11	Chambeshi	Mbesuma ferry	Chambeshi	To be determined
12	Chambeshi	Shiwang'andu	Manshya	1,000

[Source: Adapted NEP 2008]

Investors are being sought to develop hydropower stations, however the high capital investment cost and remoteness of some sites is an obstacle. Furthermore, the sites above are situated in areas with little economic activity to encourage private investment.

5.1.3 Solar energy

There are also several scattered solar photovoltaic systems installed mostly in rural schools, hospitals and some households. The government through the Ministry of Energy and Water development (MEWD) has been disseminating solar energy technology in its rural electrification program. Notably, 400 households have been installed with solar photovoltaic (PV) systems under the Energy Service companies (ESCOs) pilot project in three districts of eastern province. In addition, several schools and Chief's palaces in various parts of the country have been electrified using solar energy (NEP 2008).

5.1.4 Wind energy

Wind is another source of energy that could be tapped in Zambia. Wind speeds in Zambia are an average of 2.5 to 3 metres/second at a height of 10 metres above the ground. These speeds are more suited for water pumping but the number of such installations is currently negligible. However, an assessment of wind speeds at higher heights has not been done and thus further elaboration is required. There are currently no government projects in wind energy although potential sites have been identified for pilot projects.

5.1.5 Geothermal energy

The country also has some geothermal energy resources. There are more than eighty (80) hot springs spread in different parts of the country. The available data shows that there is potential for power generation from most of these springs. Kapisya hot spring was developed to the extent that 2 x120 kW turbines were installed in 1987 at Rufunsa with an initiative of the Italian government. However, the Kapisya installation is not operational but efforts are being made to revive the plant. There are no other exploited geothermal sites and most of them are currently just tourist attractions.

5.1.6 Biofuels

Biofuels from energy crops like sugar cane, sweet sorghum, cassava and jatropha cocas have just been recognized as an effective way of fulfilling the countries energy requirements considering the instability of petroleum supply which is experienced from time to time on the international energy markets (NEP 2008). Several conferences have so far been held to look in to ways of enhancing biofuels production and utilisation. The new national energy policy is focused on promoting the production and utilisation of biofuels, of which, biodiesel, bioethanol, biogas, gel fuel and briquettes are included as

a way of ensuring energy security and as alternative fuel. However, there is still insignificant use of biofuels in the country.

The government intends to set aside some land for the production of energy crops and is encouraging private public partnerships investments in renewable energy technology (NEP 2008). Table 5.3 below gives a summary of the resources in terms of potential, availability (were available) and utilization of renewable energy sources and technologies.

Table 5.3: Renewable energy resource statistics data

Renewable energy source /Technology	Opportunities/Use	Resource availability	Potential energy output
Solar	Thermal(Water heating), Electricity(water pumping,lighting and refrigeration)	6 - 8 hours sunshine	5.5kWh/m2/day
Wind	Electricity,Mechanical (Water pumping)	Average 3m/s at height of 10m above ground	Good potential for irrigation
Micro/Mini – hydro	Rural electrification	Reasonably extensive	Requires further assessment
Biomass(Combustion and gasification)	Electricity generation	Agro wastes(300,000 tons/year-Bellard-Tremer 2002) Sawmill wastes (112,000m3/year)	Requires further elaboration and quantification
Biomass (For household energy)	Improved charcoal production	Sawmill waste and indigenous trees from sustainable forest management	Requires to be elaborated but reasonably extensive
Biofuel (Bioethanol and Biodiesel)	Bioethanol for blending with petrol and replacement of diesel for transportation	Sugar cane Sweet sorghum Palm oil Jatropha	Potential requires elaboration
Biogas	Electricity, heating and cooking	Animal waste Municipal and sewerage waste	Potential requires elaboration

[Source:Adapted NEP 2008]

5.1.7 Fossil fuels

Petroleum is the only energy resource that is wholly imported to Zambia and is important for the transport and mining sectors. Currently, per capita oil consumption is estimated to be 172 toe/day per 1000 people (ECZ Report 2006). Most of the crude oil is imported from Asia although Zambia is a neighbour to Angola which is currently the largest producer of oil in Africa. This is because the refinery technology at INDENI is not capable of processing Angolan crude oil into finished products besides the fact that the country has been politically unstable till recently. Furthermore, Zambia has experienced recurrent fuel shortages due to frequent down times of the INDENI processing plant whose equipment has become obsolete.

The oil refinery with design capacity of 1.1 million Metric tonnes per annum was established in Ndola, and is supplied via a 1,706 kilometre pipe line that runs from Dar - es - Salaam in Tanzania known as the TAZAMA pipe line. This is owned by the government of Zambia (63% shares) and government of Tanzania (37%). The feedstock for processing is procured through international competitive bidding, while private Oil Marketing Companies (OMCs) are involved in the marketing and distribution of finished products. The transport sector consumes 53% of petroleum products followed by the mining industry taking 27% (Figure 5.5).

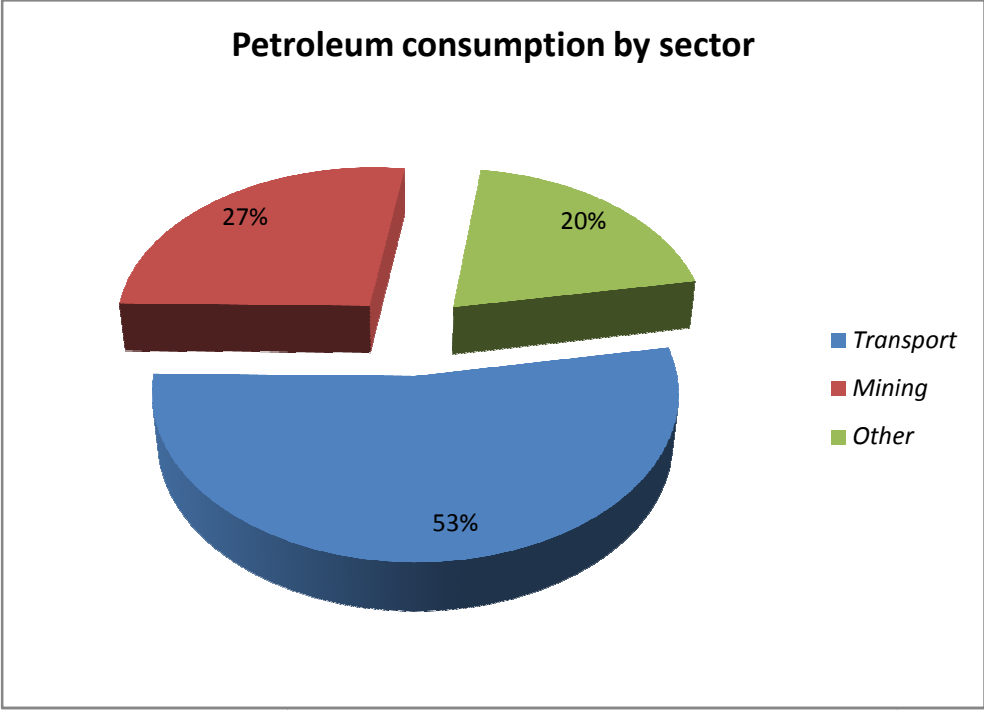


Figure 5.5: Petroleum consumption by sector

[Source: Adapted NEP 2008]

The government currently spends a considerable amount of the foreign exchange earnings on importation of crude oil and finished petroleum products. Most of the petroleum products used in the country is obtained from imported crude oil. There are several fees that are paid in acquiring the petroleum along the supply chain. Among these are harbour dues at 1.25% of the CIF Dar es Salaam value of cargo which are paid to Tanzanian authorities. A TIPER pipe charges at US\$0.75/MT of crude oil and a shipping agent charge approximately US\$50,000,000. TAZAMA also charges pumping and agent fees at INDENI of US\$39/MT and US\$15/MT respectively. The financing cost at 4% of CIF Dar es Salaam value is also paid. Furthermore, there is an import duty at 5% CIF Ndola value of petroleum feedstock and insurance at 0.15% of CIF Dar es salaam (DOE, 2010). These costs are transferred to consumers through taxes on pump price.

The refinery was owned by the government of Zambia and Total with each holding 50% shares until the year 2010. Total Oil Company (Limited) released its shares and the

government has been looking for a partner. In Table 5.4 below monthly uplifts from Indeni are listed when it is fully operational.

Table 5.4: Average monthly uplifts from Indeni oil refinery

TYPE OF PETROLEUM PRODUCT	AVERAGE MONTHLY UPLIFT(litres)
Gasoline	12,000,000
Diesel/Gas oil	30,000,000
Liquid petroleum gas	190,000
JET – A – 1	2,900,000
Heavy Fuel oil	5,800,000
Kerosene	918,000
Total	51,808,808

[Source: Adapted DOE Website]

The refinery has not been very reliable as it has had many down times which lead to fuel shortages now and again. This usually prompts government to spend on importing finished petroleum products which have to be transported by road through neighbouring countries since Zambia is landlocked. TAZAMA also takes part in the procurement, transportation and storage of finished petroleum products during periods when Indeni refinery is shut for maintenance and repair.

The refinery is situated next to Ndola fuel terminal which was established in 1973 to provide storage and act as a distribution point of finished products. It is thus mainly made up of storage tanks, road and rail loading facilities. The terminal is currently being managed by TAZAMA Pipelines limited. It has storage capacity of 50 million litres.

Most of the petroleum products are consumed by the mining and urban areas while rural areas have very little or completely no access to these products. The actual statistics of fuel consumption are not complete for most of the years as the energy regulation board relies on data provided by oil marketing companies for use in

compilation of the information. Furthermore the ERB experiences difficulties of data collection as some OMC's do not give their sales data. In the table below, the ERB statistics show fuel sales as given by some marketing company. However, It should be noted that some OMCs did not give data in some instances, but that the information gives a general reflection (Table 5.5).

Table 5.5: Petroleum consumption evolution data

YEAR	2002	2003	2004	2005	2006	2007	2008	2009
GASOLINE(Million lt)	139	155	159	151	107	165	177	213
DIESEL(Million lt)	229	266	314	329	263	414	508	534

[Source: Adapted from ERB Website]

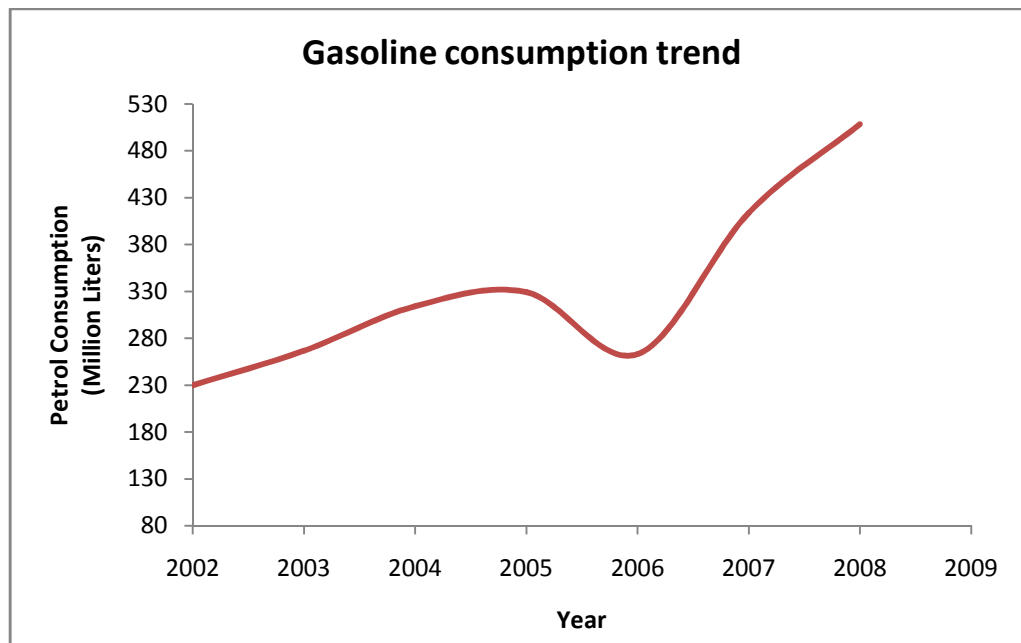


Figure 5.6: Gasoline consumption trend

[Source: Adapted ERB Website]

It is clearly seen in Figure 5.6 that the petrol consumption appears to have been rising from 2002 to 2004 reaching a peak and declines towards 2005 to a much lower

consumption in 2006 due to the global economic crunch during that period. This also indicates the dependence of Zambia on the global economy. After the economic crunch, we see the consumption rising from 2007 onwards due to increased mining activities.

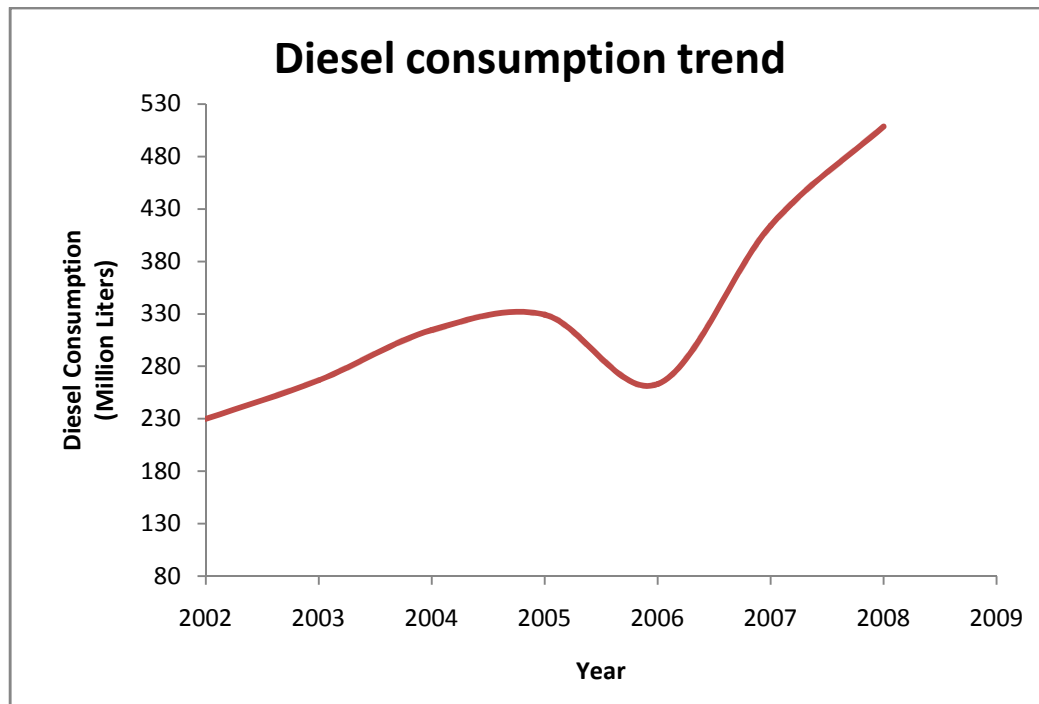


Figure 5.7: Diesel consumption trend

[Source: Adapted ERB Web]

A trend similar to that of gasoline is observed for diesel consumption, an increase from 2002 to 2005 as illustrated in Figure 5.7. This has largely been due to increased mining activities which, however major consumer of petroleum diesel.

The transport sector in Zambia is very important for social and economic activities. This is composed of largely road networks, railway and airways. Inputs and finished products in the mining and agricultural sectors are mainly transported by road and are thus greatly affected by disruptions in supply of fuel. The road network runs across the country and is the main link between urban and most of the rural country side. The main means of transport for people within towns and across the country is by road using buses and cars.

5.2 Energy pricing

The price of energy in Zambia is different depending on the energy form. The prices of firewood, charcoal, coal and renewable energy are market driven. Electricity and petroleum are priced under different mechanisms. Cost of the bulk trade of electricity is determined by bilateral contracts in which the parties to the contract negotiate prices that are subject to approval by the Energy Regulation Board (ERB). Supply to households, commercial entities and industry are governed by the Electricity Act. The retail price of electricity includes costs of generation, transmission, distribution and supply and is uniform across the country. These are regulated according to the Electricity and Energy Regulation Acts. The electricity industry at the moment is monopolized by ZESCO, the main producer and distributor of electrical power. Although being regulated by ERB, there is much to be done in this sector for improved provision of services. The government's energy regulation body does not much control on ZESCO which is wholly owned by the state. Uniformity of generation, transmission, distribution and supply costs make it difficult for development of further infrastructure by other investors

For petroleum, currently the pricing consists of the refinery gate cost, government taxes, ERB fees, fuel levy, transport as well as oil marketing companies and dealer margins. Some of the notable taxes are listed in the table below. Petroleum pricing is liberalised with the role of ERB being that of setting the pricing formula and monitoring compliance. The Energy Regulation Board sets the wholesale prices due to the monopolistic nature of the refinery. It uses the Import Parity Pricing (IPP) mechanism after abandoning the cost – plus pricing model which was noted to have had some flaws. The IPP works on how much it would cost to bring in finished petroleum products into Zambia, however it gives a discount for the existing infrastructure which is the TAZAMA pipe line.

Other petroleum products such as liquefied petroleum gas (LPG), heavy fuel oil (HFO), light fuel oil (LFO), bitumen and low sulphur oil (LSFO) have their prices determined by the petroleum refinery. Significant price variances between bulk supply and retail LPG are of major concern.

Table 5.6: Taxes on selected fossil fuel products

Description	Petrol	Diesel	Domestic Kerosene	JET 1	HFO
Import duty %	5	5	5	5	5
Exercise duty %	45	15	15	0	30
Road Levy %	15	15	0	0	0
Value added tax %	17.5	17.5	0	17.5	17.5

[Source: Ministry of water and energy Affairs website]

5.3 Analysis of energy situation in Zambia

The overview of current energy situation in Zambia shows that there is need to increase the provision of energy services on the whole for improved social – economic development of the country. The current practices in biomass use are having negative impacts on the environment and health of the population of the nation.

The fast rate at which the woodland is being depleted leading to deforestation will have adverse effects on the climate as carbon sinks are being depleted. Already, we have most land near urban areas turned into waste land and hence the need for greening. Measures that will ensure sustainable development have to be put in place for the sake of future generations.

To date, most of the country is not connected to the grid and electricity is not available to the majority of the rural population. Thus, if standard of living is to be improved, there is needed to increase provision of energy services at a local level. The main economic activity, agriculture in rural Zambia requires products to be transported to urban areas which provide the major markets. For improved production through automation, there is a need for energy. However, the current state of the electricity provision is unlikely to meet the needs of the isolated rural areas. Localised solutions thus have to be sought and implemented to ensure security of energy supply.

The large number of taxes and other charges imposed on petroleum products make them expensive for the end user. Some of these costs could be reduced by local production of biofuels given some incentives. It is possible that some policies can make biofuels competitive to fossil fuel in terms of economic, environmental and social benefits. The carbon emission tax that was recently introduced if restructured could just as well help to make biofuels more attractive. However, it is not certain whether biofuels can reduce dependence on petroleum imports without increasing other costs.

In the vision 2030 for Zambia, the energy vision is to provide universal access to clean, reliable and affordable energy at lowest total economic, financial, social and environmental costs consistent with national development goals by 2030. The targets set in this vision such as provision of abundant and reliable supply of affordable energy to both urban and rural areas can only be achieved by careful implementation of NEP 2008. At the moment there is limited use of alternative renewable energy and the potential resources have not even been estimated. Zambia is a net importer of energy and depends solely on imported petroleum for the country's transportation needs.

Clearly from above, biofuels have a place in the energy mix of Zambia, but some questions still remain unanswered. For example;

- What are the feedstock options available for biofuels production?
- What technological options are available for Zambia?
- What are the prospective potential future economic, environmental and social impacts?
- What policy measures can be put in place for the success of the biofuels industry in Zambia?

In order to answer these questions, there is need for careful analysis of the availability of feedstock, technology, environmental impacts and appropriate policies in Zambia. It is for this reason that this study uses analyses of the global, regional and local situation

to suggest possible answers to the questions above. Furthermore, the conclusions from the analysis are used to develop an illustrative model for future scenarios for selected policies in Zambia which can be used to project, monitor, analyse and evaluate developments in the biofuel industry.

5.4 State of biofuel industry in Zambia

In Zambia, biofuels have only recently been recognized as a viable option for meeting some of the country's energy requirements considering the disruptions in petroleum supply which are often experienced on the international markets (NEP 2008). Just as world attention has turned towards biofuels as an alternative to petroleum to ensure security of supply and instability in fossil fuel import prices, Zambia has also been motivated by the need for security of energy supply.

Although biofuels were only included in the national energy policy of 2008, there have been some activities before that. According to Climate Change Mitigation of Southern Africa: Zambia Country Study that was funded by DANIDA and collaborating partners between 1996 and 1998 for the Ministry of Environment and Natural Resources in Zambia on biomass availability and utilisation for transport, three feedstocks are identified for the production of biofuel which can be used for blending with petroleum fuels (Gordon A. Mackenzie, 1999). These include sugarcane and sweet sorghum for ethanol production while *Jatropha cocas* are mentioned for biodiesel production.

Several activities are already being carried out by government, NGO's and private companies in the biofuels sector but this has not been on commercial level and the impact is yet to be seen. These activities include cultivation and production of biofuel for possible blending with petroleum. Biofuel activities can be traced back to 2005 before the approval of the current energy policy. Several sectors of society and government had already taken a lot of interest in biofuels and their inclusion in the energy mix.

5.4.1 Government interventions before NEP 2008

The Government of the republic of Zambia facilitated the formation of the Biofuels Association of Zambia (BAZ) in 2005. This organisation holds stakeholder meetings, lobby's government to support biofuels production and utilisation. It is also involved in educational campaigns on biofuel.

In vision 2030, it is aimed that a strategy is developed for the implementation of a biofuel policy which at the time of its formulation no policies on biofuels existed. In the same document, there are plans to prioritise the biofuels sector in Zambia so that biofuels projects can qualify for incentives under the Zambia Development Agency Act. The Energy Regulation Act should be amended to incorporate biofuels in the definition of energy. Furthermore the need to take a national energy crop resource assessment was identified as a requirement if biofuel was to be incorporated in the energy mix. Additionally, in the vision 2030, the need for the development of an Environment Framework for biofuels production and use in Zambia was identified.

There are several government sectors and units that are involved in the biofuel program for the country. The Ministry of Energy and water development has been in the forefront of foreseeing development of biofuels. It is responsible for energy and water resources in terms of policy formulation. The ministry responsible for agriculture has also played a part in identifying suitable energy crops and has a role to play in research involving agricultural productivity. For monetary issues in the country the Ministry in charge of Finance and National Planning expected to play an important role in formulating financing mechanisms for biofuels. The Ministry of Tourism, Environment and Natural Resources is responsible for the environment and natural resources which include forests, water and others. The Ministry of Commerce Trade and Industry has the policy formulation and regulating roles within its various line departments and has a role to play in the biofuels trade. Land issues are looked at by the ministry of lands and the roles range from allocation of land to authorising use of land and will thus play an important role in allocating land resources for biofuels production. Further, Ministry of

Education, Science, Technology, and Early Childhood is responsible for issues pertaining to scientific and technological developments in the country as well as technical training.

There are also some private companies that have embarked on biofuel projects. For instance, Oval Biofuels has started planting Jatropha for biodiesel production and Thomro Biofuels another private company has acquired land for production of sugar cane with plans of using the cane as stock feed for biodiesel production.

Oval Biofuels and Thomro Biofuels are some of the notable private investors in the biofuel sector among many others. The private investor Oval Biofuels has embarked on growing jatropha cocas and processing it into biodiesel. Currently, 100,000 hectares of plantation has been cultivated with jatropha cocas and a pilot processing plant of the Ageratic type has been established with a capacity of 3,000,000 litres per annum. The company has managed to obtain tax concessions for five years under the Zambia Development Agency conditions for capital investors. Local people are already involved with both commercial plantations as well as small scale out grower schemes, both of which have manual labour in crop maintenance and harvesting. The company is also carrying out tests with biodiesel from jatropha cocas on their vehicle fleet according to the company website (<http://www.ovalbiofuels.com>). However, there are no records about the success of the trials and the out grower schemes.

Thomro biofuels is currently producing biofuel and a range of by products from Jatropha seedlings such as organic fertiliser, glycerine, briquettes, biogas and Soap. They also provide services in the sense of training, consultancy and public seminars according to the company website (<http://www.thomrobiofuels.com>).

In 2006 to 2007, DAPP (Development Aid from People to People), an NGO worked on a project in cooperation with the Gaia Movement (A Swiss based NGO) and Global Village Energy Partnership (GVEP) on a project to research on the use of Jatropha Oil in stationary engines using a dual fuel system. The "Towards biofuel self supply" project

started with small holder production of jatropha and biodiesel. About 600 farmers have been trained in jatropha and biodiesel production. A market in jatropha oil in dual fuel systems for stationary diesel engines for power production has been developed (<http://gaia-movement.org>).

The Technical Development and Advisory Unit (TDAU) at the University of Zambia is an engineering research and development unit established in 1975 with a mission to promote sustainable human development through the development, acquisition and dissemination of technology, production processes and management know-how appropriate to society. It has been developing and testing of Jatropha seed oil press of the screw press for extraction of oil from jatropha seed on behalf of Agribusiness in sustainable African plant products (ASNAP). It has also participated in conducting Biomass resource audit for small scale rural farmers in Chongwe and Mumbwa for AREED/SEMP (Country Report –Agrofuel crop production, 2007). Their work has ranged from designing and development of mechanical crushers, presses and filters for extraction of oil from various oil seeds for production of biofuels.

Marli investment is a project promoter and developer of jatropha for production of biodiesel. Among some projects the company is undertaking is the management of an outgrower scheme with over 25,000 farmers and 8,500 ha cultivation of Jatropha.

These activities and those of NGOs and the private sector have led to the realisation of the need to include biofuels in the energy mix.

5.4.2 National Energy Policy 2008

The energy policy was reviewed in 2008 to include biofuels among others. In the reviewed National Energy Policy (NEP 2008), there are several policies that have been stated which are intended to establish biofuels as part of the energy sector. These include expanding the role of biofuels in the national energy mix through the use of ethanol as a blend with petrol and use of biodiesel as a motor fuel or blend with diesel

and other productive uses like power generation. The policy intends to ensure security of supply and stabilization of prices of fuels by promoting the utilisation of biofuels for transportation as an alternative to petroleum. This is to be achieved through supporting the growing of energy crops and investment in biofuels of Zambia through appropriate support. Furthermore, Zambians are to be supported so that they participate in the biofuel industry as shareholders and ensure that the use of biofuels for the market is given priority without compromising food security and environmental sustainability.

There is also emphasis on ensuring that data and information on market demand, resource assessment and applications of biofuels is made available by undertaking economic studies on the economic feasibility of using biofuels. This is to be done through undertaking studies on needs/demand, resource and technology assessments on biofuels and building capacity to monitor and regulate biofuel production and use. The need to build capacity in institutions involved in the biofuels programme is also identified in the policy document.

The policy is to provide a legal and institutional framework for the biofuels sub – sector by establishing a regulatory framework, defining rules of entry and exit from the industry. Also local people involved in the subsector are to be protected against exploitation by investors. It is intended that a cost line in the price of petroleum products is provided and a proposed 0.5% is to be channelled towards a biofuel development fund. Laws to protect the environment against invasive species are also supposed to be enacted. It is envisaged that all the existing and proposed biofuels ventures undergo environmental impact assessments (EIA's) and that allocation of land for biofuels is done with regards to possible negative impacts of the energy crops and alien species on the environment. A regulation to ensuring that production of biofuels is done under correct environment management frameworks should be put in place.

The government intends to support investment in the biofuels industry through appropriate incentives, standards and research. Some tailor made incentives for the biofuels industry are to be introduced including those of standards for energy crops,

biofuels quality and blending ratios which should be mandatory. Furthermore, research to determine which type/species of energy crops would give the best quality and yield of biofuel is being encouraged. Innovation and development of appropriate local technology for extraction and processing of biofuel is also being prompted.

5.4.3 Analysis of the biofuel policy

As can be noted from the section above, the policies as stated in NEP 2008 are in outline form and no actual targets are set. The analysis of global and regional policies clearly shows that these influence and affect policies at national level. It is therefore necessary that these policies are given dimensions in technical, environmental, socio-economic and sustainability which are in line with the global, regional and national agenda. In this section, the policies for Zambia are looked at in terms of the quantitative and qualitative dimensions set by global and regional policies and with respect to strategies associated with biofuels. Stated as plainly as they are, the policy statements are not specific and do not give a clue of what strategies shall be put in place to achieve the various objectives embedded in them.

The first policy statement is the goal to expand the role of biofuels in the national energy mix through the use of ethanol as a blend with petrol as well as the biodiesel use in neat form or as a blend with diesel for motor vehicle fuel and uses such as stationery power generation in engines. As biofuels are just being introduced, a realistic approach will be to start with lower blends and gradual increments to meet set targets. A lesson from most of the world biofuel consuming countries is that blending targets of 2% rising to 5% and so on are set for different subsequent years and 20% target for 2030. In the sub region a target of 10% has been set for 2030 by some countries. It will be necessary to set targets which can be measured and evaluated.

Secondly, the policy aims at ensuring security of supply and stabilisation of prices of fuels by promoting utilisation of biofuels for transportation as an alternative to petroleum

fuel. This is to be achieved by support to production, investment, participation of Zambians and ensuring that the use of biofuels is given priority in the market. The implementation of this policy objective has a lot of implications on the socio – economic and environmental sustainability. Several approaches can be used in achieving the policy statement as has been done in other parts of the world. These include giving farmers loans, tax incentives, guarantying markets and educating farmers on best farming practices. It is important that the appropriate crops are grown in correct regions and the environmental consequences well established. Furthermore, it should be taken into consideration that growing of biofuels will definitely have an impact on food security as has been observed in other regions of the world. The sustainability criteria that no land used for production should be the benchmark in the support of biofuel projects.

Thirdly, the national energy policy aims at ensuring availability of data and information on market demand, resource assessment and application of biofuels. This entails undertaking studies on the economic feasibility, needs/demand, resource and technology assessments on biofuels. The economic feasibility aspect of biofuels has to be evaluated with consideration of the full well - to - wheels cycle for a comprehensive view. This is the only way in which the real benefits of the inclusion of biofuels can be evaluated although it has been observed that some of the costs associated with these fuels are difficult to account. The needs versus demand of biofuels are closely related to petroleum fuel demands in the case of blending requirements. Thus, in this regard assessments of such a nature can be done if the future fossil fuel requirements are correctly projected. A projection of this demand can then be used to analyse the resource requirement which in turn can be used to measure the available resources.

The fourth policy statement is the provision of a legal and institutional framework for the biofuels. This will give guidelines on the rules of entry and exit in the sub sector, protect local people, promote environmental sustenance and uphold laws of the country. This policy objective is paramount in ensuring the sustainability of the biofuels sector. Among the things that should be taken into account must be the environment, laws regarding labour and participation of local communities in biofuel projects. Indeed all existing

biofuel projects must undergo environmental impact assessments (EIA's), so should new projects and those that are to be decommissioned to ascertain the impacts. The welfare of locals involved in biofuel projects should also be given priority to ensure that the benefits are shared.

Finally, the energy policy on biofuels intends to support investment in the industry through appropriate incentives, standards and research. This is to be achieved through tailor made incentives, standards for energy crops, biofuel quality and blending ratios. Further, it is intended to promote research on quality and yields as well as innovative and appropriate technology. However, the incentives are not clearly outlined in the biofuel policy. In some regions of the world incentives on exercise duty, new technologies and direct funding on research have been used.

6. ANALYSIS AND RESULTS; POLICIES AND PROSPECTS FOR INTRODUCING BIOFUELS IN ZAMBIA TRANSPORT SECTOR

6.1 Analysis and Results

In Zambia, like many other countries, there is a general lack of data on the actual resource availability of biomass for production of Biofuels. This is partly because it is difficult to account for the total biomass resources, more so bio-waste which can be used as feedstock in biofuel production. This is largely due to unorganised collection and disposal systems (NEP 2008). Comprehensive assessments of the biomass resource have not been undertaken to quantify the actual potential. Information on the availability of land for cultivation of biofuel feedstock is not readily available and government interventions are not solidly in place.

In this study, software simulations are used to assess needs/demand, resources, environmental and socio - economical consequences of introducing biofuels in Zambia under different policy interventions. The input data is collected from various sources, mostly government documents, webpages and assumptions made based on critical analysis of existing literature (Appendix I). Long – range Energy Analysis and planning software (LEAP) is used to analyse this data under selected policy implementations. Policies considered successful in sustaining biofuels industry in other countries are selected and used to model the future of biofuels in Zambia.

6.1.1 Method used for scenario analysis

Several scenarios for the introduction of biofuels in the energy mix are considered in this analysis. The two main players in the biofuels sector, bioethanol and biodiesel are analysed with respect to blending quantities required for different future scenarios using estimates of demand from various sources. The quantities required to displace 5% and 10% by 2015 and 2030 respectively are estimated together with the production requirements in terms of feedstock by quantity and land where possible. Several

assumptions on crop and fuel yield were used to estimate the land requirement. The analysis does not take into account improvements in crop yield due to future technological development but rather uses averages assumed to be currently achievable in Zambia. The possible environmental consequences of fossil fuel displacement are simulated for the study period. A demand to cost benefit analysis for the scenarios is also carried out using estimates of final demand costs for different fuels.

The Long - range Energy Alternatives Planning System (LEAP) is used in analysing various mandatory blending scenarios for implementation and introduction of biofuels in the transportation fuels of Zambia. LEAP is a scenario based energy – environmental modelling tool that requires less data but has considerable built in analysis capabilities. Zambia, like many other developing countries has very segmented energy data thus making LEAP a more appropriate tool for modelling the prospects of introducing biofuel in transportation fuel mix. A careful selection of correlating data was finally selected and used as key parameters in LEAP for the baseline and business as usual scenarios. The input data and parameters for the scenario analysis are given in appendix I.

Long – range Energy Alternatives Planning System software has a standard tree structure which can be used to analyse the transformation of an energy system such as the transport sector in the case of this study. However, the data collected in this study was not very compatible to the default structures of LEAP. Analysis of the transport sector transformation in LEAP requires input data such as either passenger or vehicle kilometres travelled. However, such data is not available for Zambia, hence LEAP tree structure was modified to use final demand units of final litres of fuel consumed rather than vehicle or passenger kilometres travelled per annum. Hence the analysis in this study uses data on final fuel consumption obtained from the energy regulation board for the year 2006 as initial input (Appendix I).

6.1.2 Business as usual scenario

In the first place an energy demand analysis under the business as usual scenario is simulated to establish the possible energy consumption trends. In this analysis, energy demand is simulated as a function of income growth of the country's population. This is at an estimated income growth rate of 3.5% per annum and base year consumption data for gasoline and diesel in the year 2006(CSO Web, ERB Web).This shows that demand for the gasoline and petroleum diesel will double by the year 2030. The graph below shows the projected energy demand pattern for gasoline and diesel in Zambia from 2006 to 2030. The results are given in final energy demand units (Million Gigajoule per annum) in Figure 6.1 below.

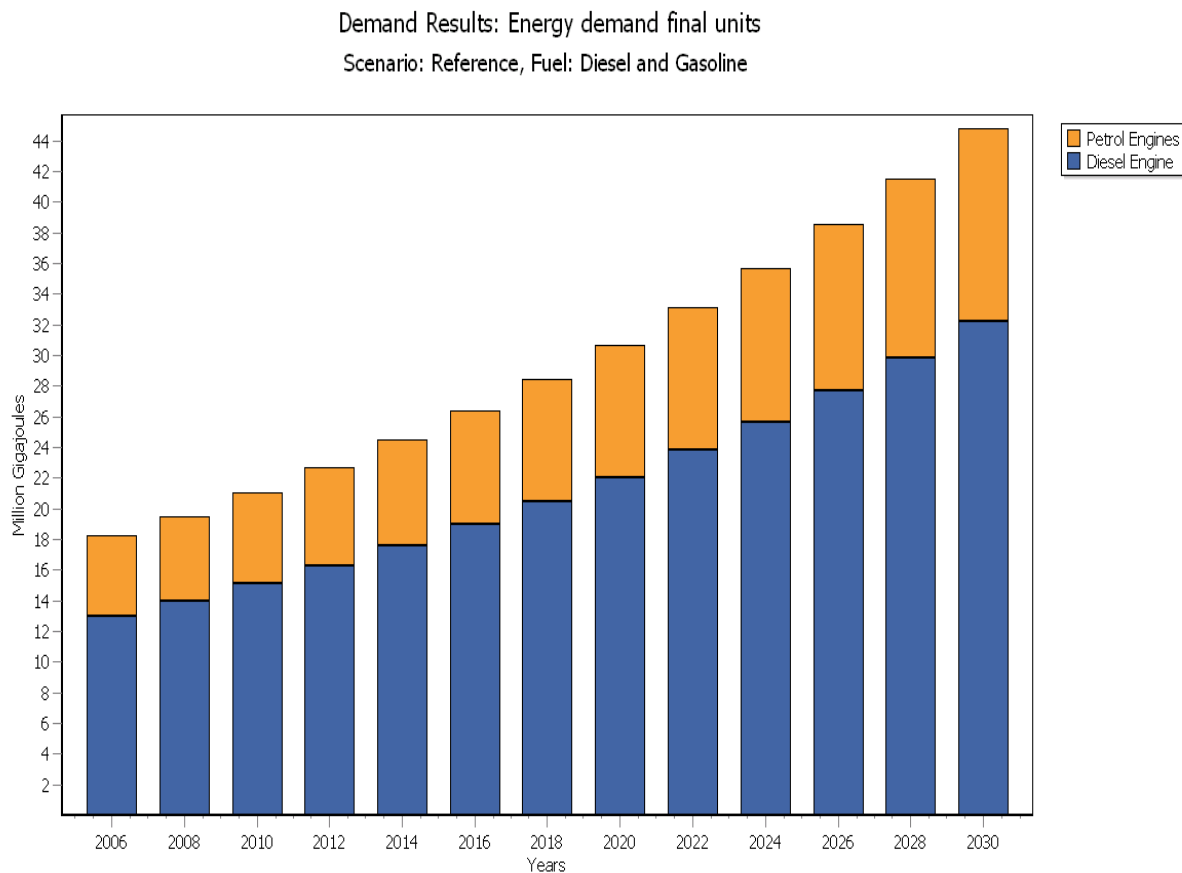


Figure 6.1: Energy demand final units for business as usual scenario (Million Gigajoules)

The environmental loadings for all GHG’s is likely to increase from 400million kilograms CO_2 equivalent to over 800million CO_2 equivalent from gasoline emissions and from about 900million kilogram CO_2 equivalent to over 2300 million kilogram CO_2 equivalent from petroleum diesel between the years 2006 and 2030. The Figure 6.2 below shows this growth in overall GHG increase over the study period. This increase is due to increased non biogenic carbon dioxide emissions.

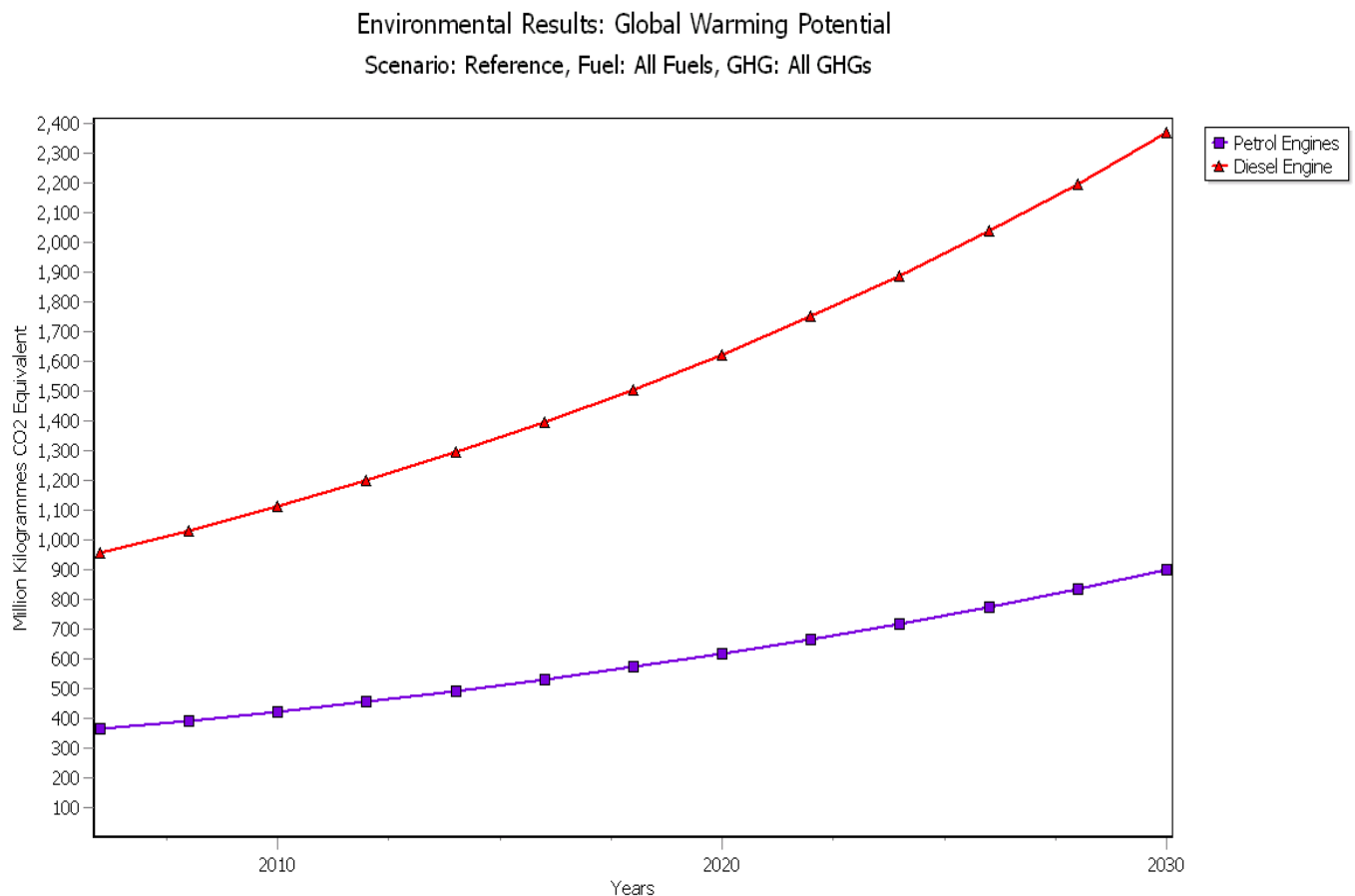


Figure 6.2: Environmental loading for global warming potential (Million Kilograms CO_2 equivalent)

6.1.3 Fuel blending scenarios

In the fuel blending scenario, analysis of the demand evolution due to the introduction of biofuel blends with petroleum is simulated and results are given in final energy demand units. The assumption taken is that blending is introduced with a mandatory 5% share of biofuel in all transportation fuel sold in Zambia by 2015 and raises to 10% in 2030.

A fuel switch in which biofuels are introduced is considered to analyse the future energy demand and consequences under specified fuel shares. A Step function is used to predict future energy demands for both petroleum products and biofuels. The biofuels considered are bioethanol and biodiesel. In all cases, the base year is taken to be 2000, with the introduction of biofuels being at 5% blending in 2015 and reaching a 10% blending capacity in 2030. The step function used in LEAP to compute this, is given below.

$$Step(2015, 95, 2030, 90) \qquad \qquad \text{Function 6.1}$$

This function increases the consumption of biofuels from 5% of total projected demand for 2015 to 10% for 2030. The reference scenario is based on data for fuel consumption in 2006 (See Appendix I). The energy intensities for the fuels can be found in appendix II.

6.1.3.1 Ethanol blending

As stated above, the results below show the demand changes in the gasoline demand due to an introduction of mandatory E5 by 2015 rising to E10 in 2030 which is interpolated by an in built step function of LEAP. In table 6.1, an extract of energy demand final units for the period 2016 to 2030 is shown to give the projected figures from the simulation. It is observed that there is an increase in demand of ethanol of 300,000 gigajoules in 2016 which increases to 900,000 gigajoules by 2030. A 5% (300000GJ) demand of ethanol in 2015 translates to about 13 million litres of ethanol

taking energy intensity of $23.4MJ/lt$ and 10% in 2030 will be about 38 million litres of ethanol.

TABLE 6.1: Energy demand final units for ethanol blending (Million gigajoules)

Year	2016	2018	2020	2022	2024	2026	2028	2030	Total
Gasoline	7.3	7.9	8.5	9.1	9.9	10.6	11.5	11.7	107.2
Ethanol	0.3	0.3	0.3	0.3	0.3	0.4	0.4	0.9	3.1

In the Figure 6.3 below the graphical representation of the changes in energy shares due to the introduction of ethanol in the energy mix is illustrated. Mandatory blending is introduced in 2015 at 5% and required to increase to 10% by 2030. In this simulation a step function 6.1 above is used to interpolate values in the period.

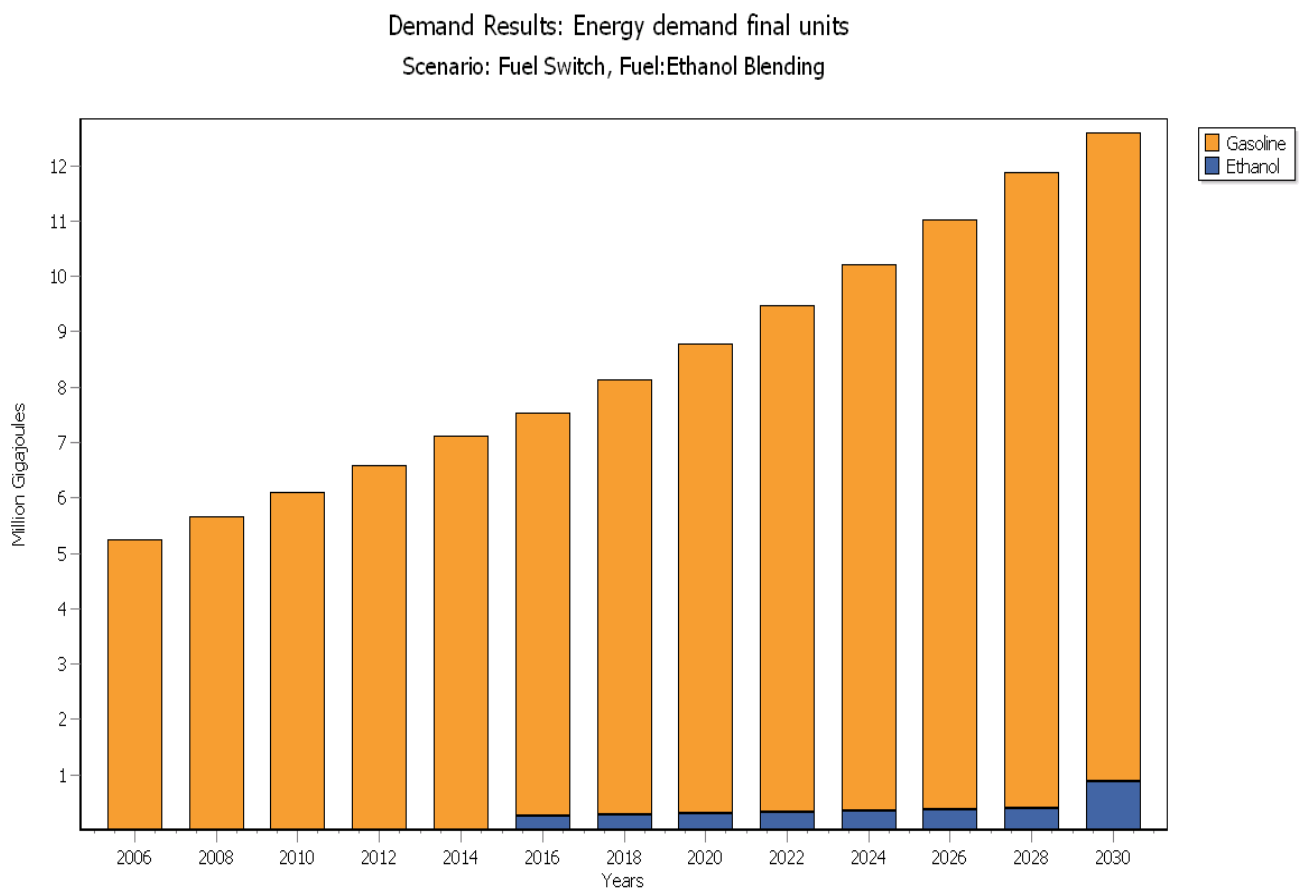


Figure 6.3: Energy demand final units for ethanol blending

A fuel switch (blending with ethanol) shows that the overall GHG emissions are reduced in comparison to the business as usual scenario (reference). There is an observable decrease of 25 million kilograms in 2016 and 90 million kilograms CO_2 equivalent emissions compared to the baseline scenario. Details of the values from the simulation are given in Table 6.2 below. The changes expected are graphically represented in Figure 6.4 given below. The line at the top represents the business as usual situation while the bottom one starting in 2016 is when ethanol is introduced in the energy mix.

TABLE 6.2: Ethanol blending, emissions for all GHG's (Million kilograms CO_2 Equivalent)

Year	2016	2018	2020	2022	2024	2026	2028	2030
Fuel Switch	504.1	543.7	586.5	632.6	682.3	736	793.8	811.2
Reference	530.6	572.4	617.4	665.9	718.3	774.7	835.6	901.3

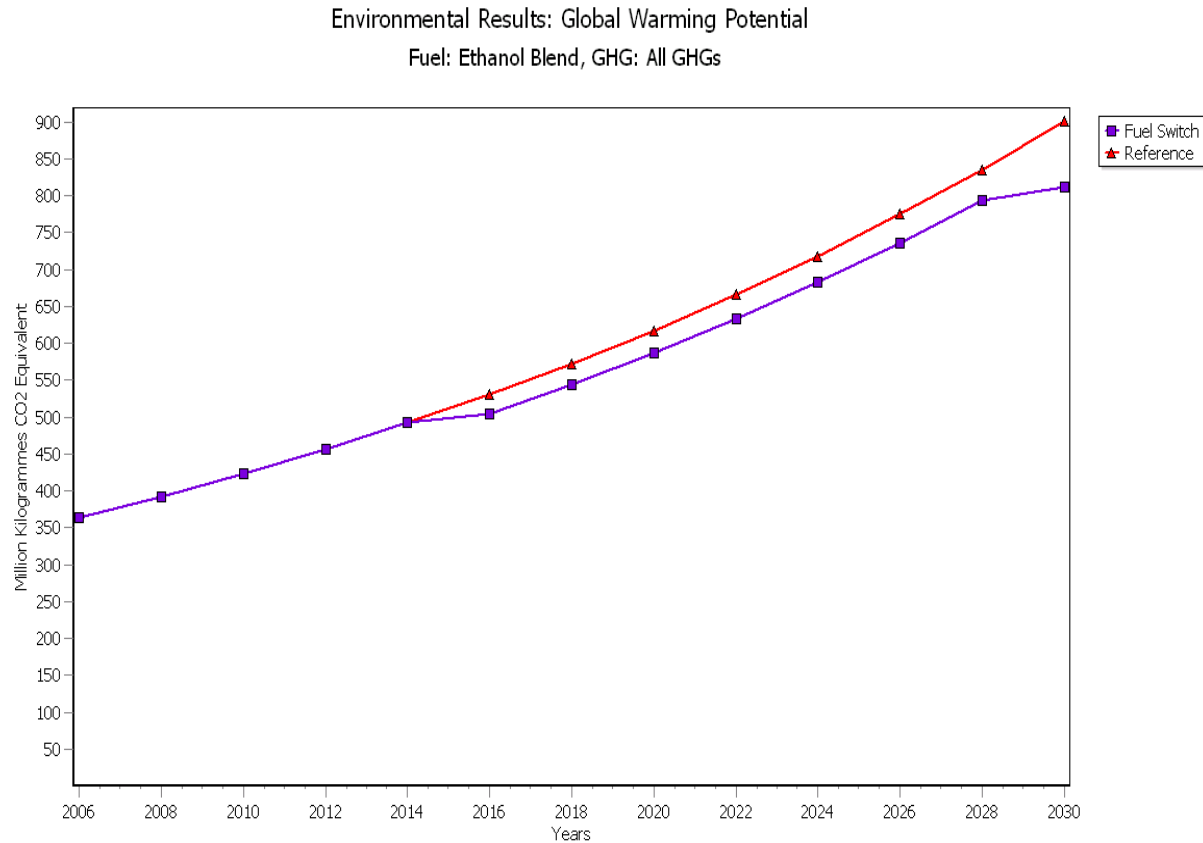


Figure 6.4: Global warming potential for ethanol blending (Million kilograms CO₂ Equivalent)

6.1.3.2 Biodiesel blending

The results for the simulated demand changes due to the introduction of biodiesel blending are given in the Figure 6.5 below. In this simulation, a mandatory displacement of 5% petroleum diesel by 2015 and 10% biodiesel is modelled function given by function 6.1. In Table 6.3, the changes in final energy demand units for the period 2016 to 2030 are given and it can be seen that in 2016, 900000 gigajoules of petroleum diesel can be displaced and this figure can be increased to 3 million gigajoules by 2030 in a mandatory 5% blending rising to 10% by 2030 as computed by function 6.1 above. The biodiesel required to displace five percent fossil diesel in 2016 is about 27 million litres at final energy intensity of 33.3 MJ/lt. This rises to 90 million litres of biodiesel for a ten percent displacement of petroleum diesel.

TABLE 6.3: Energy demand final units for biodiesel blending (Million Gigajoules)

Year	2016	2018	2020	2022	2024	2026	2028	2030	Total
Diesel	18	19.4	21	22.6	24.4	26.3	28.4	29	265.1
Biodiesel	0.9	1	1	1.1	1.2	1.3	1.4	3	10.9

An overall graphical representation of fuel shares with the introduction of biodiesel blending is shown in Figure 6.5 below.

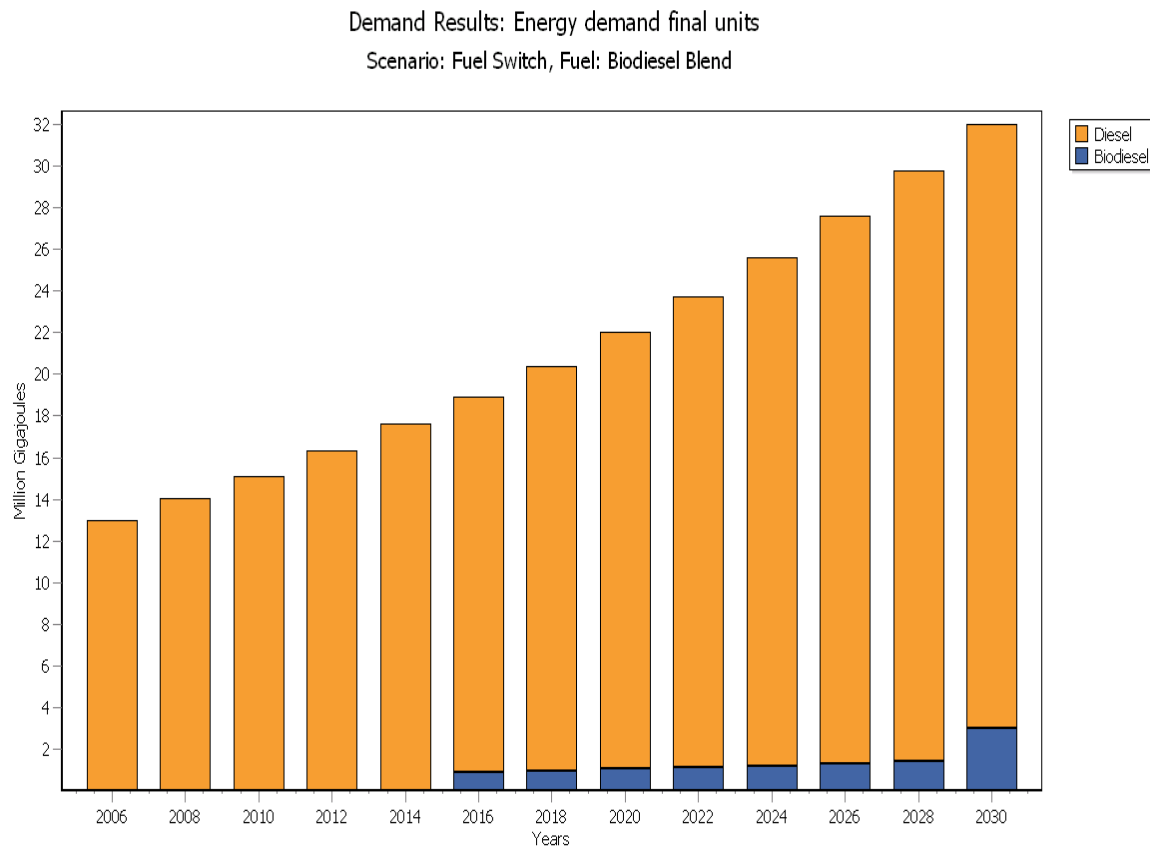


Figure 6.5: Energy demand final units for biodiesel blending (Million Gigajoules)

The environmental consequences of introducing biodiesel in the fuel energy mix were also simulated. This shows a reduction of 69.8 million kilograms in 2016 and 237 million kilograms of CO_2 in the biodiesel fuel blending scenario compared to the business as

usual scenario. Table 6.4 shows values for the period 2016 to 2030 in million kilograms CO_2 equivalent. The global warming potential in represented in Figure 6.6 for biodiesel.

TABLE 6.4: Biodiesel blending, emissions for all GHG's (Million kilograms CO_2 Equivalent)

Year	2016	2018	2020	2022	2024	2026	2028	2030
Fuel Switch	1325.7	1429.9	1542.3	1663.6	1794.4	1935.5	2087.6	2133.2
Reference	1395.5	1505.2	1623.5	1751.1	1888.8	2037.3	2197.5	2370.3

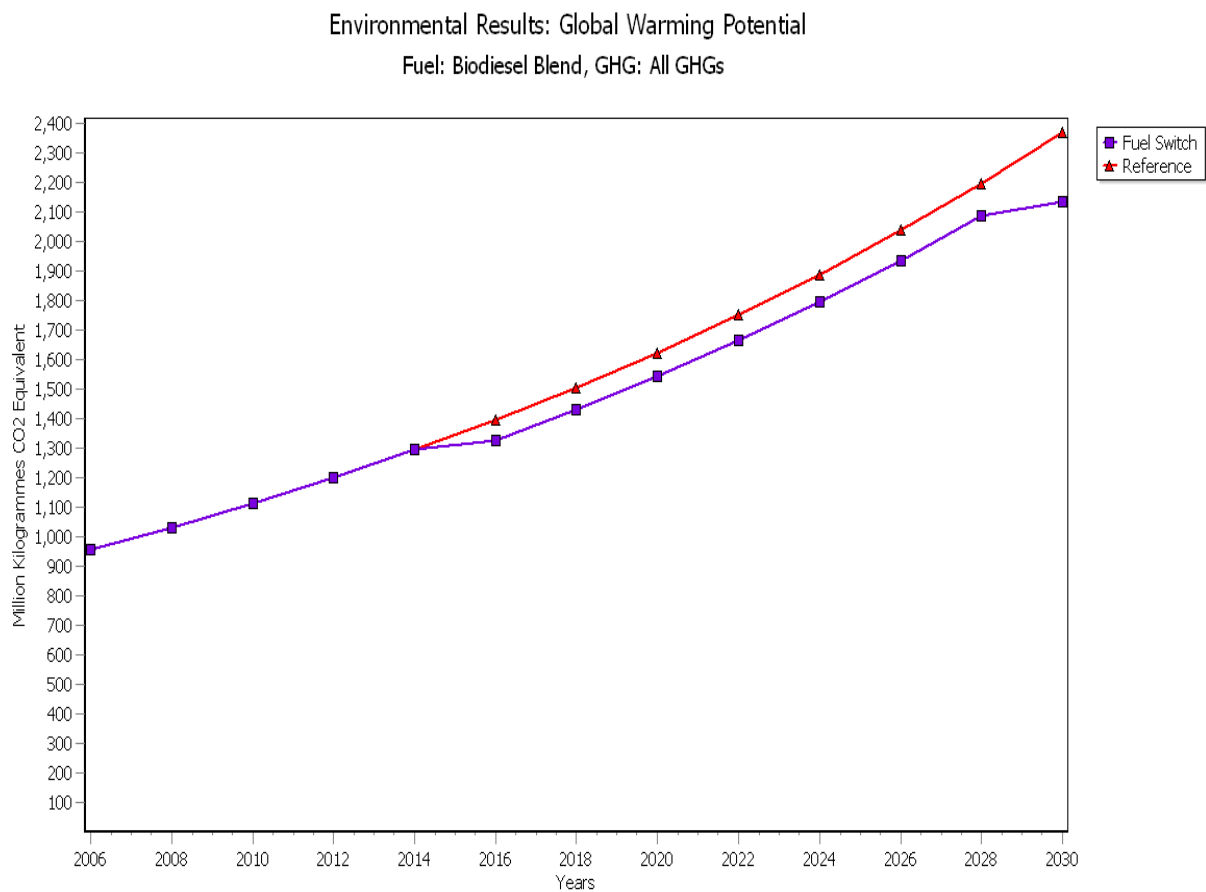


Figure 6.6: Global warming potential for biodiesel blending (Million kilograms CO_2 Equivalent)

6.1.4 Land resources

Land required to meet the different biofuel demands was estimated from average yields of litres per hectare for proposed biofuel feedstock currently available in Zambia. In this analysis the feedstock considered for bioethanol production are sugar molasses and cane sucrose while jatropha cocas is selected for biodiesel production in this model.

It is found that to meet the ethanol demand for 2016 at approximate yields of 8480 *lt/ha* of cane, 1500 *Ha* of sugar cane has to be cultivated. Assuming no changes in yields, the amount of land required to meet the 2030 target of 10% ethanol for use in gasoline engines, 4480 *Ha* of land is required for cultivating sugar cane. This is 1% of irrigable land available in Zambia which is 420, 000*Ha* in total (FAO 2005).

In order to meet the biodiesel target for 2016, at average yields of 1890 *lt/ha* requires 14,270 *Ha* of jatropha plantation. This would have to be increased to 47,570 *Ha* by 2030 for the 10% target to be achieved assuming that there are no changes in crop yields. The amount of land required to achieve these demands is very minimal considering that the country over 16 million *ha* of untapped arable land (ECZ, 2008).

6.1.4 Demand cost analysis

Finally, a cost benefit analysis based on final demand cost was carried out using LEAP. In this analysis, estimates of final energy demand cost for each fuel type are used to compute the possible cost benefits that would arise from the different blending scenarios. Due to limited data on production costs of biofuels in Zambia, the cost prices for these fuels are those considered logical from the average international cost prices. Furthermore, the environmental costs are not accounted for in this analysis.

6.1.4.1 Ethanol blending scenario

The introduction of bioethanol in the fuel sector for petrol engines, assuming the cost price per litre of diesel increases from the current \$1.40 in 2010 and reaches \$1.80 in 2030 while ethanol price falls from \$1.60 to \$1.20 by 2030 due to improved production techniques and government support, it is seen from Figure 6.7 that there could be a slight reduction on final demand cost. This is done by interpolation carried out using function 6.2 given below.

Interp(2010,1.40,2030,1.80)

Function 6.2

The monetary saving by 2030 translates to about \$22 million which is not much compared to the total demand costs given. The changes due are graphically represented in Figure 6.7.

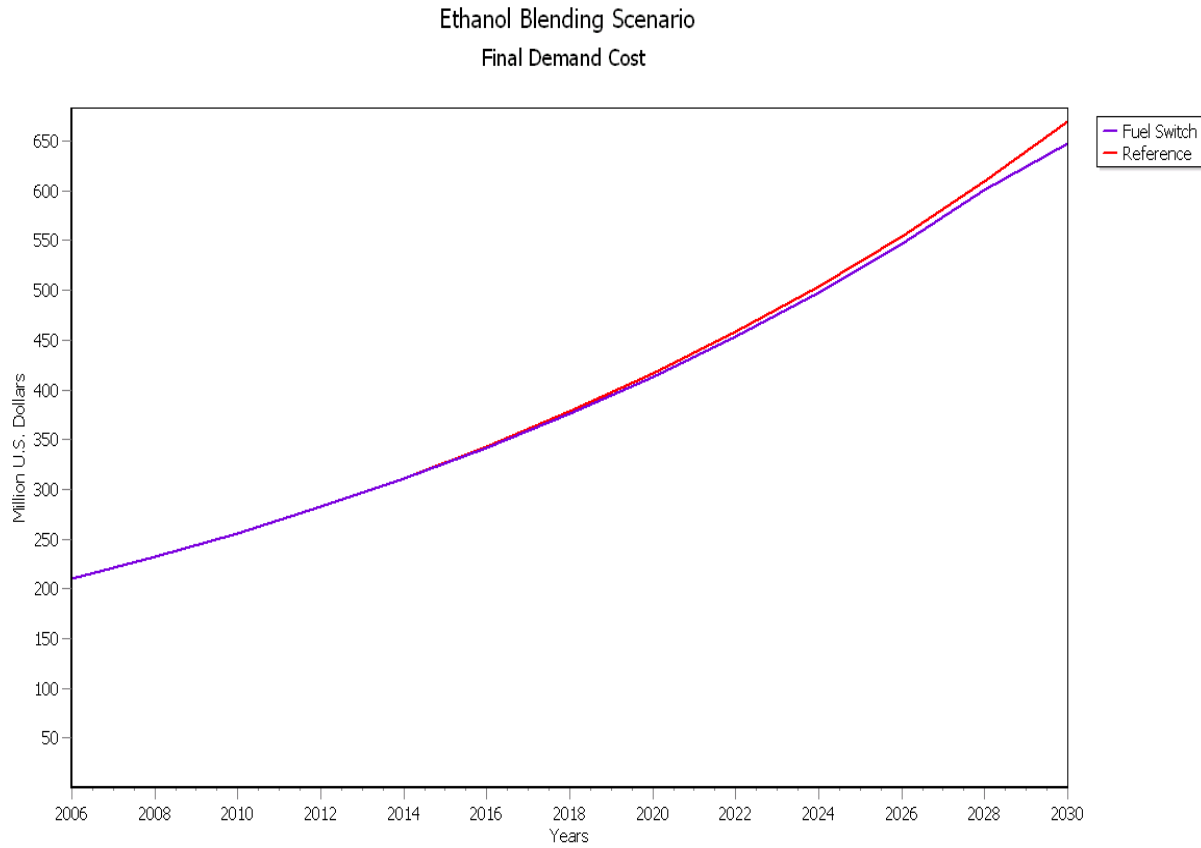


Figure 6.7: Final demand cost of fuel for ethanol scenario (Million US Dollars)

The cost of diesel is also assumed to increase in the study period due to increased demand for transportation fuel and depletion of petroleum oil reserves. Under almost similar assumptions, with the cost of diesel assumed to increase from the current \$1.30 per litre to \$1.70 by 2030, the simulation gives a saving of only 13 million dollars for the projected demand. An interpolation similar to that for ethanol above is used for biodiesel cost transformation as given by function 6.3 below.

$$\text{Interp}(2010, 1.30, 2030, 1.70)$$

Function 6.3

The savings are quiet marginal although this could translate to a considerable savings in terms of foreign exchange. The analysis is however not comprehensive due to unavailability of reliable data on current costs for biofuel as well as projections on future demand costs. The projected savings are illustrated in Figure 6.8 below.

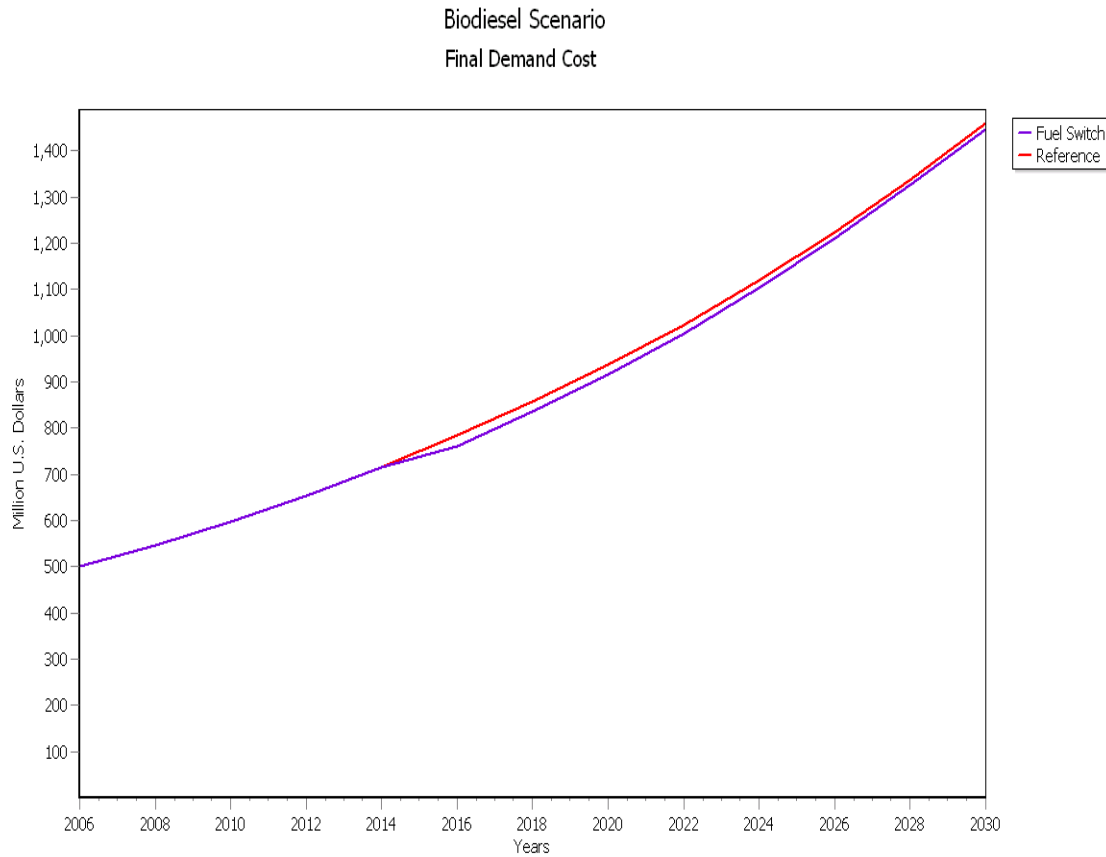


Figure 6.8: Final demand cost of fuel for diesel scenario (Million US Dollars)

6.1.5 Limitations of the model

The data that was used in this study comes from several different sources and was selected and synthesized using engineering manipulation and judgement (Appendix I). This is because data and information on energy resources in Zambia is scanty and incomplete in many cases. Furthermore the units of measurement vary from different sources thus requiring elaborate and logical conversions in order to come up with appropriate data for use with LEAP. Generally, the data units had to be converted and the LEAP demand branch structures modified for the simulations to be carried out. This was because the LEAP default data structures for the required analysis needed data which was either not available or found in an inappropriate form. Further, LEAP

analyses energy requirements in equivalent energy units and not volumes. In this case tonnes of oil equivalent were used.

Furthermore, the results from the scenario analysis on projected fuel consumption is close to the results obtained by Yamba et al for Zambia in a Country study that was carried out in 1999 although the reference years and other data varied slightly (Zambia country Report on climate mitigation in southern Africa, 1999). The demand projections simulated in this study give results in agreement with those obtained in the referred country study in that both show that the blending targets can be achieved. It is observed that the differences in results are due to changes in data input as the data used here is more recent and might use different functions of LEAP and projections.

6.2 Policies and prospects for biofuel in the Zambia transport Sector

6.2.1 Potential Feedstock for Biofuel

There are several feedstocks available for biofuel production in Zambia, however records do not give the actual potential figures in most cases. The potential feedstock with the exclusion of food crops include biodegradable municipal waste, molasses, agriculture and forest residue which, however need to be adequately assessed and quantified.

There is also potential for bioethanol from sugar cane and sweet sorghum although these are food crops to a greater extent. With the current sugar demand which is less than current production levels and the expected expansion in production of cane sugar makes it an attractive feedstock option for bioethanol. The projected demand for bioethanol from sugar cane from the simulation are met for blending ratios of 5% and 10% for the most years period if all molasses used to produce bioethanol. The projected demand can also be highly supplemented by use of the by – products from the sugar industry for ethanol production.

Biodiesel feedstock potential is mostly from jatropha which can be grown in most parts of the country. There is also potential to produce biodiesel from palm oil. This can be done by utilising waste lands and some of the uncultivated land area. Cotton seed is also available, but the potential needs to be elaborated.

Although not well elaborated, there is some potential as well as for the production of second generation biofuels from biodegradable municipal and timber industry waste. The existence of several landfills in highly populated cities and the forest waste gives an opportunity for production of biogas and other synthetic fuels from advanced technology. However, the actual potential must be established in order for comprehensive feasibility studies to be made.

The scenario analysis shows that the projected demand for biofuels to meet the selected blending ratios for 2015 and 2030 respectively can be achieved with available resources. It clearly shows that the projected demands can be met without the inclusion of second generation biofuels. However, a balance has to be made between the availability of land for food crops and biofuels.

6.2.2 Technology

A look at biofuel production also indicates that Zambia is capable of adopting most of the mature technology for producing both bioethanol and biodiesel. The production of ethanol by fermentation of sugars is not new in this country, thus the skills required are locally available. Since sugar cane is the feedstock with the highest potential for bio ethanol production, the sugar to ethanol production path appears to be more appropriate as the country does not yet have capacity to implement more advanced technology immediately. Similarly, bio – oil is already being produced for food and only to a less extent for fuel. It is found that currently, there is capacity to produce biodiesel on both small and large scale from this study as stated earlier on. The transesterification

process for biodiesel production can easily be adopted in the case of jatropha feedstock and used oil from food industry.

From the survey and analysis of technology used in the production of second generation biofuels fuels, it is observed that Zambia does not yet have capacity to successfully produce these fuels. Firstly, the technologies are not very well established even in places where they are implemented and the country could be lacking in technical capacity more so in the case of gasification. However, there could be capacity to sequester methane from landfills for use as a fuel.

Most of the vehicles in Zambia are imported from around the world with most of them being manufactured in Japan. There have been no conclusive studies carried out as to find out whether these vehicles are compatible with the different biofuels. However, since studies done in most of the other countries show that low blends have very little known negative impacts, blending with these(5% and 10%) would thus be the most realistic starting point for Zambia. The compatibility issues can then be analysed and revealed for higher blends with consideration of incoming equipment.

6.2.3 Social – economic aspects

Biofuels are expected to have an impact on the social – economic structure of a country in which they are implemented as observed in the different regions. These include social activities in terms of change in daily occupations and economic activities of local communities. The occupation of the majority of people in Zambia is agriculture, hence the introduction of biofuels is expected to increase agricultural activities. Out grower schemes would create jobs for the many people that can not manage to invest as individuals and increase their income. The minimal requirements set by the International Labour Organization should also be considered when investors are given opportunities to establish plantations in the country.

In rural areas where the prices of fossil fuel are comperatively higher than in urban areas due to distance from the distribution centers, thus localised production of biofuels in these areas is more cost competitive. The possibility of using biofuels for motive and stationery power is another consequence that would lead to improved standard of life in rural areas that are not connected to the grid.

Considerable tax system should be put in place to encourage production and use of biofuels. The cost competitiveness of other products from biofuel producing crops such as sugar from sugar cane compared to bioethanol for transportation should also be taken into account. This requires a review of the tax and investment incentives to be given to biofuel industry to make it cost competitive. Furthermore, a complete cost benefit analysis for the biofuels that can be produced in the country should be carried out with consideration of environmental consequences.

6.2.4 Environmental aspects

The environmental impacts both positive and negative are going to be experienced in the event that biofuels are introduced on a large scale in Zambia. For now, these impacts can only be anticipated and analysed in the environmental impact assessments of biofuels projects. Some expected benefits are re - vegetation of deforested regions in case that these areas are used for growing biofuel crops. The blending of biofuels with fossil fuels is likely to reduce carbon dioxide emissions coupled with other positive environmental effects which accompany this practice such as improved vehicle performance.

As seen from the results in the scenario analysis, there is a reduction in emissions is achieved with introduction of biofuels blending. This brings about the possibility of selling carbon credits if the biofuels are grown in an environmentally sustainable way. Furthermore, there is the possibility of project funding under the concept of clean development mechanisms. This has to be further explored to identify which projects can qualify in the biofuels sector.

6.2.5 Sustainability

It is clear from the literature survey that sustainability of biofuels is still being debated. However, there seem to be prospects for sustainable biofuel production from feedstock such as biodegradable municipal and timber industry wastes. In the event of embarking on the biofuel path, measures as proposed by various studies on sustainability of bioenergy should be taken into account and implemented.

The land requirements to achieve the blending requirements from the results of the scenarios analysis show that the required land resource is very minimal in comparison to the uncultivated land area. This gives an opportunity to develop the biofuels industry with very little effect on agricultural land for food production. However, selection of areas to be used for growing of biofuels should be done with caution to avoid negative impacts. If land resource is set, identified and assessed to establish the potential for biofuel crop production, it would be much easier to carry out a comprehensive feasibility study.

The scenario analysis indicates that at the current fuel consumption levels, it is possible to meet blending levels with both bioethanol and biodiesel with respect to land and other resources.

6.2.6 Policy

The establishment of biofuels industry in almost all countries has been largely influenced by national policies. A biofuels industry can not be established without government support due to competition from the well established fossil industry which has a better economy. Therefore, Zambia has to put in place policies that will take advantage of the current strengths of possible benefits and also overcome the weaknesses. Some policy measures that can stimulate the development of the biofuel industry in the country include tax concessions, mandatory blending, government financial support, encouraging use of alternative fuel by giving incentives.

Some concessions that can be implemented include tax exemptions on biofuels related activities. The recent carbon tax based on engine size capacity on vehicles rather than carbon dioxide emission should be revised to exempt alternative clean fuels and be based on emissions per litre of fuel. The government should facilitate the sourcing of funds for biofuels projects that include infrastructure development. Standards that mandate blending of certain percentages of biofuels with fossil fuels should be as adopted by Energy Regulation Board and must be enforced with appropriate legislation.

The policies analysed in this study are mandatory blending with assumption that other legislation and support structures are in place to achieve the set targets. As seen in the results, these ratios can be attained under these assumptions. However, in reality other factors such as droughts, shift in policy have to be considered.

6.2.7 Propects of biofuels in Zambia

There is considerable potential for biofuels in Zambia as shown by the results in this study. This potential entails that Zambia there is the possibility of further increasing production and blending targets. The minimal land requirements for meeting low blending targets entail that even greater targets can be set and possibly meet. However, transformation of industry has to be done to provide for the growth of a strong biofuel industry. In the bioethanol sector, the preference of grain sugar to bioethanol to be taken into account and a balance has to be reached. In the event that grain sugar production will continue to be more profitable, policies to encourage bioethanol production will have to be implemented. These should include tax incentives, support of co – production of grain sugar and bioethanol and technology transfer that improves of production. Similar, policies should be considered on a case by case basis for biofuels.

The possibility of producing more bioethanol than the projected demands in the simulation open the possibilities of more ambitious blending targets. In the event that these targets are surpassed, there arise possibilities of exporting to international

markets. Furthermore, higher blends would mean that the current vehicle technologies would not be able to run on these higher blends. This makes it necessary to consider implementing policies that encourage transfer of technology that can use higher blends such as flex fuel vehicles in the case of bioethanol and other well proven technologies for higher biodiesel blends. Policies such as zero tax on flex fuel vehicles and other biofuel compatible technologies will have to be implemented to encourage the development of the industry.

The biofuels industry will only develop with investment as is currently in its infancy. The government has to encourage investment in the sector by offering incentives to investors and creating an enabling environment. This can be enhanced by giving tax rebates and guaranteeing markets and reasonable returns on biofuel to producers in the early stages of development of the industry. The government as well as private entities should be allowed to invest as the country is a freemarket economy. Public private partnerships could be the most appropriate kind of investment due to the nature of stakehold. The need for incentives is evident in the development of the biofuel industry in countries around the world from the review of the global situation analysed earlier.

A well developed biofuels industry will entail enhanced quality of life provided sustainability criteria are abided to with regard to policy implementation. It is very important that International Labour Organization (ILO) standards and norms regarding working conditions in the biofuels are followed. This will assist in distributingthe benefits to the entire population.

7. CONCLUSIONS AND FUTURE WORK

7.1 Conclusion

The review of biofuels in countries around the world shows that most of them are searching for alternatives to fossil fuel due to the declining reserves and the need to mitigate climate change. Furthermore, the need to secure supply of energy and possibilities of exporting biofuel to areas that have high energy demand are among the main factors driving countries in different parts of the world. With the current uncertainties in petroleum prices and the instability in petroleum producing countries, it is eminent that countries search for alternatives. Zambia has made a good decision to embark on a path to produce and utilise biofuels for transportation like many other countries.

Literature shows that there are several feedstock options in Zambia that can be used to produce biofuel. Although actual quantification has not been made, the main feedstock are sugarcane and sweet sorghum together with their by products for the production of bioethanol. Jatropha can be used on large scale together with palm oil for production of biodiesel. All the crops mentioned above are already being produced in Zambia although none of them is for commercial production of biofuels. The results from this study show that Zambia has considerable potential to produce required feedstock to meet blending demands in the simulated period. There exists potential to increase the blending requirements in the event that the simulated targets are surpassed.

The main technologies for producing biofuels are fermentation for bioethanol and transesterification for biodiesel from the global perspective. Other technologies such as gasification are not well established in production of biofuel for transport. The technologies for alcohol production such as fermentation are already in use in other industries. Oil as with transesterification processes is similarly being produced by other industries. Technologies in these associated industries can thus be adopted. In case of

utilisation, the studies show that low biofuels blends do not have adverse effects on most of the vehicles manufactured after 1990. Thus, there should be no great concern on the compartibility of vehicles operating in Zambia although special cases have to be taken into account. Further, policies to facilitate technology transfer can be formulated and implemented to overcome these concerns.

The LEAP simulations indicate that blending targets of 5% in 2016 rising to 10% in 2030 can be achieved for both bioethanol and biodiesel assuming that assumptions on fuel consumption will not increase so much compared to actual future demand. Projected requirement of 10% in 2030 for either fuel is less than the base year demand for 2008 for either fossil gasoline or diesel according to the simulation (Table 5.5). This entails that it would be possible to achieve 10% of the national fuel needs under the assumptions in the simulations. The country can also reduce on greenhouse gas emissions although this is also dependent on the other conditions, such as technology used for production of the biofuels. These include meeting the sustainability criteria for producing environmentally friendly biofuels.

Experiences from other countries show that economic and social benefits from biofuel greatly depend on the international and local conditions in a specific country. The local and international costs of petroleum fuel affect biofuels either negatively or positively. The social - economic aspects of the country is further dependant on local policies that relate to biofuels. Considering that Zambia is just starting the production and use of biofuels, it is important to select policies that will enhance the development of industry. Mandatory blending starting with low blends and increasing eventually would assist in achieving an authentic set target. Tax incentives and rebates on biofuels would also assist in increasing production as seen from other countries. Policies on land, water and environment that avoid adverse effects of biofuels should be implemented to satisfy the sustainability criteria of biofuels. Social aspects such as human rights of workers in large plantations shall have to taken in to consideration.

Biofuels are among the alternatives to fossil transportation fuels, however they have to be supported by appropriate policies even when other physical and social factors support there production. It is thus important to implement the appropriate policies in Zambia and to monitor the effects of each and every policy.

7.2 Future work

Currently, data on many biomass resources for use as biofuel feedstock is either incomplete or not available in Zambia. Thus a resources assessment must be carried out to establish the actual potential for biofuel by feedstock type. The yields for various feedstock and the land for its production must be identified and quantified. This includes quantification of biodegradable municipal solid and timber waste.

Indigenous research on production technologies of both first generation and second generation fuels should be encouraged so that the most appropriate technologies are identified and adopted. The development of indigenous technology will assist in enhancing the sustainability of biofuels since it will reduce the problem of technical skills usually encountered with exotic technologies.

Comprehensive information on production costs for different biofuels, well to wheels is not available for Zambia. This is required to establish the possible benefit of biofuels in the local conditions. It is important to account for the total input cost in terms of the capital investments and environmental costs in the production line of biofuels.

Finally, work on the environmental impacts of mass production of certain oil plant species, in this case *Jatropha* which is already being grown although there are some environmental concerns over the effects of increased production. Research on the sustainability of biofuels in Zambia according to set standards should be carried out for the different feedstocks.

REFERENCES

- Abdullar A.Z., Razali N, Mootabadi H, Salamatinia B, 2007. Critical technical areas in future improvement in biodiesel technologies, *Environmental Resource Lett.* 2007:2:1 -6
- A.E. Farrell, R.J. Plevin, B.T. Turner, A.D. Jones, M. O'Hare and D.M. Kammen, Supporting online material to ethanol can contribute to energy and environmental goals, *Science* **311** (2006), pp. 506–508
- Biofuels and Sustainability Report, Ecological Society of America, available at www.dft.gov.uk/rfa/reportspublications/reviewofindirecteffectsofbiofuels.cfm as accessed on 9 october 2010
- Cane plantation photo_57605581 accessed at WWW.shutterstock.com
- Charlse Y. Wereko-Brobby and Essel B. Hagan, *Biomass conversion and Technology*, John Wiley & Sons Ltd, 1996
- Country Energy Statistics - Zambia
www.nationmaster.com/country/za-zambia/ene-energy as accessed September 2011
- Demirbas A. Biodiesel from wood oils in compressed methanol, *Energy Source Part A* 2009;31:1530 – 6
- D. Pimentel, Ethanol fuels: energy balance, economics, and environmental impacts are negative, *Natural Resources Research* 12 (2003), pp. 127–134.

D. Pimentel, Ethanol fuels: energy, economics and environmental impact, *International Sugar Journal* 103 (2001), p. 491.

Edgard Gnansounou, Luis Panicheri and Juan David Villegas, 2007, Sustainable Liquid biofuels for Transport, The context of Southern Africa Development Community, working paper

E. Leibtag, Corn prices near record high, but what about food costs? *AmberWaves* 6 (2008), pp. 10–15.

ECZ, 2008, Zambia Environment Outlook Report 3, A report of the Environmental Council of Zambia.

ECZ, 2004, State of the Environment Report, A report of the Environmental Council of Zambia.

FAO, 2005 FAO/WFP Crop and Food Supply Assessment Mission to Zambia, 24 June 2005

Foley Gerald and Barnard Geoffrey, 1983, Biomass gasification in developing countries: Technical Report No. 1, Earthscan, IIED, Russel Press limited, London

Franz J. Kaltner et al, 2006. Liquid biofuels for Transportation: Brazil Country Study on potential and implications for sustainable agriculture and energy in the 21st Century. Available on line at <http://www.eia.doe.gov/oiaf/ieo/world.html> as accessed on 12 July 2009

Giovanni Sorda, Martin Banse, Claudia Kemfert,,2010.An overview of Biofuel policies across the world, *Energy Policy* 38, 2010 , pg 6977 - 6988

Gnansounou E. Et al., 2007. Sustainable Liquid Biofuels for Transport: The Context of Southern Africa development Community (SADC).

Goldon A. Mackenzie "Climate change mitigation in Southern Africa: Zambia Country Study" Riso National Laboratory, Denmark ISBN 87 - 550 – 2430 – 0 available at <http://www.risoe.dk/sys-ucc> accessed 25 October 2011

GTZ, 2006. Liquid biofuels for Transportation: Tanzania Country Study on potential and implications for sustainable agriculture and energy in the 21st Century, 2006. Available on line at <http://www.gtz.de/de/dokumente/en-biofuels-for-transportation-in-Tanzania-2005.pdf> as accessed on 12 July 2009

H. Shapouri and A. McAloon, The 2001 net energy balance of corn–ethanol, US Department of Agriculture, Washington, DC (2004).

H. Shapouri and A. McAloon, 2004, The 2001 net energy balance of corn–ethanol, US Department of Agriculture, Washington, DC (2004).

H. von Blottnitz and M.A. Curran, A review of assessments conducted on bio-ethanol as a transportation fuel from a net energy, greenhouse gas, and environmental life cycle perspective, *Journal of Cleaner Production* **15** (2007), pp. 607–619.

Gaia movement, 2007, Annual Report 2007 accessed at <http://www.gaia-movement.org> 12 April 2012

GVEP, 2007 Gap Fund Project in Zambia, GVEP Report on implementation of the Towards Biofuel Self - Supply in Dual fuel systems. [http://www.gvepinternational.org/where we are working/africa/zambia/gapfund project in zambia](http://www.gvepinternational.org/where_we_are_working/africa/zambia/gapfund_project_in_zambia) as accessed 15 March 2011

I.E.A.2004.Sustainable International bio - energy trade: Securing supply and demand, www.fairbiotrade.org as accessed April 2009

Jatropha photo accessed at
www.jatrpha.org/photogallery.php on 15 June 2012

J.A. Sheehan, K. Aden, K. Paustian, J. Kilian, Brenner, M. Walsh and R. Nelson, Energy and environmental aspects of using corn stover for fuel ethanol, *Journal of Industrial Ecology* 7 (2004), pp. 117–146. View Record in Scopus | Cited By in Scopus (106)

Johnson, F., Matsika, E. 2006, Bioenergy Trade and regional development: The case of ethanol in Southern Africa. Energy for Sustainable development. Volume X No. 1 . Pages 45 - 53

Johnson, F., Rosillo – Calle, Frank 2007. Biomass livelihoods and international trade: Challenges and Opportunities for the EU and Southern Africa. Stockholm Environmental Institute. Available on line at: <http://www.sei.se>

Lew Fulton, Tom Howes and Jeffrey Hardy, 2004. Biofuels for Transport, An international perspective, IEA 2004 Available on line at: <http://www/iea.org/textbase/npdf/free/2004biofuels2004.pdf>

Ma F. Hanna MA Biodiesel production: a review, Bioresource Technology 1999:70:1 – 15

Macedo I, Leal M, Silva J. da. Assessment of greenhouse gas emissions in the production and use of fuel ethanol in Brazil, 2004.
www.unica.com.br/i_pages/files/pdf_ingles.pdf. as accessed 7 July 2010

Mozaffarian M. Zwart, R.W.R., 2003, “Feasibitlity of Biomass/Waste – related SNG production technologies”, report nr; ECN – C -03 – 066, Dutch Ministry of Economic affairs.

National Academy of Science—NAS. Water implications of biofuels production in the United States. 2007. Report in brief. Available at

http://dels.nas.edu/dels/rpt_briefs/biofuels_brief_final.pdf as accessed on 17 October 2010

National energy Policy 2008 for Zambia, Policy Document Government of the Republic of Zambia (GRZ)

National Non – Food Crops Centre, “GHG benefits from use of vegetable oils for electricity, heat, transport and industrial purposes, NNFCC 10 – 016” Accessed 2011 -06 -27

Ohlström M. Et al, 2001, “New concepts for biofuels in transportation”, VTT research notes 2074, Technical research centre of Finland, Espoo 2001

P.R. Adler, S.J. Del Grosso and W.J. Parton, Life-cycle assessment of net Greenhouse - gas flux for bioenergy cropping systems, *Ecological Applications* 17 (2007), pp. 675–691.

P.J. Crutzen, A.R. Mosier, K.A. Smith and W. Winiwarter, N₂O release from agro-biofuel production negates global warming reduction by replacing fossil fuels, *Atmospheric Chemistry and Physics Discussion* 7 (2007), pp. 11191–11205

Rogério Cezar de Cerqueira Leite et al 2009, Energy Volume 34, Issue 5, May 2009, “Can Brazil replace 5% of the 2025 gasoline world demand with ethanol?” , Pages 655-661

Sivan Kath, Gerald Leach and Sadhir Chella Rajan, Advancing bioenergy for sustainable development: Guidelines for Policy makers Volume I,II and III,2005

Smeets E, Junginger M, Faaij A, Walter A, Dolzan P. Sustainability of Brazilian bio-ethanol. Universiteit Utrech. Copernicus Institute, Department of Science, Technology and Society. Report NWS. 2006, E-2006-110. Available at <http://igitur-archive.library.uu.nl/chem/2007-0628-202408/UUindex.html>

Stryer, Lubert, 1975. Biochemistry W.H., Freeman and company, ISBN -7167-0174-X

Stockholm Environmental Institute (SEI), (2008). “LEAP- Long range Energy Alternatives Planning System, Training Exercises”, U.S. Center, MA 02144 – 1224 ,Somerville, USA

Web: www.energycommunity.org and www.sei-us.org as accessed 10 September 2011.

The energy and resources Institute, India Habitat Centre, Liquid biofuels for Transportation: India Country Study on potential and implications for sustainable agriculture and energy, 2005. Available at <http://www.lab-biokraftstoffe.de/downloads/PDF/fachinformationen/biofuels-for-transportation-in-india.pdf>

T.W. Patzek and D. Pimentel, Thermodynamics of energy production from biomass, *Critical Reviews in Plant Sciences* **24** (2005), pp. 327–364.

Wang M. Development and use of GREET 1.6 fuel-cycle model for transportation fuels and vehicle technologies. Technical report no. ANL/ESD/TM-163. Argonne National Laboratory, Center for Transportation Research, Argonne, IL, 2001.

Available at online

http://www.transportation.anl.gov/modeling_simulation/GREET/publications.html.

Zhao lixin etal, 2005. Liquid biofuels for Transportation: China Country Study on potential and implications for sustainable agriculture and energy in the 21st Centuary, 2006. Available online at <http://www.gtz.de/de/dokumente/en-biofuels-for-transportation-in-china-2005.pdf> as accessed 12 July 2009

Appendix I

LEAP simulation

A scenario analysis for Zambia using Long - range Energy Alternatives Planning system was used to simulate the transformation of the transportation fuels sector in Zambia. The data for the LEAP simulations was obtained from government documents and web pages(DOE Web, ECZ Report 2006, ERB Website, CSO Web). The data had to be critically selected due to the inconsistencies and incompleteness in most cases.

Key Parameters

Income	\$1364
Population	11.2 Million
GDP	Population *Income
Number of Households	2.4 Million
Household_size	5
Income_Growth_rate	3.5% per annum
Population_growth_rate	3% per annum
End_year_urbanisation	45%

Demand input data

A demand branch was created and named “Transport” on the assumption that all the gasoline and diesel is consumed by vehicles. Two technology branches were then created under this demand branch namely Gasoline and Diesel Engines independently. Then the consumed fuel for each branch was created as gasoline and ethanol under the Gasoline engines and diesel and biodiesel under the diesel engine branch. Final Energy intensities were taken on Mega joule per liter basis (MJ/lt). Input data for the year 2006 is used as base year while 2015 is the first scenario year in which 5% to petroleum fuels

are displaced by biofuels and this rises to 10% in 2030. Table below gives the gasoline and diesel consumption data for 2006 in liters which are used in the LEAP simulation.

Table : Petroleum consumption evolution data

YEAR	2002	2003	2004	2005	2006	2007	2008	2009
GASOLINE(Million lt)	139	155	159	151	107	165	177	213
DIESEL(Million lt)	229	266	314	329	263	414	508	534

[Source: Adapted from ERB Website]

Results from LEAP simulations

Using the step function 1 below the following tables were generated. The base year for the simulation data is for the year 2006 obtained from the energy regulation website.

The results below are as obtained from LEAP.

Step(2015,95,2030,90) function 1

Demand Results: Energy demand final units

Scenario: Fuel Switch, Fuel: All Fuels

Branch: Demand\Transport\Passenger\Road\Petrol

Engines

Units: Million Gigajoules

Year	2016	2018	2020	2022	2024	2026	2028	2030	Total
Gasoline	7.3	7.9	8.5	9.1	9.9	10.6	11.5	11.7	107.2
Ethanol	0.3	0.3	0.3	0.3	0.3	0.4	0.4	0.9	3.1

Environmental Results: Global Warming Potential

Fuel: Bioethanol Blending, GHG: All GHGs

Branch: Demand\Transport\Passenger\Road\Petrol Engines

Units: Million Kilogram's CO2 Equivalent

Year	2016	2018	2020	2022	2024	2026	2028	2030
Fuel Switch	504.1	543.7	586.5	632.6	682.3	736	793.8	811.2
Reference	530.6	572.4	617.4	665.9	718.3	774.7	835.6	901.3

Demand Results: Energy demand final units: energy only

Scenario: Fuel Switch, Fuel: All Fuels

Branch: Demand\Transport\Passenger\Road\Diesel Engine

Units: Million Gigajoules

Year	2016	2018	2020	2022	2024	2026	2028	2030	Total
Diesel	18	19.4	21	22.6	24.4	26.3	28.4	29	265.1
Biodiesel	0.9	1	1	1.1	1.2	1.3	1.4	3	10.9

Environmental Results: Global Warming Potential

Fuel: Biodiesel Blend, GHG: All GHGs

Branch: Demand\Transport\Passenger\Road\Diesel

Engine

Units: Million Kilograms CO2 Equivalent

Year	2016	2018	2020	2022	2024	2026	2028	2030
Fuel Switch	1325.7	1429.9	1542.3	1663.6	1794.4	1935.5	2087.6	2133.2
Reference	1395.5	1505.2	1623.5	1751.1	1888.8	2037.3	2197.5	2370.3

Corresponding graphical presentation of the above results are also obtained in LEAP.

The projections of the cost for biofuels in future are generated using the functions 2 and 3 for bioethanol and biodiesel respectively.

Interp(2010, 1.40,2030,1.80) function 2

Interp(2010, 1.30,2030,1.70) function 3

Appendix II

This is a quick-reference list of conversion factors used by the Bioenergy Feedstock Development Programs at ORNL. It was compiled from a wide range of sources, and is designed to be concise and convenient rather than all-inclusive. Most conversion factors and data are given to only 3 significant figures. Users are encouraged to consult other original sources for independent verification of these numbers. The following are links to Web sites we have found useful (many universities worldwide maintain good guides and conversion calculator pages):

- [U.S. National Institute of Standards and Technology \(NIST\)](#)
- [Centre for Innovation in Mathematics Teaching, University of Exeter, U.K.](#)
- [Department of Geological Sciences, University of Michigan](#)
- [Convertit.com Measurement Converter](#)

Energy contents are expressed here as Lower Heating Value (LHV) unless otherwise stated (this is closest to the actual energy yield in most cases). Higher Heating Value (HHV, including condensation of combustion products) is greater by between 5% (in the case of coal) and 10% (for natural gas), depending mainly on the hydrogen content of the fuel. For most biomass feedstocks this difference appears to be 6-7%. The appropriateness of using LHV or HHV when comparing fuels, calculating thermal efficiencies, etc. really depends upon the application. For stationary combustion where exhaust gases are cooled before discharging (e.g. power stations), HHV is more appropriate. Where no attempt is made to extract useful work from hot exhaust gases (e.g. motor vehicles), the LHV is more suitable. In practice, many European publications report LHV, whereas North American publications use HHV.

Energy units

Quantities

- 1.0 joule (J) = one Newton applied over a distance of one meter (= 1 kg m²/s²).
- 1.0 joule = 0.239 calories (cal)

- 1.0 calorie = 4.187 J
- 1.0 gigajoule (GJ) = 10^9 joules = 0.948 million Btu = 239 million calories = 278 kWh
- 1.0 British thermal unit (Btu) = 1055 joules (1.055 kJ)
- 1.0 Quad = One quadrillion Btu (10^{15} Btu) = 1.055 exajoules (EJ), or approximately 172 million barrels of oil equivalent (boe)
- 1000 Btu/lb = 2.33 gigajoules per tonne (GJ/t)
- 1000 Btu/US gallon = 0.279 megajoules per liter (MJ/l)

Power

- 1.0 watt = 1.0 joule/second = 3.413 Btu/hr
- 1.0 kilowatt (kW) = 3413 Btu/hr = 1.341 horsepower
- 1.0 kilowatt-hour (kWh) = 3.6 MJ = 3413 Btu
- 1.0 horsepower (hp) = 550 foot-pounds per second = 2545 Btu per hour = 745.7 watts = 0.746 kW

Some common units of measure

- 1.0 U.S. ton (short ton) = 2000 pounds
- 1.0 imperial ton (long ton or shipping ton) = 2240 pounds
- 1.0 metric tonne (tonne) = 1000 kilograms = 2205 pounds
- 1.0 US gallon = 3.79 liter = 0.833 Imperial gallon
- 1.0 imperial gallon = 4.55 liter = 1.20 US gallon
- 1.0 liter = 0.264 US gallon = 0.220 imperial gallon
- 1.0 US bushel = 0.0352 m^3 = 0.97 UK bushel = 56 lb, 25 kg (corn or sorghum) = 60 lb, 27 kg (wheat or soybeans) = 40 lb, 18 kg (barley)

Areas and crop yields

- 1.0 hectare = $10,000 \text{ m}^2$ (an area 100 m x 100 m, or 328 x 328 ft) = 2.47 acres

- $1.0 \text{ km}^2 = 100 \text{ hectares} = 247 \text{ acres}$
- $1.0 \text{ acre} = 0.405 \text{ hectares}$
- $1.0 \text{ US ton/acre} = 2.24 \text{ t/ha}$
- $1 \text{ metric tonne/hectare} = 0.446 \text{ ton/acre}$
- $100 \text{ g/m}^2 = 1.0 \text{ tonne/hectare} = 892 \text{ lb/acre}$
 - for example, a "target" bioenergy crop yield might be: 5.0 US tons/acre
 $(10,000 \text{ lb/acre}) = 11.2 \text{ tonnes/hectare} (1120 \text{ g/m}^2)$

Biomass energy

- **Cord:** a stack of wood comprising 128 cubic feet (3.62 m^3); standard dimensions are 4 x 4 x 8 feet, including air space and bark. One cord contains approx. 1.2 U.S. tons (oven-dry) = 2400 pounds = 1089 kg
 - 1.0 metric tonne **wood** = 1.4 cubic meters (solid wood, not stacked)
 - Energy content of **wood fuel** (HHV, bone dry) = 18-22 GJ/t (7,600-9,600 Btu/lb)
 - Energy content of **wood fuel** (air dry, 20% moisture) = about 15 GJ/t (6,400 Btu/lb)
- Energy content of **agricultural residues** (range due to moisture content) = 10-17 GJ/t (4,300-7,300 Btu/lb)
- Metric tonne **charcoal** = 30 GJ (= 12,800 Btu/lb) (but usually derived from 6-12 t air-dry wood, i.e. 90-180 GJ original energy content)
- Metric tonne **ethanol** = 7.94 petroleum barrels = 1262 liters
 - ethanol energy content (LHV) = 11,500 Btu/lb = 75,700 Btu/gallon = 26.7 GJ/t = 21.1 MJ/liter. HHV for ethanol = 84,000 Btu/gallon = 89 MJ/gallon = 23.4 MJ/liter
 - ethanol density (average) = 0.79 g/ml (= metric tonnes/ m^3)
- Metric tonne **biodiesel** = 37.8 GJ (33.3 - 35.7 MJ/liter)
 - biodiesel density (average) = 0.88 g/ml (= metric tonnes/ m^3)

Fossil fuels

- **Barrel of oil** equivalent (boe) = approx. 6.1 GJ (5.8 million Btu), equivalent to 1,700 kWh. "Petroleum barrel" is a liquid measure equal to 42 U.S. gallons (35 Imperial gallons or 159 liters); about 7.2 barrels oil are equivalent to one tonne of oil (metric) = 42-45 GJ.
- **Gasoline:** US gallon = 115,000 Btu = 121 MJ = 32 MJ/liter (LHV). HHV = 125,000 Btu/gallon = 132 MJ/gallon = 35 MJ/liter
 - Metric tonne gasoline = 8.53 barrels = 1356 liter = 43.5 GJ/t (LHV); 47.3 GJ/t (HHV)
 - gasoline density (average) = 0.73 g/ml (= metric tonnes/m³)
- **Petro-diesel** = 130,500 Btu/gallon (36.4 MJ/liter or 42.8 GJ/t)
 - petro-diesel density (average) = 0.84 g/ml (= metric tonnes/m³)
- Note that the energy content (heating value) of petroleum products per unit mass is fairly constant, but their density differs significantly – hence the energy content of a liter, gallon, etc. varies between gasoline, diesel and kerosene.
- Metric tonne **coal** = 27-30 GJ (bituminous/anthracite); 15-19 GJ (lignite/sub-bituminous) (the above ranges are equivalent to 11,500-13,000 Btu/lb and 6,500-8,200 Btu/lb).
 - Note that the energy content (heating value) per unit mass varies greatly between different "ranks" of coal. "Typical" coal (rank not specified) usually means bituminous coal, the most common fuel for power plants (27 GJ/t).
- **Natural gas:** HHV = 1027 Btu/ft³ = 38.3 MJ/m³; LHV = 930 Btu/ft³ = 34.6 MJ/m³
 - Therm (used for natural gas, methane) = 100,000 Btu (= 105.5 MJ)

Carbon content of fossil fuels and bioenergy feedstocks

- **coal** (average) = 25.4 metric tonnes carbon per terajoule (TJ)
 - 1.0 metric tonne **coal** = 746 kg carbon
- **oil** (average) = 19.9 metric tonnes carbon / TJ
- 1.0 US gallon **gasoline** (0.833 Imperial gallon, 3.79 liter) = 2.42 kg carbon

- 1.0 US gallon **diesel/fuel oil** (0.833 Imperial gallon, 3.79 liter) = 2.77 kg carbon
- **natural gas (methane)** = 14.4 metric tonnes carbon / TJ
- 1.0 cubic meter **natural gas (methane)** = 0.49 kg carbon
- carbon content of **bioenergy feedstocks**: approx. 50% for woody crops or wood waste; approx. 45% for graminaceous (grass) crops or agricultural residues